

T3 intermediate-EFT one-loop RGE term comparison: EFT\_1 after  $F$

This stage has active field content  $\text{SM}+S_1+S_2$ . The report combines the renormalisable RGE running with the dimension-five  $\psi^2\phi^2$  Wilson-coefficient running generated after integrating out  $F$ . Throughout,  $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$ . For the Wilson sector only the tree-generated stage-1 coefficient  $C^{(0)}$  is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Renormalisable EFT1 couplings

$\beta_{g_Y}^{(1)}$

$$16\pi^2\mu\frac{dg_Y}{d\mu} = \beta_{g_Y}^{(1)}$$

| Column Representative term structure |                     |
|--------------------------------------|---------------------|
| T1                                   | $\frac{43g_Y^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{43g_Y^3}{6}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{43g_Y^3}{6}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{47g_Y^3}{6}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{55g_Y^3}{6}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{67g_Y^3}{6}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{15g_Y^3}{2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{43g_Y^3}{6}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{15g_Y^3}{2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{17g_Y^3}{2}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{61g_Y^3}{6}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{15g_Y^3}{2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{43g_Y^3}{6}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{15g_Y^3}{2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{17g_Y^3}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{61g_Y^3}{6}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{47g_Y^3}{6}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{43g_Y^3}{6}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{43g_Y^3}{6}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{47g_Y^3}{6}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{55g_Y^3}{6}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{47g_Y^3}{6}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{22g_Y^3}{3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{47g_Y^3}{6}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{28g_Y^3}{3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{71g_Y^3}{6}$ |

$$\beta_{g_2}^{(1)}$$

$$16\pi^2\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column Representative term structure

|    |                     |
|----|---------------------|
| T1 | $-\frac{5g_2^3}{2}$ |
|----|---------------------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5g_2^3}{2}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5g_2^3}{2}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5g_2^3}{2}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5g_2^3}{2}$  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{5g_2^3}{2}$  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{17g_2^3}{6}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{17g_2^3}{6}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{17g_2^3}{6}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{17g_2^3}{6}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{17g_2^3}{6}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{17g_2^3}{6}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{17g_2^3}{6}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{17g_2^3}{6}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{17g_2^3}{6}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{17g_2^3}{6}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5g_2^3}{2}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5g_2^3}{2}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5g_2^3}{2}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5g_2^3}{2}$  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{5g_2^3}{2}$  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11g_2^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $-\frac{11g_2^3}{6}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $-\frac{11g_2^3}{6}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{11g_2^3}{6}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{11g_2^3}{6}$ |

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column Representative term structure

|    |           |
|----|-----------|
| T1 | $-7g_3^3$ |
|----|-----------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-7g_3^3$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-7g_3^3$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-7g_3^3$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-7g_3^3$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-7g_3^3$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-7g_3^3$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-7g_3^3$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-7g_3^3$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-7g_3^3$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-7g_3^3$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-7g_3^3$ |

$$\beta_{Y_u}^{(1)}$$

$$16\pi^2 \mu \frac{dY_u}{d\mu} = \beta_{Y_u}^{(1)}$$

## Terms 1 of 3

| Column | Representative term structure                                                            |
|--------|------------------------------------------------------------------------------------------|
| T1     | $3 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right)\left(Y_u\right)_{i j}$   |
| T2     | $+ \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right)\left(Y_u\right)_{i j}$   |
| T3     | $+ 3 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right)\left(Y_u\right)_{i j}$ |

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                          | T2                                                                                          | T3                                                                                            |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) \left( Y_u \right)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) \left( Y_u \right)_{ij}$ | $+ 3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) \left( Y_u \right)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) \left( Y_u \right)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) \left( Y_u \right)_{ij}$ | $+ 3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) \left( Y_u \right)_{ij}$ |

## Terms 2 of 3

| Column | Representative term structure                                             |  |
|--------|---------------------------------------------------------------------------|--|
| T1     | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ |  |
| T2     | $-\frac{9}{4}g_2^2(Y_u)_{ij}$                                             |  |
| T3     | $-8g_3^2(Y_u)_{ij}$                                                       |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                        | T2                            | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------|-------------------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                        | T2                            | T3                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|---------------------------------------------------------------------------|-------------------------------|---------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2(Y_u)_{ij}$ | $-8g_3^2(Y_u)_{ij}$ |

# Terms 3 of 3

Column Representative term structure

|    |                                 |
|----|---------------------------------|
| T1 | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
|----|---------------------------------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|-----------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $-\frac{17}{12}g_Y{}^2(Y_u)_{ij}$ |

$$\beta_{Y_d}^{(1)}$$

$$16\pi^2 \mu \frac{dY_d}{d\mu} = \beta_{Y_d}^{(1)}$$

## Terms 1 of 3

| Column | Representative term structure                                                 |
|--------|-------------------------------------------------------------------------------|
| T1     | $\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$    |
| T2     | $+3 \text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |
| T3     | $+3 \text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                         | T2                                                                           | T3                                                                           |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |

## Terms 2 of 3

| Column | Representative term structure                                             |  |
|--------|---------------------------------------------------------------------------|--|
| T1     | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ |  |
| T2     | $-\frac{9}{4}g_2^2(Y_d)_{ij}$                                             |  |
| T3     | $-8g_3^2(Y_d)_{ij}$                                                       |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                        | T2                            | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------|-------------------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$ | $-\frac{9}{4}g_2^2(Y_d)_{ij}$ | $-8g_3^2(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                                                 | T2                                        | T3                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$ | $-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$ | $-8 g_3^2\left(Y_d\right)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$ | $-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$ | $-8 g_3^2\left(Y_d\right)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$ | $-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$ | $-8 g_3^2\left(Y_d\right)_{ij}$ |

# Terms 3 of 3

Column Representative term structure

|    |                                |
|----|--------------------------------|
| T1 | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
|----|--------------------------------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                               |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $-\frac{5}{12}g_Y{}^2(Y_d)_{ij}$ |

$$\beta_{Y_e}^{(1)}$$

$$16\pi^2 \mu \frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                                                           |
|--------|-----------------------------------------------------------------------------------------|
| T1     | $3 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right)\left(Y_e\right)_{i j}$  |
| T2     | $+3 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right)\left(Y_e\right)_{i j}$ |
| T3     | $+ \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right)\left(Y_e\right)_{i j}$  |

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                          | T2                                                                                           | T3                                                                                          |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) \left( Y_e \right)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) \left( Y_e \right)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) \left( Y_e \right)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) \left( Y_e \right)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) \left( Y_e \right)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) \left( Y_e \right)_{ij}$ |

## Terms 2 of 2

| Column | Representative term structure                                     |
|--------|-------------------------------------------------------------------|
| T1     | $+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{ij}$ |
| T2     | $-\frac{9}{4} g_2^2\left(Y_e\right)_{ij}$                         |
| T3     | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{ij}$                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $T_1$                                                | $T_2$                         | $T_3$                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-------------------------------|--------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}\left(Y_e(Y_e)^\dagger Y_e\right)_{ij}$ | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                       | T2                              | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----------------------------------------------------------|---------------------------------|----------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $+\frac{3}{2} \left( Y_e (Y_e)^\dagger Y_e \right)_{ij}$ | $-\frac{9}{4} g_2^2 (Y_e)_{ij}$ | $-\frac{15}{4} g_Y^2 (Y_e)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $+\frac{3}{2} \left( Y_e (Y_e)^\dagger Y_e \right)_{ij}$ | $-\frac{9}{4} g_2^2 (Y_e)_{ij}$ | $-\frac{15}{4} g_Y^2 (Y_e)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $+\frac{3}{2} \left( Y_e (Y_e)^\dagger Y_e \right)_{ij}$ | $-\frac{9}{4} g_2^2 (Y_e)_{ij}$ | $-\frac{15}{4} g_Y^2 (Y_e)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $+\frac{3}{2} \left( Y_e (Y_e)^\dagger Y_e \right)_{ij}$ | $-\frac{9}{4} g_2^2 (Y_e)_{ij}$ | $-\frac{15}{4} g_Y^2 (Y_e)_{ij}$ |

$$\beta_{m_{S_1}^2}^{(1)}$$

$$16\pi^2\mu\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure |  |
|--------|-------------------------------|--|
| T1     | $-6g_Y^2m_{S_1}^2$            |  |
| T2     | $+6\lambda_{12}m_{S_2}^2$     |  |
| T3     | $+4\lambda_{S_1}m_{S_1}^2$    |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                 | T2                        | T3                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------|---------------------------|----------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-6g_Y^2m_{S_1}^2$ | $+6\lambda_{12}m_{S_2}^2$ | $+4\lambda_{S_1}m_{S_1}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                  | $+6\lambda_{12}m_{S_2}^2$ | $+4\lambda_{S_1}m_{S_1}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                  | $6\lambda_{12}m_{S_2}^2$  | $+4\lambda_{S_1}m_{S_1}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                  | $+6\lambda_{12}m_{S_2}^2$ | $+4\lambda_{S_1}m_{S_1}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-6g_Y^2m_{S_1}^2$ | $+6\lambda_{12}m_{S_2}^2$ | $+4\lambda_{S_1}m_{S_1}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-6g_Y^2m_{S_1}^2$ | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                  | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                  | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                  | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-6g_Y^2m_{S_1}^2$ | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-6g_Y^2m_{S_1}^2$ | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                  | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                  | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                  | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-6g_Y^2m_{S_1}^2$ | $+4\lambda_{12}m_{S_2}^2$ | $+6\lambda_{S_1}m_{S_1}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-6g_Y^2m_{S_1}^2$ | $+2\lambda_{12}m_{S_2}^2$ | —                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                  | $+2\lambda_{12}m_{S_2}^2$ | —                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                  | $+2\lambda_{12}m_{S_2}^2$ | —                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                  | $+2\lambda_{12}m_{S_2}^2$ | —                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-6g_Y^2m_{S_1}^2$ | $+2\lambda_{12}m_{S_2}^2$ | —                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-6g_Y^2m_{S_1}^2$ | —                         | —                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                  | —                         | —                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                  | —                         | —                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                  | —                         | —                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-6g_Y^2m_{S_1}^2$ | —                         | —                          |

## Terms 2 of 3

| Column | Representative term structure |  |
|--------|-------------------------------|--|
| T1     | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ |  |
| T2     | $-\frac{9g_2^2 m_{S_1}^2}{2}$ |  |
| T3     | $-12g_2^2 m_{S_1}^2$          |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                            | T2                            | T3                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------|-------------------------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                             | —                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —                             | —                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                             | —                             | —                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —                             | —                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                             | —                             | —                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                             | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                             | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                             | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                             | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                             | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                             | $-\frac{9g_2^2 m_{S_1}^2}{2}$ | —                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                             | $-12g_2^2 m_{S_1}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —                             | $-12g_2^2 m_{S_1}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                             | —                             | $-12g_2^2 m_{S_1}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —                             | $-12g_2^2 m_{S_1}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                             | —                             | $-12g_2^2 m_{S_1}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                             | $-12g_2^2 m_{S_1}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —                             | $-12g_2^2 m_{S_1}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                            | T2 | T3                   |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-------------------------------|----|----------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —                             | —  | $-12g_2^2 m_{S_1}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —  | $-12g_2^2 m_{S_1}^2$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —                             | —  | $-12g_2^2 m_{S_1}^2$ |



# Terms 3 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ |
| T2     | $+m_{S_2}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                         | T2                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------|------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                          | —                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                          | —                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                          | —                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                          | —                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                          | —                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                          | —                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                          | —                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                          | —                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                          | —                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                          | —                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | $+m_{S_2}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | $+m_{S_2}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | $+m_{S_2}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | $+m_{S_2}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+m_{S_1}^2(8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ | $+m_{S_2}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

$$\beta_{m_{S_2}^2}^{(1)}$$

$$16\pi^2\mu\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                            |  |
|--------|----------------------------------------------------------|--|
| T1     | $-12g_2^2m_{S_2}^2$                                      |  |
| T2     | $+2\lambda_{12}m_{S_1}^2$                                |  |
| T3     | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                        | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|---------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12g_2^2m_{S_2}^2$ | $+2\lambda_{12}m_{S_1}^2$ | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12g_2^2m_{S_2}^2$ | $+2\lambda_{12}m_{S_1}^2$ | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12g_2^2m_{S_2}^2$ | $+2\lambda_{12}m_{S_1}^2$ | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12g_2^2m_{S_2}^2$ | $+2\lambda_{12}m_{S_1}^2$ | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12g_2^2m_{S_2}^2$ | $+2\lambda_{12}m_{S_1}^2$ | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                   | $+4\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | $6\lambda_{12}m_{S_1}^2$  | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                   | $+6\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | $+6\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                   | $+6\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | $+6\lambda_{12}m_{S_1}^2$ | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12g_2^2m_{S_2}^2$ | —                         | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12g_2^2m_{S_2}^2$ | —                         | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12g_2^2m_{S_2}^2$ | —                         | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12g_2^2m_{S_2}^2$ | —                         | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12g_2^2m_{S_2}^2$ | —                         | $+m_{S_2}^2(8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |

## Terms 2 of 3

| Column | Representative term structure |  |  |  |  |  |  |
|--------|-------------------------------|--|--|--|--|--|--|
| T1     | $-\frac{3g_Y^2 m_{S_2}^2}{2}$ |  |  |  |  |  |  |
| T2     | $-6g_Y^2 m_{S_2}^2$           |  |  |  |  |  |  |
| T3     | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                             | T2                   | T3                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------|----------------------|-------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                              | —                    | —                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_2}^2}{2}$  | —                    | —                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                              | $-6g_Y^2 m_{S_2}^2$  | —                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{27g_Y^2 m_{S_2}^2}{2}$ | —                    | —                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                              | $-24g_Y^2 m_{S_2}^2$ | —                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                              | —                    | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_Y^2 m_{S_2}^2}{2}$  | —                    | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                              | $-6g_Y^2 m_{S_2}^2$  | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{27g_Y^2 m_{S_2}^2}{2}$ | —                    | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                              | $-24g_Y^2 m_{S_2}^2$ | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                              | —                    | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3g_Y^2 m_{S_2}^2}{2}$  | —                    | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                              | $-6g_Y^2 m_{S_2}^2$  | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{27g_Y^2 m_{S_2}^2}{2}$ | —                    | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                              | $-24g_Y^2 m_{S_2}^2$ | $-\frac{9g_2^2 m_{S_2}^2}{2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                              | —                    | —                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_2}^2}{2}$  | —                    | —                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                              | $-6g_Y^2 m_{S_2}^2$  | —                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{27g_Y^2 m_{S_2}^2}{2}$ | —                    | —                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                              | $-24g_Y^2 m_{S_2}^2$ | —                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                              | —                    | —                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_2}^2}{2}$  | —                    | —                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                              | T2                    | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|---------------------------------|-----------------------|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —                               | $-6g_Y{}^2m_{S_2}^2$  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{27g_Y{}^2m_{S_2}^2}{2}$ | —                     | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —                               | $-24g_Y{}^2m_{S_2}^2$ | —  |

# Terms 3 of 3

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $+6\lambda_{S_2} m_{S_2}^2$                                 |
| T2     | $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                          | T2                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                           | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                           | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                           | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                           | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                           | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+6\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{S_2} m_{S_2}^2$ | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                           | $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                           | $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                           | $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                           | $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                           | $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

$$\beta_{\lambda_H}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_H}{d\mu} = \beta_{\lambda_H}^{(1)}$$

## Terms 1 of 7

| Column | Representative term structure                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_H (12 \text{Tr} (Y_d (Y_d)^\dagger) + 4 \text{Tr} (Y_e (Y_e)^\dagger) + 12 \text{Tr} (Y_u (Y_u)^\dagger))$ |
| T2     | $-12 \text{Tr} (Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$                                                               |
| T3     | $-4 \text{Tr} (Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$                                                                |

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                        | T2                                                | T3                                               |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\lambda_H(12\text{Tr}(Y_d(Y_d)^\dagger) + 4\text{Tr}(Y_e(Y_e)^\dagger) + 12\text{Tr}(Y_u(Y_u)^\dagger))$ | $-12\text{Tr}(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger)$ | $-4\text{Tr}(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger)$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\lambda_H(12\text{Tr}(Y_d(Y_d)^\dagger) + 4\text{Tr}(Y_e(Y_e)^\dagger) + 12\text{Tr}(Y_u(Y_u)^\dagger))$ | $-12\text{Tr}(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger)$ | $-4\text{Tr}(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger)$ |

## Terms 2 of 7

| Column | Representative term structure                                                                        |
|--------|------------------------------------------------------------------------------------------------------|
| T1     | $-12 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger Y_u \left( Y_u \right)^\dagger \right)$ |
| T2     | $+4 \lambda_{T3} \left( \lambda_{T3} \right)^*$                                                      |
| T3     | $+\frac{9 g_2^4}{4}$                                                                                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $T_1$                                                              | $T_2$                             | $T_3$               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|-----------------------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                 | T2                                | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|--------------------------------------------------------------------|-----------------------------------|---------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |

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| Column | Representative term structure                         |  |  |  |  |  |  |  |  |  |
|--------|-------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ |  |  |  |  |  |  |  |  |  |
| T2     | $-9g_2^2 \lambda_H$                                   |  |  |  |  |  |  |  |  |  |
| T3     | $+\frac{3g_Y^4}{4}$                                   |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                  | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|---------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                  | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|---------------------|---------------------|
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |

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| Column | Representative term structure |  |
|--------|-------------------------------|--|
| T1     | $+12\lambda_H^2$              |  |
| T2     | $+2\lambda_{H1}^2$            |  |
| T3     | $+6\lambda_{H2}^2$            |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1               | T2                 | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------|--------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |

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| Column | Representative term structure            |  |  |  |  |  |  |  |  |  |
|--------|------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $+3\lambda_{H2}^{\text{Adj}^2}$          |  |  |  |  |  |  |  |  |  |
| T2     | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ |  |  |  |  |  |  |  |  |  |
| T3     | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                              | T2                                       | T3                                       |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------------|------------------------------------------|------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                        | —                                        |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                               | —                                        | —                                        |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                               | —                                        | —                                        |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                               | —                                        | —                                        |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                               | —                                        | —                                        |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                              | T2 | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|---------------------------------|----|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | —  | —  |

# Terms 6 of 7

| Column | Representative term structure                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$             |
| T2     | $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |
| T3     | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$                                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                               | T2                                                                                                                                           | T3                                                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |

| Column |  | Representative term structure   |
|--------|--|---------------------------------|
| T1     |  | $+3\lambda_{H1}^{\text{Adj}^2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |



$$\beta_{\lambda_{S_1}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$$

# Terms 1 of 9

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $12g_Y^4$                     |
| T2     | $-12g_Y^2\lambda_{S_1}$       |
| T3     | $+6\lambda_{12}^2$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1         | T2                      | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------|-------------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $12g_Y^4$  | $-12g_Y^2\lambda_{S_1}$ | $+6\lambda_{12}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | $-3g_Y^2\lambda_{S_1}$  | $+6\lambda_{12}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | —                       | $6\lambda_{12}^2$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | $-3g_Y^2\lambda_{S_1}$  | $+6\lambda_{12}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $12g_Y^4$  | $-12g_Y^2\lambda_{S_1}$ | $+6\lambda_{12}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+12g_Y^4$ | —                       | $+4\lambda_{12}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —          | —                       | $+4\lambda_{12}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —          | —                       | $+4\lambda_{12}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —          | —                       | $+4\lambda_{12}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+12g_Y^4$ | —                       | $+4\lambda_{12}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+12g_Y^4$ | —                       | $+4\lambda_{12}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —          | —                       | $+4\lambda_{12}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —          | —                       | $+4\lambda_{12}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —          | —                       | $+4\lambda_{12}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+12g_Y^4$ | —                       | $+4\lambda_{12}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+12g_Y^4$ | —                       | $+2\lambda_{12}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —          | —                       | $+2\lambda_{12}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —          | —                       | $+2\lambda_{12}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —          | —                       | $+2\lambda_{12}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+12g_Y^4$ | —                       | $+2\lambda_{12}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12g_Y^4$ | —                       | $+6\lambda_{12}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                       | $+6\lambda_{12}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | —                       | $+6\lambda_{12}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                       | $+6\lambda_{12}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12g_Y^4$ | —                       | $+6\lambda_{12}^2$ |

## Terms 2 of 9

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+4\lambda_{H1}^2$            |
| T2     | $+10\lambda_{S1}^2$           |
| T3     | $\frac{3g_Y^4}{4}$            |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                 | T2                  | T3                  |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|--------------------|---------------------|---------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H1}^2$ | $+10\lambda_{S1}^2$ | —                   |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H1}^2$ | $+10\lambda_{S1}^2$ | $\frac{3g_Y^4}{4}$  |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H1}^2$ | $+10\lambda_{S1}^2$ | —                   |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H1}^2$ | $+10\lambda_{S1}^2$ | $\frac{3g_Y^4}{4}$  |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H1}^2$ | $+10\lambda_{S1}^2$ | —                   |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | —                   |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | —                   |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | —                   |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | —                   |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | —                   |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{H1}^2$ | $+12\lambda_{S1}^2$ | —                   |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | —                   |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | —                   |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | —                   |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | —                   |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | —                   |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H1}^2$ | $+14\lambda_{S1}^2$ | —                   |

# Terms 3 of 9

| Column | Representative term structure      |
|--------|------------------------------------|
| T1     | $\frac{9g_2^4}{4}$                 |
| T2     | $+g_Y^2(6g_2^2 - 12\lambda_{S_1})$ |
| T3     | $-9g_2^2\lambda_{S_1}$             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                                 | T3                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|------------------------------------|------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                                  | —                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                                  | —                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                                  | —                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                                  | —                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                                  | —                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{9g_2^4}{4}$  | $+g_Y^2(6g_2^2 - 12\lambda_{S_1})$ | $-9g_2^2\lambda_{S_1}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{9g_2^4}{4}$ | —                                  | $-9g_2^2\lambda_{S_1}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{9g_2^4}{4}$  | —                                  | $-9g_2^2\lambda_{S_1}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{9g_2^4}{4}$  | —                                  | $-9g_2^2\lambda_{S_1}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{9g_2^4}{4}$  | $+g_Y^2(6g_2^2 - 12\lambda_{S_1})$ | $-9g_2^2\lambda_{S_1}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{9g_2^4}{4}$  | $+g_Y^2(6g_2^2 - 12\lambda_{S_1})$ | $-9g_2^2\lambda_{S_1}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{9g_2^4}{4}$ | —                                  | $-9g_2^2\lambda_{S_1}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{9g_2^4}{4}$  | —                                  | $-9g_2^2\lambda_{S_1}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{9g_2^4}{4}$  | —                                  | $-9g_2^2\lambda_{S_1}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{9g_2^4}{4}$  | $+g_Y^2(6g_2^2 - 12\lambda_{S_1})$ | $-9g_2^2\lambda_{S_1}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                                  | —                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                   | —                                  | —                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | —                                  | —                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                   | —                                  | —                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | —                                  | —                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                                  | —                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                                  | —                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                                  | —                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                                  | —                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                                  | —                      |

# Terms 4 of 9

| Column | Representative term structure                                                                                                   |  |  |  |  |  |  |
|--------|---------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|
| T1     | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$                                                                                        |  |  |  |  |  |  |
| T2     | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$                                                                                        |  |  |  |  |  |  |
| T3     | $8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                       | T2                                       | T3                                                                                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                        | —                                                                                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                        | —                                                                                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                        | —                                                                                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                        | —                                                                                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                        | —                                                                                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                        | —                                                                                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                        | —                                                                                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                        | —                                                                                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                        | —                                                                                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                        | —                                                                                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                        | —                                                                                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                        | —                                                                                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                        | —                                                                                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                        | —                                                                                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                        | —                                                                                                                               |

# Terms 5 of 9

| Column | Representative term structure                                                                         |
|--------|-------------------------------------------------------------------------------------------------------|
| T1     | $+8\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left( \lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^*$      |
| T2     | $+24\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*$ |
| T3     | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$                                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                               | T2                                                                                                    | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                | —                                                                                                     | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                | —                                                                                                     | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                | —                                                                                                     | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                | —                                                                                                     | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left( \lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^*$ | $+24\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                | —                                                                                                     | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                | —                                                                                                     | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left( \lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^*$ | $+24\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                | —                                                                                                     | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                | —                                                                                                     | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                | —                                                                                                     | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                | —                                                                                                     | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                | —                                                                                                     | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                | —                                                                                                     | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                | —                                                                                                     | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                | —                                                                                                     | —                                                         |

# Terms 6 of 9

| Column | Representative term structure                                                   |  |
|--------|---------------------------------------------------------------------------------|--|
| T1     | $21g_2^4$                                                                       |  |
| T2     | $+g_Y^2 \left( 12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$ |  |
| T3     | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$                      |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1        | T2                                                                              | T3                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------|---------------------------------------------------------------------------------|------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —         | —                                                                               | —                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —         | —                                                                               | —                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —         | —                                                                               | —                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —         | —                                                                               | —                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —         | —                                                                               | —                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —         | —                                                                               | —                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —         | —                                                                               | —                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —         | —                                                                               | —                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —         | —                                                                               | —                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —         | —                                                                               | —                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —         | —                                                                               | —                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —         | —                                                                               | —                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —         | —                                                                               | —                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —         | —                                                                               | —                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —         | —                                                                               | —                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $21g_2^4$ | $+g_Y^2 \left( 12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $21g_2^4$ | —                                                                               | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $21g_2^4$ | —                                                                               | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $21g_2^4$ | —                                                                               | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $21g_2^4$ | $+g_Y^2 \left( 12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $21g_2^4$ | $+g_Y^2 \left( 12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $21g_2^4$ | —                                                                               | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $21g_2^4$ | —                                                                               | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $21g_2^4$ | —                                                                               | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $21g_2^4$ | $+g_Y^2 \left( 12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$ |

# Terms 7 of 9

| Column | Representative term structure              |  |
|--------|--------------------------------------------|--|
| T1     | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$  |  |
| T2     | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |  |
| T3     | $+26\lambda_{S1}^{\text{Adj}^2}$           |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                        | T2                                         | T3                               |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-------------------------------------------|--------------------------------------------|----------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                          | —                                |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                          | —                                |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                          | —                                |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                          | —                                |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                          | —                                |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                         | —                                          | —                                |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                         | —                                          | —                                |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                         | —                                          | —                                |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                         | —                                          | —                                |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                         | —                                          | —                                |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                         | —                                          | —                                |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                         | —                                          | —                                |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                         | —                                          | —                                |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                         | —                                          | —                                |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                         | —                                          | —                                |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ | $+26\lambda_{S1}^{\text{Adj}^2}$ |

# Terms 8 of 9

| Column | Representative term structure                                                           |  |
|--------|-----------------------------------------------------------------------------------------|--|
| T1     | $+g_Y^2 \left( 3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$ |  |
| T2     | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$                                             |  |
| T3     | $+6\lambda_{12}^{\text{Adj}^2}$                                                         |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                      | T2                                          | T3                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                       | —                                           | —                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                       | —                                           | —                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                       | —                                           | —                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                       | —                                           | —                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                       | —                                           | —                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                       | —                                           | —                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                       | —                                           | —                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                       | —                                           | —                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                       | —                                           | —                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                       | —                                           | —                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                       | —                                           | —                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                       | —                                           | —                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                       | —                                           | —                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                       | —                                           | —                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                       | —                                           | —                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                       | —                                           | —                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( 3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$ | —                                           | —                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                       | —                                           | —                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( 3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$ | —                                           | —                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                       | —                                           | —                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( 3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$ | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( 3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$ | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |



| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+6\lambda_{12}^{\text{Cross}^2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}^{\text{Cross}^2}$ |

$$\beta_{\lambda_{S_2}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

# Terms 1 of 9

| Column | Representative term structure                           |  |  |  |  |  |  |  |  |  |
|--------|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $21g_2^4$                                               |  |  |  |  |  |  |  |  |  |
| T2     | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ |  |  |  |  |  |  |  |  |  |
| T3     | $+2\lambda_{12}^2$                                      |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1        | T2                                                      | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------|---------------------------------------------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —         | —                                                       | $+4\lambda_{12}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —         | —                                                       | $6\lambda_{12}^2$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —         | —                                                       | $+6\lambda_{12}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —         | —                                                       | $+6\lambda_{12}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —         | —                                                       | $+6\lambda_{12}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —         | —                                                       | $+6\lambda_{12}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $21g_2^4$ | $+g_2^2(-24\lambda_{S_2}-12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |

## Terms 2 of 9

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $+4\lambda_{H2}^2$                        |
| T2     | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ |
| T3     | $+14\lambda_{S2}^2$                       |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                 | T2                                        | T3                  |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|--------------------|-------------------------------------------|---------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | —                                         | $+12\lambda_{S2}^2$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | —                                         | $+10\lambda_{S2}^2$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}^2$ | —                                         | $+10\lambda_{S2}^2$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | —                                         | $+10\lambda_{S2}^2$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}^2$ | —                                         | $+10\lambda_{S2}^2$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | —                                         | $+10\lambda_{S2}^2$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S2}^2$ |

# Terms 3 of 9

| Column | Representative term structure                                                        |  |
|--------|--------------------------------------------------------------------------------------|--|
| T1     | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$                                         |  |
| T2     | $+26\lambda_{S_2}^{\text{Adj}^2}$                                                    |  |
| T3     | $+g_Y{}^2\left(3g_2{}^2-3\lambda_{S_2}-\frac{3\lambda_{S_2}^{\text{Adj}}}{2}\right)$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                | T3                                                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y{}^2\left(3g_2{}^2-3\lambda_{S_2}-\frac{3\lambda_{S_2}^{\text{Adj}}}{2}\right)$    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y{}^2\left(27g_2{}^2-27\lambda_{S_2}-\frac{27\lambda_{S_2}^{\text{Adj}}}{2}\right)$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                 | —                                                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                 | —                                                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                 | —                                                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                 | —                                                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                 | —                                                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                 | —                                                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                 | —                                                                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                 | —                                                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                 | —                                                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                 | —                                                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                 | —                                                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                 | —                                                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                 | —                                                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                 | —                                                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                 | —                                                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y{}^2\left(3g_2{}^2-3\lambda_{S_2}-\frac{3\lambda_{S_2}^{\text{Adj}}}{2}\right)$    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y{}^2\left(27g_2{}^2-27\lambda_{S_2}-\frac{27\lambda_{S_2}^{\text{Adj}}}{2}\right)$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+14\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                       |

# Terms 4 of 9

| Column | Representative term structure                                            |  |  |  |  |  |  |  |  |  |
|--------|--------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $+\frac{3g_Y^4}{4}$                                                      |  |  |  |  |  |  |  |  |  |
| T2     | $+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$ |  |  |  |  |  |  |  |  |  |
| T3     | $+12g_Y^4$                                                               |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                    | T2                                                                        | T3          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------|---------------------------------------------------------------------------|-------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                         | —           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{4}$   | —                                                                         | —           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | $+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$  | $+12g_Y^4$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{243g_Y^4}{4}$ | —                                                                         | —           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | $+g_Y^2\left(48g_2^2-48\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}}\right)$ | $+192g_Y^4$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                     | —                                                                         | —           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{3g_Y^4}{4}$   | —                                                                         | —           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                     | —                                                                         | $+12g_Y^4$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{243g_Y^4}{4}$ | —                                                                         | —           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                     | —                                                                         | $+192g_Y^4$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                     | —                                                                         | —           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{3g_Y^4}{4}$   | —                                                                         | —           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                     | —                                                                         | $+12g_Y^4$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{243g_Y^4}{4}$ | —                                                                         | —           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                     | —                                                                         | $+192g_Y^4$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                         | —           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3g_Y^4}{4}$    | —                                                                         | —           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                         | $12g_Y^4$   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{243g_Y^4}{4}$  | —                                                                         | —           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                         | $192g_Y^4$  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                         | —           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{4}$   | —                                                                         | —           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | $+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$  | $+12g_Y^4$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{243g_Y^4}{4}$ | —                                                                         | —           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | $+g_Y^2\left(48g_2^2-48\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}}\right)$ | $+192g_Y^4$ |

# Terms 5 of 9

| Column | Representative term structure           |
|--------|-----------------------------------------|
| T1     | $\frac{9g_2^4}{4}$                      |
| T2     | $-9g_2^2\lambda_{S_2}$                  |
| T3     | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                     | T3                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|------------------------|-----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                      | —                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                      | —                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                      | —                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                      | —                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                      | —                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{9g_2^4}{4}$  | $-9g_2^2\lambda_{S_2}$ | $+\frac{\lambda_{12}^{\text{Adj}2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                      | —                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                   | —                      | —                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | —                      | —                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                   | —                      | —                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | —                      | —                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                      | —                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                      | —                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                      | —                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                      | —                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                      | —                                       |

# Terms 6 of 9

| Column | Representative term structure                                                                            |
|--------|----------------------------------------------------------------------------------------------------------|
| T1     | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$                                                                 |
| T2     | $8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$          |
| T3     | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                       | T2                                                                                              | T3                                                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                                                                               | —                                                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                                                                               | —                                                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                                                                               | —                                                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                                                                               | —                                                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                                                                               | —                                                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | $8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | $8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                               | —                                                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                                                                               | —                                                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                                                                               | —                                                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                                                                               | —                                                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                                                                               | —                                                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                                                                               | —                                                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                                                                               | —                                                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                                                                               | —                                                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                                                                               | —                                                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                                                                               | —                                                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                                                                               | —                                                                                                        |

# Terms 7 of 9

| Column | Representative term structure                                                                         |  |
|--------|-------------------------------------------------------------------------------------------------------|--|
| T1     | $+24\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*$ |  |
| T2     | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$                                             |  |
| T3     | $+g_Y^2 \left( 6g_2^2 - 12\lambda_{S_2} \right)$                                                      |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                    | T2                                                          | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                     | —                                                           | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                     | —                                                           | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                     | —                                                           | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                     | —                                                           | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                     | —                                                           | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                     | —                                                           | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+24\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$   | —                                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                     | —                                                           | $+g_Y^2 \left( 6g_2^2 - 12\lambda_{S_2} \right)$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                     | $+g_Y^2 \left( \frac{27g_2^2}{2} - 27\lambda_{S_2} \right)$ | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                     | —                                                           | $+g_Y^2 \left( 24g_2^2 - 48\lambda_{S_2} \right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                     | —                                                           | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+24\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$   | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                     | —                                                           | $+g_Y^2 \left( 6g_2^2 - 12\lambda_{S_2} \right)$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                     | $+g_Y^2 \left( \frac{27g_2^2}{2} - 27\lambda_{S_2} \right)$ | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                     | —                                                           | $+g_Y^2 \left( 24g_2^2 - 48\lambda_{S_2} \right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                     | —                                                           | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                     | —                                                           | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                     | —                                                           | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                     | —                                                           | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                     | —                                                           | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                     | —                                                           | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                     | —                                                           | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                     | —                                                           | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                     | —                                                           | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                     | —                                                           | —                                                 |



# Terms 8 of 9

| Column | Representative term structure               |  |
|--------|---------------------------------------------|--|
| T1     | $-3g_Y^2\lambda_{S_2}$                      |  |
| T2     | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |  |
| T3     | $+6\lambda_{12}^{\text{Adj}^2}$             |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                      | T2                                          | T3                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------|---------------------------------------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                       | —                                           | —                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                       | —                                           | —                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                       | —                                           | —                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                       | —                                           | —                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                       | —                                           | —                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                       | —                                           | —                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                       | —                                           | —                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                       | —                                           | —                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                       | —                                           | —                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                       | —                                           | —                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                       | —                                           | —                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                       | —                                           | —                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                       | —                                           | —                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                       | —                                           | —                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                       | —                                           | —                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                       | —                                           | —                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-3g_Y^2\lambda_{S_2}$  | —                                           | —                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-12g_Y^2\lambda_{S_2}$ | —                                           | —                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-27g_Y^2\lambda_{S_2}$ | —                                           | —                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-48g_Y^2\lambda_{S_2}$ | —                                           | —                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+6\lambda_{12}^{\text{Adj}^2}$ |

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+6\lambda_{12}^{\text{Cross}^2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}^{\text{Cross}^2}$ |

$$\beta_{\lambda_{H1}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$$

## Terms 1 of 7

| Column | Representative term structure                                                                                                              |  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |  |
| T2     | $+12\lambda_{T3}\,(\lambda_{T3})^*$                                                                                                        |  |
| T3     | $-\frac{9g_2^2\lambda_{H1}}{2}$                                                                                                            |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                         | T2                                  | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ | $-\frac{9g_2^2\lambda_{H1}}{2}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ | $-\frac{9g_2^2\lambda_{H1}}{2}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ | $-\frac{9g_2^2\lambda_{H1}}{2}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ | $-\frac{9g_2^2\lambda_{H1}}{2}$  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ | $-\frac{9g_2^2\lambda_{H1}}{2}$  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\lambda_{H1}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  | —                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                          | $+4\lambda_{T3}\,(\lambda_{T3})^*$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                               | T3                               |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|----------------------------------|----------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $+4\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $+4\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $+4\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ |

## Terms 2 of 7

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $+3g_Y^4$                        |
| T2     | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3     | $+6\lambda_{12}\lambda_{H2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1        | T2                               | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------|----------------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+6\lambda_{12}\lambda_{H2}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —         | —                                | $+6\lambda_{12}\lambda_{H2}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  | $+6\lambda_{12}\lambda_{H2}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —         | —                                | $+6\lambda_{12}\lambda_{H2}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+6\lambda_{12}\lambda_{H2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+4\lambda_{12}\lambda_{H2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —         | —                                | $+4\lambda_{12}\lambda_{H2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  | $+4\lambda_{12}\lambda_{H2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —         | —                                | $+4\lambda_{12}\lambda_{H2}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+4\lambda_{12}\lambda_{H2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+4\lambda_{12}\lambda_{H2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —         | —                                | $+4\lambda_{12}\lambda_{H2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  | $+4\lambda_{12}\lambda_{H2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —         | —                                | $+4\lambda_{12}\lambda_{H2}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+4\lambda_{12}\lambda_{H2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+2\lambda_{12}\lambda_{H2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —         | —                                | $+2\lambda_{12}\lambda_{H2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  | $+2\lambda_{12}\lambda_{H2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —         | —                                | $+2\lambda_{12}\lambda_{H2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | $+2\lambda_{12}\lambda_{H2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | —                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —         | —                                | —                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  | —                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —         | —                                | —                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ | —                            |

# Terms 3 of 7

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+6\lambda_H\lambda_{H1}$     |
| T2     | $+4\lambda_{H1}^2$            |
| T3     | $+4\lambda_{H1}\lambda_{S1}$  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                        | T2                 | T3                           |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------|--------------------|------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H1}\lambda_{S1}$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H1}\lambda_{S1}$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H1}\lambda_{S1}$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H1}\lambda_{S1}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H1}\lambda_{S1}$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+6\lambda_{H1}\lambda_{S1}$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ | $+8\lambda_{H1}\lambda_{S1}$ |

# Terms 4 of 7

| Column | Representative term structure |  |  |  |  |  |  |
|--------|-------------------------------|--|--|--|--|--|--|
| T1     | $+\frac{3g_Y^4}{4}$           |  |  |  |  |  |  |
| T2     | $-3g_Y^2\lambda_{H1}$         |  |  |  |  |  |  |
| T3     | $+\frac{9g_2^4}{4}$           |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                    | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|-----------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | —                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | —                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                   | —                     | $+\frac{9g_2^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                   | —                     | $+\frac{9g_2^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                   | —                     | $+\frac{9g_2^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                   | —                     | $+\frac{9g_2^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                   | —                     | $+\frac{9g_2^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                   | —                     | $+\frac{9g_2^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | —                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | —                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | —                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ | —                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----|----|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | —  | —  | —  |



# Terms 5 of 7

| Column | Representative term structure                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-9g_2^2\lambda_{H1}$                                                                                                          |
| T2     | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$                                                                                      |
| T3     | $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                    | T2                                        | T3                                                                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                         | —                                                                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                         | —                                                                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                         | —                                                                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                         | —                                                                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                         | —                                                                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ | —                                                                                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                         | —                                                                                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                         | —                                                                                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                         | —                                                                                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                         | —                                                                                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                         | —                                                                                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                         | —                                                                                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                         | —                                                                                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                         | —                                                                                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                         | —                                                                                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                         | —                                                                                                                              |

# Terms 6 of 7

Column Representative term structure

|    |                                                                                                                                                                                                              |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1 | $+3\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$                                                                                     |
| T2 | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*)$ |
| T3 | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                       | T2                                                                                                                                                                                                           | T3                                                                                                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+3\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*)$ | —                                                                                                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+3\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (4 \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*)$ | —                                                                                                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | —                                                                                                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                                            | $\lambda_{H1} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4 \lambda_{S_1}^{\text{Adj}})$ |

# Terms 7 of 7

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $+6g_2^4$                                                     |
| T2     | $+6\lambda_{H1}^{\text{Adj}^2}$                               |
| T3     | $+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1        | T2                              | T3                                                            |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-----------|---------------------------------|---------------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —         | —                               | —                                                             |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —         | —                               | —                                                             |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —         | —                               | —                                                             |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —         | —                               | —                                                             |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —         | —                               | —                                                             |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —         | —                               | —                                                             |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —         | —                               | —                                                             |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —         | —                               | —                                                             |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —         | —                               | —                                                             |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —         | —                               | —                                                             |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —         | —                               | —                                                             |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —         | —                               | —                                                             |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —         | —                               | —                                                             |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —         | —                               | —                                                             |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —         | —                               | —                                                             |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                             |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                             |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                             |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                             |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                             |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6g_2^4$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

$$\beta_{\lambda_{H2}}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_{H2}}{d\mu} = \beta_{\lambda_{H2}}^{(1)}$$

## Terms 1 of 7

| Column | Representative term structure                                                                                                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ |
| T2     | $+4\lambda_{T3} (\lambda_{T3})^*$                                                                                                              |
| T3     | $+6g_2^4$                                                                                                                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                             | T2                                 | T3        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                              | $+6\lambda_{T3} (\lambda_{T3})^*$  | —         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                              | $+12\lambda_{T3} (\lambda_{T3})^*$ | —         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                              | $+12\lambda_{T3} (\lambda_{T3})^*$ | —         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                              | $+12\lambda_{T3} (\lambda_{T3})^*$ | —         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                              | $+12\lambda_{T3} (\lambda_{T3})^*$ | —         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                              | $+12\lambda_{T3} (\lambda_{T3})^*$ | —         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H2}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ | $+4\lambda_{T3} (\lambda_{T3})^*$  | $+6g_2^4$ |

## Terms 2 of 7

| Column | Representative term structure    |  |  |  |  |  |  |  |  |  |
|--------|----------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $-\frac{33g_2^2\lambda_{H2}}{2}$ |  |  |  |  |  |  |  |  |  |
| T2     | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |  |  |  |  |  |  |  |  |  |
| T3     | $+2\lambda_{12}\lambda_{H1}$     |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                               | T2                               | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------|----------------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{3g_Y^2\lambda_{H2}}{2}$  | $+2\lambda_{12}\lambda_{H1}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                | $+2\lambda_{12}\lambda_{H1}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{15g_Y^2\lambda_{H2}}{2}$ | $+2\lambda_{12}\lambda_{H1}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                | $+2\lambda_{12}\lambda_{H1}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{51g_Y^2\lambda_{H2}}{2}$ | $+2\lambda_{12}\lambda_{H1}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                | $-\frac{3g_Y^2\lambda_{H2}}{2}$  | $+4\lambda_{12}\lambda_{H1}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                | —                                | $+4\lambda_{12}\lambda_{H1}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                | $-\frac{15g_Y^2\lambda_{H2}}{2}$ | $+4\lambda_{12}\lambda_{H1}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                | —                                | $+4\lambda_{12}\lambda_{H1}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                | $-\frac{51g_Y^2\lambda_{H2}}{2}$ | $+4\lambda_{12}\lambda_{H1}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                | $-\frac{3g_Y^2\lambda_{H2}}{2}$  | $+4\lambda_{12}\lambda_{H1}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                | —                                | $+4\lambda_{12}\lambda_{H1}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                | $-\frac{15g_Y^2\lambda_{H2}}{2}$ | $+4\lambda_{12}\lambda_{H1}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                | —                                | $+4\lambda_{12}\lambda_{H1}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                | $-\frac{51g_Y^2\lambda_{H2}}{2}$ | $+4\lambda_{12}\lambda_{H1}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{9g_2^2\lambda_{H2}}{2}$  | $-\frac{3g_Y^2\lambda_{H2}}{2}$  | $+6\lambda_{12}\lambda_{H1}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{9g_2^2\lambda_{H2}}{2}$  | —                                | $+6\lambda_{12}\lambda_{H1}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{9g_2^2\lambda_{H2}}{2}$  | $-\frac{15g_Y^2\lambda_{H2}}{2}$ | $+6\lambda_{12}\lambda_{H1}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{9g_2^2\lambda_{H2}}{2}$  | —                                | $+6\lambda_{12}\lambda_{H1}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9g_2^2\lambda_{H2}}{2}$  | $-\frac{51g_Y^2\lambda_{H2}}{2}$ | $+6\lambda_{12}\lambda_{H1}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{3g_Y^2\lambda_{H2}}{2}$  | —                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                | —                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                               | T2                               | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----------------------------------|----------------------------------|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{15g_Y^2\lambda_{H2}}{2}$ | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{51g_Y^2\lambda_{H2}}{2}$ | —  |

# Terms 3 of 7

| Column | Representative term structure |  |
|--------|-------------------------------|--|
| T1     | $+6\lambda_H\lambda_{H2}$     |  |
| T2     | $+4\lambda_{H2}^2$            |  |
| T3     | $+8\lambda_{H2}\lambda_{S2}$  |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                        | T2                 | T3                           |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------|--------------------|------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+6\lambda_{H2}\lambda_{S2}$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+4\lambda_{H2}\lambda_{S2}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+4\lambda_{H2}\lambda_{S2}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+4\lambda_{H2}\lambda_{S2}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+4\lambda_{H2}\lambda_{S2}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+4\lambda_{H2}\lambda_{S2}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ | $+8\lambda_{H2}\lambda_{S2}$ |

# Terms 4 of 7

| Column | Representative term structure   |  |
|--------|---------------------------------|--|
| T1     | $+6\lambda_{H2}^{\text{Adj}^2}$ |  |
| T2     | $+\frac{3g_Y^4}{4}$             |  |
| T3     | $-3g_Y^2\lambda_{H2}$           |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                              | T2                   | T3                     |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------------|----------------------|------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    | —                      |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{3g_Y^4}{4}$  | $-3g_Y^2\lambda_{H2}$  |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    | —                      |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{27g_Y^4}{4}$ | $-15g_Y^2\lambda_{H2}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    | —                      |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                               | —                    | —                      |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                               | $+\frac{3g_Y^4}{4}$  | $-3g_Y^2\lambda_{H2}$  |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                               | —                    | —                      |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                               | $+\frac{27g_Y^4}{4}$ | $-15g_Y^2\lambda_{H2}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                               | —                    | —                      |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                               | —                    | —                      |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                               | $+\frac{3g_Y^4}{4}$  | $-3g_Y^2\lambda_{H2}$  |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                               | —                    | —                      |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                               | $+\frac{27g_Y^4}{4}$ | $-15g_Y^2\lambda_{H2}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                               | —                    | —                      |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                               | —                    | —                      |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                               | $+\frac{3g_Y^4}{4}$  | $-3g_Y^2\lambda_{H2}$  |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                               | —                    | —                      |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                               | $+\frac{27g_Y^4}{4}$ | $-15g_Y^2\lambda_{H2}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                               | —                    | —                      |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    | —                      |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{3g_Y^4}{4}$  | $-3g_Y^2\lambda_{H2}$  |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    | —                      |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{27g_Y^4}{4}$ | $-15g_Y^2\lambda_{H2}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    | —                      |



# Terms 5 of 7

| Column | Representative term structure                                                                              |
|--------|------------------------------------------------------------------------------------------------------------|
| T1     | $+3g_Y^4$                                                                                                  |
| T2     | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ |
| T3     | $+\frac{9g_2^4}{4}$                                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1         | T2                                                                                                         | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------|------------------------------------------------------------------------------------------------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | —                                                                                                          | —                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                                                                                                          | —                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3g_Y^4$  | —                                                                                                          | —                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                                                                                                          | —                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12g_Y^4$ | —                                                                                                          | —                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+3g_Y^4$  | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+12g_Y^4$ | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+3g_Y^4$  | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+12g_Y^4$ | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | —                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | —                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+3g_Y^4$  | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | —                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —          | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | —                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+12g_Y^4$ | $\lambda_{H2}(6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$ | —                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | —                                                                                                          | —                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                                                                                                          | —                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3g_Y^4$  | —                                                                                                          | —                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                                                                                                          | —                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12g_Y^4$ | —                                                                                                          | —                   |

# Terms 6 of 7

| Column | Representative term structure                                                                                           |  |
|--------|-------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $-9g_2^2\lambda_{H2}$                                                                                                   |  |
| T2     | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$                                                                               |  |
| T3     | $+3\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                    | T2                                        | T3                                                                                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                         | —                                                                                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                         | —                                                                                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                         | —                                                                                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                         | —                                                                                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                         | —                                                                                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | $+3\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | $+3\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ | —                                                                                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                         | —                                                                                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                         | —                                                                                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                         | —                                                                                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                         | —                                                                                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                         | —                                                                                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                         | —                                                                                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                         | —                                                                                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                         | —                                                                                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                         | —                                                                                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                         | —                                                                                                                       |

# Terms 7 of 7

| Column | Representative term structure                                                                                                                                                                       |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)$ |  |
| T2     | $+24\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$                                                                                                      |  |
| T3     | $+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$                                                                                                                                         |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                              | T2 | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)+24\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ | —  | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*)+24\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ | —  | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                               | —  | $+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                               | —  | $+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | $+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                               | —  | $+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                               | —  | $+\lambda_{H1}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

$$\beta_{\lambda_{12}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$$

# Terms 1 of 10

| Column | Representative term structure   |  |
|--------|---------------------------------|--|
| T1     | $4\lambda_{T3}(\lambda_{T3})^*$ |  |
| T2     | $-12g_2^2\lambda_{12}$          |  |
| T3     | $-6g_Y^2\lambda_{12}$           |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                               | T2                     | T3                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------|------------------------|------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-3g_Y^2\lambda_{12}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-15g_Y^2\lambda_{12}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-30g_Y^2\lambda_{12}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+3\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{12}$  | $-3g_Y^2\lambda_{12}$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-15g_Y^2\lambda_{12}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-30g_Y^2\lambda_{12}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+3\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{12}$  | $-3g_Y^2\lambda_{12}$  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-15g_Y^2\lambda_{12}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $3\lambda_{T3}(\lambda_{T3})^*$  | $-9g_2^2\lambda_{12}$  | $-30g_Y^2\lambda_{12}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-3g_Y^2\lambda_{12}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-15g_Y^2\lambda_{12}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $4\lambda_{T3}(\lambda_{T3})^*$  | $-12g_2^2\lambda_{12}$ | $-30g_Y^2\lambda_{12}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                | —                      | —                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                | —                      | —                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                | —                      | —                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                | —                      | —                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                | —                      | —                      |

# Terms 2 of 10

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+4\lambda_{12}^2$            |
| T2     | $+4\lambda_{12}\lambda_{S_1}$ |
| T3     | $+8\lambda_{12}\lambda_{S_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                 | T2                            | T3                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------|-------------------------------|-------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}^2$ | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{12}^2$ | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{12}^2$ | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{12}^2$ | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{12}^2$ | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{12}^2$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}^2$ | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{12}^2$ | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{12}^2$ | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{12}^2$ | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{12}^2$ | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}^2$ | —                             | —                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{12}^2$ | —                             | —                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{12}^2$ | —                             | —                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{12}^2$ | —                             | —                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{12}^2$ | —                             | —                             |

# Terms 3 of 10

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$ |
| T2     | $+4\lambda_{H1}\lambda_{H2}$               |
| T3     | $+\frac{3g_Y^4}{4}$                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                           | T3                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|------------------------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$ | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$ | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$ | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$ | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{12}\lambda_{S_2}^{\text{Adj}}$ | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | $+4\lambda_{H1}\lambda_{H2}$ | —                    |

# Terms 4 of 10

| Column | Representative term structure             |  |
|--------|-------------------------------------------|--|
| T1     | $+48g_Y^4$                                |  |
| T2     | $+\frac{9g_2^4}{4}$                       |  |
| T3     | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1         | T2                  | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------|---------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | —                   | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                   | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | —                   | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                   | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+48g_Y^4$ | —                   | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+48g_Y^4$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —          | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+48g_Y^4$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —          | —                   | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —          | —                   | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —          | —                   | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —          | —                   | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+48g_Y^4$ | —                   | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | —                   | —                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                   | —                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | —                   | —                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                   | —                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+48g_Y^4$ | —                   | —                                         |

# Terms 5 of 10

| Column | Representative term structure                                                                                                                                         |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$                           |  |
| T2     | $+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$                                                             |  |
| T3     | $+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \left( \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* + \left( \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                          | T2                                                                                                        | T3                                                                                                                                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ | $+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \left( \left( \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* + \left( \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^* \right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ | $+9\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \left( \left( \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* + \left( \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^* \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                           | —                                                                                                         | —                                                                                                                                                                                    |



# Terms 6 of 10

| Column | Representative term structure                                                                                                                                                  |  |  |  |  |  |  |  |  |  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right)$ |  |  |  |  |  |  |  |  |  |
| T2     | $+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$                                                                                                                                     |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{4}{3}\lambda_{T3} (\lambda_{T3})^*$                                                                                                                                     |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                             | T2                                         | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                              | —                                          | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                              | —                                          | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                              | —                                          | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                              | —                                          | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                              | —                                          | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                              | —                                          | —                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right)$ | —                                          | —                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                              | —                                          | —                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                              | —                                          | —                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                              | —                                          | —                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                              | —                                          | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+9\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right)$ | —                                          | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                              | —                                          | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                              | —                                          | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                              | —                                          | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                              | $+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$ | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                              | $+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$ | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                              | $+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$ | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                              | $+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$ | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                              | $+4\lambda_{12}\lambda_{S_1}^{\text{Adj}}$ | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                              | —                                          | $\frac{4}{3}\lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                              | —                                          | $\frac{4}{3}\lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                              | —                                          | $\frac{4}{3}\lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                              | —                                          | $\frac{4}{3}\lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                              | —                                          | $\frac{4}{3}\lambda_{T3} (\lambda_{T3})^*$ |

# Terms 7 of 10

| Column | Representative term structure                             |  |
|--------|-----------------------------------------------------------|--|
| T1     | $+16g_2^4$                                                |  |
| T2     | $+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$ |  |
| T3     | $+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$   |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1         | T2                                                        | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------|-----------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | —                                                         | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                                                         | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | —                                                         | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                                                         | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —          | —                                                         | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —          | —                                                         | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —          | —                                                         | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —          | —                                                         | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —          | —                                                         | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —          | —                                                         | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —          | —                                                         | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —          | —                                                         | —                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —          | —                                                         | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —          | —                                                         | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —          | —                                                         | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —          | —                                                         | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —          | —                                                         | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —          | —                                                         | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —          | —                                                         | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —          | —                                                         | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+16g_2^4$ | $+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$ | $+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+16g_2^4$ | $+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$ | $+g_Y^2(-3\lambda_{12} - 2\lambda_{12}^{\text{Cross}})$   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+16g_2^4$ | $+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$ | $+g_Y^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+16g_2^4$ | $+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$ | $+g_Y^2(-15\lambda_{12} - 10\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+16g_2^4$ | $+g_2^2(-24\lambda_{12} - 16\lambda_{12}^{\text{Cross}})$ | $+g_Y^2(-30\lambda_{12} - 20\lambda_{12}^{\text{Cross}})$ |

# Terms 8 of 10

Column Representative term structure

|    |                                                                                                                                    |
|----|------------------------------------------------------------------------------------------------------------------------------------|
| T1 | $+\lambda_{S_1} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$                                            |
| T2 | $+\lambda_{12} \left( \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ |
| T3 | $+\lambda_{S_2} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                      | T2                                                                                                                                 | T3                                                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                       | —                                                                                                                                  | —                                                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\lambda_{S_1} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ | $+\lambda_{12} \left( \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ | $+\lambda_{S_2} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\lambda_{S_1} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ | $+\lambda_{12} \left( \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ | $+\lambda_{S_2} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\lambda_{S_1} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ | $+\lambda_{12} \left( \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ | $+\lambda_{S_2} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\lambda_{S_1} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ | $+\lambda_{12} \left( \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ | $+\lambda_{S_2} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\lambda_{S_1} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ | $+\lambda_{12} \left( \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ | $+\lambda_{S_2} \left( 8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3} \right)$ |

# Terms 9 of 10

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $+16\lambda_{12}^{\text{Adj}2}$                                     |
| T2     | $+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$                         |
| T3     | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              | T2                                          | T3                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|---------------------------------------------|---------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                           | —                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                           | —                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                           | —                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                           | —                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                           | —                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               | —                                           | —                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               | —                                           | —                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               | —                                           | —                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               | —                                           | —                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               | —                                           | —                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                               | —                                           | —                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               | —                                           | —                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                               | —                                           | —                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                               | —                                           | —                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                               | —                                           | —                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                           | —                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                           | —                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                           | —                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                           | —                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                           | —                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+16\lambda_{12}^{\text{Adj}2}$ | $+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+16\lambda_{12}^{\text{Adj}2}$ | $+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+16\lambda_{12}^{\text{Adj}2}$ | $+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+16\lambda_{12}^{\text{Adj}2}$ | $+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+16\lambda_{12}^{\text{Adj}2}$ | $+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |

# Terms 10 of 10

Column Representative term structure

|    |                                                                     |
|----|---------------------------------------------------------------------|
| T1 | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
|----|---------------------------------------------------------------------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |

$$\beta_{\lambda_{T3}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                                                                                                                                   |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\lambda_{T3}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{H2}^{\text{Adj}})$ |  |  |  |  |  |  |  |  |  |
| T2     | $-\frac{21g_2^2\lambda_{T3}}{2}$                                                                                                                                |  |  |  |  |  |  |  |  |  |
| T3     | $-\frac{9g_Y^2\lambda_{T3}}{2}$                                                                                                                                 |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                              | T2                               | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{T3}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{T3}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{T3}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{T3}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{T3}(6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                               | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                               | —                                | —                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                               | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                               | —                                | —                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                               | —                                | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                               | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                               | —                                | —                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                               | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                               | —                                | —                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                               | —                                | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                               | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                               | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                               | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                               | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                               | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1 | T2                               | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----|----------------------------------|----------------------------------|
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |

## Terms 2 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+2\lambda_{12}\lambda_{T3}$  |
| T2     | $+2\lambda_H\lambda_{T3}$     |
| T3     | $+4\lambda_{H1}\lambda_{T3}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                           | T2                        | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|---------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |



**Terms 3 of 5**

| Column | Representative term structure                                                                                                                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+4\lambda_{H2}\lambda_{T3}$                                                                                                                                                                                                                                                            |
| T2     | $-3g_Y{}^2\lambda_{T3}$                                                                                                                                                                                                                                                                 |
| T3     | $\lambda_{T3}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $T_1$                        | $T_2$                   | $T_3$                                                                                                                                                                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y{}^2\lambda_{T3}$ | —                                                                                                                                                                                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y{}^2\lambda_{T3}$ | —                                                                                                                                                                                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                       | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y{}^2\lambda_{T3}$ | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y{}^2\lambda_{T3}$ | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                       | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y{}^2\lambda_{T3}$ | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y{}^2\lambda_{T3}$ | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | $\lambda_{T3}\left(6\text{Tr}\left(Y_d(Y_d)^\dagger\right)+2\text{Tr}\left(Y_c(Y_c)^\dagger\right)+6\text{Tr}\left(Y_u(Y_u)^\dagger\right)+\frac{\lambda_{12}^{Adj}}{6}+\frac{2\lambda_{H1}^{Adj}}{3}+2\lambda_{H2}^{Adj}\right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y{}^2\lambda_{T3}$ | —                                                                                                                                                                                                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y{}^2\lambda_{T3}$ | —                                                                                                                                                                                                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y{}^2\lambda_{T3}$ | —                                                                                                                                                                                                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y{}^2\lambda_{T3}$ | —                                                                                                                                                                                                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                       | —                                                                                                                                                                                                                                 |

# Terms 4 of 5

| Column | Representative term structure                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-9g_2^2\lambda_{T3}$                                                                                                               |
| T2     | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3     | $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                    | T2                                                                                                                                  | T3                                                                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{T3}$ | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ | $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{T3}$ | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ | $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                                   | —                                                                                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                                                                                                                   | —                                                                                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                                                                                   | —                                                                                                                 |

# Terms 5 of 5

Column Representative term structure

|    |                                                                                                                                                                                                                                                                                 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1 | $+4 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* - \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \right)$                                                              |
| T2 | $\lambda_{T3} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - 4 \lambda_{H1}^{\text{Adj}})$                                                                                                                              |
| T3 | $\lambda_{T3} \left( 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - \frac{4 \lambda_{12}^{\text{Adj}}}{3} - 2 \lambda_{12}^{\text{Cross}} - 2 \lambda_{H1}^{\text{Adj}} - \frac{4 \lambda_{H2}^{\text{Adj}}}{3} \right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                 | T2                                                                                                                                                 | T3                                                                                                                                                                                                                                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* - \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \right)$ | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* - \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \right)$ | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | —                                                                                                                                                                                                                                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                  | $\lambda_{T3} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - 4 \lambda_{H1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                  | $\lambda_{T3} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - 4 \lambda_{H1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                  | $\lambda_{T3} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - 4 \lambda_{H1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                  | $\lambda_{T3} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - 4 \lambda_{H1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                  | $\lambda_{T3} (6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - 4 \lambda_{H1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                  | —                                                                                                                                                  | $\lambda_{T3} \left( 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - \frac{4 \lambda_{12}^{\text{Adj}}}{3} - 2 \lambda_{12}^{\text{Cross}} - 2 \lambda_{H1}^{\text{Adj}} - \frac{4 \lambda_{H2}^{\text{Adj}}}{3} \right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                  | —                                                                                                                                                  | $\lambda_{T3} \left( 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - \frac{4 \lambda_{12}^{\text{Adj}}}{3} - 2 \lambda_{12}^{\text{Cross}} - 2 \lambda_{H1}^{\text{Adj}} - \frac{4 \lambda_{H2}^{\text{Adj}}}{3} \right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | $\lambda_{T3} \left( 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - \frac{4 \lambda_{12}^{\text{Adj}}}{3} - 2 \lambda_{12}^{\text{Cross}} - 2 \lambda_{H1}^{\text{Adj}} - \frac{4 \lambda_{H2}^{\text{Adj}}}{3} \right)$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                  | —                                                                                                                                                  | $\lambda_{T3} \left( 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - \frac{4 \lambda_{12}^{\text{Adj}}}{3} - 2 \lambda_{12}^{\text{Cross}} - 2 \lambda_{H1}^{\text{Adj}} - \frac{4 \lambda_{H2}^{\text{Adj}}}{3} \right)$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                  | —                                                                                                                                                  | $\lambda_{T3} \left( 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) - \frac{4 \lambda_{12}^{\text{Adj}}}{3} - 2 \lambda_{12}^{\text{Cross}} - 2 \lambda_{H1}^{\text{Adj}} - \frac{4 \lambda_{H2}^{\text{Adj}}}{3} \right)$ |

$$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

# Terms 1 of 6

| Column | Representative term structure                        |  |
|--------|------------------------------------------------------|--|
| T1     | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ |  |
| T2     | $-9g_2^2\lambda_{12}^{\text{Adj}}$                   |  |
| T3     | $-6g_Y^2\lambda_{12}^{\text{Adj}}$                   |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                                  | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|-------------------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                   | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                   | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                   | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                   | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                   | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                   | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                   | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                   | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                   | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                   | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{2}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | —                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                         | T2                                    | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|--------------------------------------------|---------------------------------------|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $-\frac{2}{9}\lambda_{T3}(\lambda_{T3})^*$ | $-24g_2{}^2\lambda_{12}^{\text{Adj}}$ | —  |

## Terms 2 of 6

| Column | Representative term structure              |  |
|--------|--------------------------------------------|--|
| T1     | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$  |  |
| T2     | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ |  |
| T3     | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                        | T2                                         | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------|--------------------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                          | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                          | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                          | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                          | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                          | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                          | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                          | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                          | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                          | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                          | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ |

# Terms 3 of 6

| Column | Representative term structure                                                                                                                                                   |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$                                                                                                                           |
| T2     | $\frac{64}{3}\lambda_{S1^\dagger 2}S_2^{\dagger 2}\left(\lambda_{S1^\dagger 2}S_2^{\dagger 2}\right)^*$                                                                         |
| T3     | $+\frac{4}{3}\lambda_{S1^\dagger S_2}S_2^{\dagger 2}\left(5\left(\lambda_{S1^\dagger S_2}S_2^{\dagger 2}\right)^*+\left(\lambda_{S1}S_1^{\dagger 2}S_2^\dagger\right)^*\right)$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                                    | T2                                                                                                      | T3                                                                                                                                                                              |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | $\frac{64}{3}\lambda_{S1^\dagger 2}S_2^{\dagger 2}\left(\lambda_{S1^\dagger 2}S_2^{\dagger 2}\right)^*$ | $+\frac{4}{3}\lambda_{S1^\dagger S_2}S_2^{\dagger 2}\left(5\left(\lambda_{S1^\dagger S_2}S_2^{\dagger 2}\right)^*+\left(\lambda_{S1}S_1^{\dagger 2}S_2^\dagger\right)^*\right)$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | $\frac{64}{3}\lambda_{S1^\dagger 2}S_2^{\dagger 2}\left(\lambda_{S1^\dagger 2}S_2^{\dagger 2}\right)^*$ | $+\frac{4}{3}\lambda_{S1^\dagger S_2}S_2^{\dagger 2}\left(5\left(\lambda_{S1^\dagger S_2}S_2^{\dagger 2}\right)^*+\left(\lambda_{S1}S_1^{\dagger 2}S_2^\dagger\right)^*\right)$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ | —                                                                                                       | —                                                                                                                                                                               |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                                                       | —                                                                                                                                                                               |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                                                       | —                                                                                                                                                                               |

# Terms 4 of 6

| Column | Representative term structure                                                                                                                                                                            |  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $+ \frac{4}{3} \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \left( \left( \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}} \right)^* + 5 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^* \right)$ |  |
| T2     | $+ g_Y^2 \left( -6 g_2^2 - 3 \lambda_{12}^{\text{Adj}} \right)$                                                                                                                                          |  |
| T3     | $+ g_Y^2 \left( 18 g_2^2 - 15 \lambda_{12}^{\text{Adj}} \right)$                                                                                                                                         |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                     | T2                                                              | T3                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \frac{4}{3} \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \left( \left( \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}} \right)^* + 5 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^* \right) + g_Y^2 \left( -6 g_2^2 - 3 \lambda_{12}^{\text{Adj}} \right)$ | —                                                               | —                                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                      | —                                                               | $+ g_Y^2 \left( 18 g_2^2 - 15 \lambda_{12}^{\text{Adj}} \right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | $+ g_Y^2 \left( 48 g_2^2 - 30 \lambda_{12}^{\text{Adj}} \right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \frac{4}{3} \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \left( \left( \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}} \right)^* + 5 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^* \right) + g_Y^2 \left( -6 g_2^2 - 3 \lambda_{12}^{\text{Adj}} \right)$ | —                                                               | —                                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                      | —                                                               | $+ g_Y^2 \left( 18 g_2^2 - 15 \lambda_{12}^{\text{Adj}} \right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | $+ g_Y^2 \left( 48 g_2^2 - 30 \lambda_{12}^{\text{Adj}} \right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                      | $+ g_Y^2 \left( -2 g_2^2 - 3 \lambda_{12}^{\text{Adj}} \right)$ | —                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | —                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                      | —                                                               | $+ g_Y^2 \left( 6 g_2^2 - 15 \lambda_{12}^{\text{Adj}} \right)$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                      | —                                                               | $+ g_Y^2 \left( 16 g_2^2 - 30 \lambda_{12}^{\text{Adj}} \right)$ |



Terms 5 of 6

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $+ \frac{16\lambda_{12}^{\text{Adj}} \lambda_{12}^{\text{Cross}}}{3}$ |
| T2     | $+16\lambda_{12}^{\text{Adj}} \lambda_{S_1}^{\text{Adj}}$             |
| T3     | $+16\lambda_{12}^{\text{Adj}} \lambda_{S_2}^{\text{Adj}}$             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | $T_1$                                                               | $T_2$                                                    | $T_3$                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                        | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                        | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                        | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                        | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                        | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                        | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                        | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                        | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                        | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                        | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                        | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                        | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                        | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{16\lambda_{12}^{\text{Adj}}\lambda_{12}^{\text{Cross}}}{3}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ |

Terms 6 of 6

| Column | Representative term structure                          |  |
|--------|--------------------------------------------------------|--|
| T1     | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |

$$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$             |  |
| T2     | $+\frac{33g_2^4}{4}$                                              |  |
| T3     | $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                   | T3                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----------------------|-------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                    | —                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                    | —                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                    | —                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                    | —                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                     | —                    | —                                                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                     | —                    | —                                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                     | —                    | —                                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                     | —                    | —                                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                     | —                    | —                                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                     | —                    | —                                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                     | —                    | —                                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                     | —                    | —                                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                    | —                                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                    | —                                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                    | —                                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                    | —                                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ |

## Terms 2 of 5

| Column | Representative term structure                                                   |
|--------|---------------------------------------------------------------------------------|
| T1     | $+g_Y^2 \left( -\frac{3\lambda_{12}}{2} - 6\lambda_{12}^{\text{Cross}} \right)$ |
| T2     | $+\lambda_{12}^2$                                                               |
| T3     | $+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                | T2                | T3                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                 | —                 | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                 | —                 | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                 | —                 | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                 | —                 | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                 | —                 | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                 | —                 | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                 | —                 | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                 | —                 | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                 | —                 | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                 | —                 | —                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                 | —                 | —                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                 | —                 | —                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                 | —                 | —                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2 \left( -\frac{3\lambda_{12}}{2} - 6\lambda_{12}^{\text{Cross}} \right)$   | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( -\frac{3\lambda_{12}}{4} - 3\lambda_{12}^{\text{Cross}} \right)$   | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2 \left( -\frac{3\lambda_{12}}{2} - 6\lambda_{12}^{\text{Cross}} \right)$   | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( -\frac{15\lambda_{12}}{4} - 15\lambda_{12}^{\text{Cross}} \right)$ | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2 \left( -\frac{15\lambda_{12}}{2} - 30\lambda_{12}^{\text{Cross}} \right)$ | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ |

# Terms 3 of 5

| Column | Representative term structure                                                                           |
|--------|---------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{12}(8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}})$ |
| T2     | $+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$                                          |
| T3     | $+4\lambda_{12}^{\text{Adj}^2}$                                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                      | T2                                                             | T3                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                       | —                                                              | —                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                       | —                                                              | —                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                       | —                                                              | —                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                       | —                                                              | —                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                       | —                                                              | —                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                       | —                                                              | —                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                       | —                                                              | —                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                       | —                                                              | —                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                       | —                                                              | —                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                       | —                                                              | —                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                       | —                                                              | —                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                       | —                                                              | —                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                       | —                                                              | —                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\lambda_{12}(8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}})$ | $+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ | $+4\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\lambda_{12}(8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}})$ | $+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ | $+4\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\lambda_{12}(8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}})$ | $+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ | $+4\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\lambda_{12}(8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}})$ | $+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ | $+4\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\lambda_{12}(8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}})$ | $+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ | $+4\lambda_{12}^{\text{Adj}^2}$ |

# Terms 4 of 5

| Column | Representative term structure                            |  |
|--------|----------------------------------------------------------|--|
| T1     | $+11\lambda_{12}^{\text{Cross}2}$                        |  |
| T2     | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ |  |
| T3     | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                                       | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                        | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                        | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                        | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                        | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                        | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                                        | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                                        | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                                        | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                                        | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                                        | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                                        | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                                        | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                                        | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                                        | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                                        | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                        | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                        | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                        | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                        | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                        | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+11\lambda_{12}^{\text{Cross}2}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+11\lambda_{12}^{\text{Cross}2}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+11\lambda_{12}^{\text{Cross}2}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+11\lambda_{12}^{\text{Cross}2}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+11\lambda_{12}^{\text{Cross}2}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |

# Terms 5 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+\lambda_{H1}\lambda_{H2}$   |
| T2     | $+\frac{3g_Y^4}{16}$          |
| T3     | $+12g_Y^4$                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                          | T2                    | T3         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------|-----------------------|------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                           | —                     | —          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                           | —                     | —          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                           | —                     | —          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                           | —                     | —          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                           | —                     | —          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                           | —                     | —          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                           | —                     | —          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                           | —                     | —          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                           | —                     | —          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                           | —                     | —          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                           | —                     | —          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                           | —                     | —          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                           | —                     | —          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                           | —                     | —          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                           | —                     | —          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                           | —                     | —          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                           | —                     | —          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                           | —                     | —          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                           | —                     | —          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                           | —                     | —          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\lambda_{H1}\lambda_{H2}$ | —                     | —          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\lambda_{H1}\lambda_{H2}$ | $+\frac{3g_Y^4}{16}$  | —          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\lambda_{H1}\lambda_{H2}$ | —                     | —          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\lambda_{H1}\lambda_{H2}$ | $+\frac{27g_Y^4}{16}$ | —          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\lambda_{H1}\lambda_{H2}$ | —                     | $+12g_Y^4$ |

$$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

## Terms 1 of 6

| Column | Representative term structure                                                                                                                                                               |  |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ |  |
| T2     | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$                                                                                                                                      |  |
| T3     | $+g_Y^2\left(-12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$                                                                                                                         |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                          | T2                                                     | T3                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                           | —                                                      | —                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                           | —                                                      | —                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                           | —                                                      | —                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                           | —                                                      | —                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                           | —                                                      | —                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+g_Y^2\left(-12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+g_Y^2\left(-12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\lambda_{H1}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right))$ | $+\frac{16}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | —                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                           | $-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | $+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                           | $-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | —                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                           | $-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | —                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                           | $-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | —                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                           | $-\frac{8}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$  | —                                                                   |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1 | T2                                         | T3                                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----|--------------------------------------------|------------------------------------------------------------------------|
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —  | $-\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$ | $+g_Y{}^2\left(-4g_2{}^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —  | $-\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$ | —                                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —  | $-\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$ | —                                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —  | $-\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$ | —                                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —  | $-\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$ | —                                                                      |

# Terms 2 of 6

| Column | Representative term structure                          |  |
|--------|--------------------------------------------------------|--|
| T1     | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$                    |  |
| T2     | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ |  |
| T3     | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$                |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  | T2                                                     | T3                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|--------------------------------------------------------|-----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                                      | —                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                                      | —                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                                      | —                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                                      | —                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                                      | —                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2 \lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                                      | $+2\lambda_H \lambda_{H1}^{\text{Adj}}$ |

# Terms 3 of 6

| Column | Representative term structure                                                                                                            |  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$                                                                                                |  |
| T2     | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$                                                                                                |  |
| T3     | $+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                        | T2                                        | T3                                                                                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                         | —                                                                                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                         | —                                                                                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                         | —                                                                                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                         | —                                                                                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                         | —                                                                                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | $+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | $+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ | —                                                                                                                                        |

# Terms 4 of 6

| Column | Representative term structure                                                                                                                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$                                                                                    |
| T2     | $-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (2 \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*)$ |
| T3     | $+g_Y^2 \left( -6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$                                                                                                                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                 | T2                                                                                                                                                                                                                                                                                | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (2 \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) + g_Y^2 \left( -6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ |                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $-\frac{4}{3}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (2 \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) + g_Y^2 \left( -6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ |                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | $+g_Y^2 \left( -2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | $+g_Y^2 \left( -2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                                                                                                                                                                 | —                                                            |

# Terms 5 of 6

| Column | Representative term structure                                      |  |
|--------|--------------------------------------------------------------------|--|
| T1     | $-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$                       |  |
| T2     | $+g_Y^2\left(6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$             |  |
| T3     | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                           | T2                                                     | T3                                                                 |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|----------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                      | —                                                                  |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                      | —                                                                  |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                      | —                                                                  |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                      | —                                                                  |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                      | —                                                                  |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                      | —                                                                  |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                      | —                                                                  |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$ | —                                                      | —                                                                  |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                            | $+g_Y^2\left(6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$ | —                                                                  |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                      | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                      | —                                                                  |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                                      | —                                                                  |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$ | —                                                      | —                                                                  |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                            | $+g_Y^2\left(6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$ | —                                                                  |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                      | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                      | —                                                                  |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                      | —                                                                  |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$ | —                                                      | —                                                                  |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                            | $+g_Y^2\left(2g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$ | —                                                                  |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                      | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                      | —                                                                  |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                      | —                                                                  |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $-\frac{3g_Y^2\lambda_{H1}^{\text{Adj}}}{2}$ | —                                                      | —                                                                  |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                            | $+g_Y^2\left(2g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$ | —                                                                  |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                      | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  |

# Terms 6 of 6

| Column | Representative term structure                                                                                                                               |  |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ |  |
| T2     | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$                                                                                                              |  |
| T3     | $+12\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$                                                                                                    |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                                                                                                                                          | T2                                             | T3                                                       |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                           | —                                              | —                                                        |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                           | —                                              | —                                                        |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                           | —                                              | —                                                        |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                           | —                                              | —                                                        |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                           | —                                              | —                                                        |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                           | —                                              | —                                                        |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                           | —                                              | —                                                        |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                           | —                                              | —                                                        |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                           | —                                              | —                                                        |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                           | —                                              | —                                                        |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                           | —                                              | —                                                        |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                           | —                                              | —                                                        |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                           | —                                              | —                                                        |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                           | —                                              | —                                                        |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                           | —                                              | —                                                        |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | —                                                        |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | —                                                        |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | —                                                        |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | —                                                        |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | —                                                        |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | $+12\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | $+12\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | $+12\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | $+12\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H1}^{\text{Adj}}(6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ | $+12\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ |

$$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

## Terms 1 of 6

| Column | Representative term                           | structure                                                                                                                                                                                |
|--------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H2}^{\text{Adj}}$                   | $(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ |
| T2     | $+\frac{8}{3}\lambda_{T3}$                    | $(\lambda_{T3})^*$                                                                                                                                                                       |
| T3     | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |                                                                                                                                                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                | T2                                         | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                 | —                                          | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S2}^{\text{Adj}})$ | $-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                                                                                                                                                                                                 | T2                                                    | T3                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $\lambda_{H2}^{\text{Adj}}(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{\text{Adj}})$ | $-\frac{8}{9}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |



## Terms 2 of 6

| Column | Representative term structure                |  |
|--------|----------------------------------------------|--|
| T1     | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ |  |
| T2     | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$       |  |
| T3     | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$    |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                           | T2                                     | T3                                        |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|----------------------------------------------|----------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                      | —                                         |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                      | —                                         |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                      | —                                         |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                      | —                                         |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                      | —                                         |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ |

# Terms 3 of 6

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$                         |  |
| T2     | $+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$            |  |
| T3     | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$ |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                        | T2                                                       | T3                                                                 |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | —                                                                  |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   | —                                                                  |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$  |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$  | —                                                                  |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(8g_2^2-\frac{51\lambda_{H2}^{\text{Adj}}}{2}\right)$  |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | —                                                                  |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   | —                                                                  |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(18g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$ | —                                                                  |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(24g_2^2-\frac{51\lambda_{H2}^{\text{Adj}}}{2}\right)$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | —                                                                  |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   | —                                                                  |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(18g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$ | —                                                                  |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(24g_2^2-\frac{51\lambda_{H2}^{\text{Adj}}}{2}\right)$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                        | —                                                                  |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                        | —                                                                  |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                        | —                                                                  |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                        | —                                                                  |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                        | —                                                                  |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | —                                                                  |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   | —                                                                  |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$  |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$  | —                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                          | T2 | T3                                                                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|---------------------------------------------|----|--------------------------------------------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $+2\lambda_{H^2}^{\text{Adj}}\lambda_{S_2}$ | —  | $+g_Y^2\left(8g_2^2-\frac{51\lambda_{H^2}^{\text{Adj}}}{2}\right)$ |

# Terms 4 of 6

| Column | Representative term structure                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$ |
| T2     | $+16\lambda_{T3} (\lambda_{T3})^*$                                                                                             |
| T3     | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$                                                                                            |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                                                                                                                                               | T2 | T3                                  |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                | —  | —                                   |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                | —  | —                                   |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                | —  | —                                   |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                | —  | —                                   |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                | —  | —                                   |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $\lambda_{H2}^{\text{Adj}} (6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger)) + 16\lambda_{T3} (\lambda_{T3})^*$ |    | $-9g_2^2 \lambda_{H2}^{\text{Adj}}$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                | —  | —                                   |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                | —  | —                                   |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                | —  | —                                   |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                | —  | —                                   |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                | —  | —                                   |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                | —  | —                                   |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                | —  | —                                   |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                | —  | —                                   |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                | —  | —                                   |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                | —  | —                                   |

# Terms 5 of 6

| Column | Representative term structure                                                                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$                                                                                                                                 |
| T2     | $-4\lambda_{S_1^\dagger S_2 S_2^\dagger} ( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} )^*$                                                                           |
| T3     | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} ( 2 ( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} )^* + ( \lambda_{S_1^\dagger S_2 S_2^\dagger} )^* )$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                                                                                                          | T3                                                                                                                                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | $-4\lambda_{S_1^\dagger S_2 S_2^\dagger} ( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} )^*$ | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} ( 2 ( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} )^* + ( \lambda_{S_1^\dagger S_2 S_2^\dagger} )^* )$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | $-4\lambda_{S_1^\dagger S_2 S_2^\dagger} ( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} )^*$ | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} ( 2 ( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} )^* + ( \lambda_{S_1^\dagger S_2 S_2^\dagger} )^* )$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ | —                                                                                                           | —                                                                                                                                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                                                           | —                                                                                                                                                                                     |

# Terms 6 of 6

| Column | Representative term structure                                                                     |  |
|--------|---------------------------------------------------------------------------------------------------|--|
| T1     | $+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |  |
| T2     | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$                                          |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                 | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                 | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |

$$\beta_{\lambda}^{(1)}_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$$

$$16\pi^2\,\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda}^{(1)}_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$$

### Terms 1 of 4

| Column | Representative term structure                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T2     | $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$                                                                                                     |
| T3     | $+2\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}(\lambda_{T3})^*$                                                                                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                  | T2                                                                              | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$ | $+2\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}(\lambda_{T3})^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$ | $+2\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}(\lambda_{T3})^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                   | —                                                                               | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                   | —                                                                               | —                                                           |

# Terms 2 of 4

| Column | Representative term structure                                      |  |
|--------|--------------------------------------------------------------------|--|
| T1     | $-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$     |  |
| T2     | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$     |  |
| T3     | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                             | T3                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                              | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                              | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                              | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                              | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                              | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                              | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                              | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                              | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                              | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                              | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                              | —                                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                              | —                                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                              | —                                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                              | —                                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                              | —                                                                  |



# Terms 3 of 4

| Column | Representative term structure                                                                |
|--------|----------------------------------------------------------------------------------------------|
| T1     | $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$                        |
| T2     | $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ |
| T3     | $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                    | T2                                                                                           | T3                                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                     | —                                                                                            | —                                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                     | —                                                                                            | —                                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                     | —                                                                                            | —                                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                     | —                                                                                            | —                                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ | $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                     | —                                                                                            | —                                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                     | —                                                                                            | —                                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ | $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S_1}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                     | —                                                                                            | —                                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                     | —                                                                                            | —                                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                     | —                                                                                            | —                                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                     | —                                                                                            | —                                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                     | —                                                                                            | —                                                                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                     | —                                                                                            | —                                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                     | —                                                                                            | —                                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                     | —                                                                                            | —                                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                     | —                                                                                            | —                                                                      |

Terms 4 of 4

| Column | Representative term structure                                                       |
|--------|-------------------------------------------------------------------------------------|
| T1     | $+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   |

$$\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{d\mu}=\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

## Terms 1 of 4

| Column | Representative term structure                                                                                                                         |  |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger))$ |  |
| T2     | $+6\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$                                   |  |
| T3     | $-6\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^*$                                                                              |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                    | T2                                                                                                                  | T3                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+6\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $-6\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger))$ | $+6\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $-6\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                     | —                                                                                                                   | —                                                                        |

# Terms 2 of 4

| Column | Representative term structure                                               |  |
|--------|-----------------------------------------------------------------------------|--|
| T1     | $-9g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$        |  |
| T2     | $-3g_Y^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$        |  |
| T3     | $+2\lambda_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                   | T2                                                                   | T3                                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                                    | —                                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                                    | —                                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                                    | —                                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                    | —                                                                    | —                                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $-3g_Y^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $+2\lambda_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                    | —                                                                    | —                                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                    | —                                                                    | —                                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $-3g_Y^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $+2\lambda_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                    | —                                                                    | —                                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                                    | —                                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                                    | —                                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                                    | —                                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                                    | —                                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                                    | —                                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                                    | —                                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                                    | —                                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                                    | —                                                                           |

# Terms 3 of 4

| Column | Representative term structure                                                                                                                                       |  |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$                                                                                            |  |
| T2     | $+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$                                            |  |
| T3     | $+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       | T2                                                                                                                       | T3                                                                                                                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$ | $+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $+\lambda_{H1}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger})$ | $+\lambda_{H1}^{\text{Adj}}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                        | —                                                                                                                                                                   |

# Terms 4 of 4

| Column | Representative term structure                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{H2}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}})$                                              |
| T2     | $+\lambda_{H2}^{\text{Adj}}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$ |
| T3     | $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{T3}$                                                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                         | T2                                                                                                                                                                      | T3                                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{H2}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}})$ | $+\lambda_{H2}^{\text{Adj}}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$ | $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{T3}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{H2}(4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}})$ | $+\lambda_{H2}^{\text{Adj}}\left(-\frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$ | $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{T3}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                          | —                                                                                                                                                                       | —                                                                      |

$$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

## Terms 1 of 3

| Column | Representative term structure                                                                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H^\dagger H^\dagger S_2 S_2} (6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ |
| T2     | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$                                                                                  |
| T3     | $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                 | T2                                                                                                                | T3                                                                                                                                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H^\dagger H^\dagger S_2 S_2} (6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ | $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H^\dagger H^\dagger S_2 S_2} (6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ | $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                  | —                                                                                                                 | —                                                                                                                                                                               |

## Terms 2 of 3

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$     |
| T2     | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$     |
| T3     | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                             | T3                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|------------------------------------------------|----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                              | —                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                              | —                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                              | —                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                              | —                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                              | —                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                              | —                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                  |



# Terms 3 of 3

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$              |
| T2     | $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3     | $+2\lambda_{H^\dagger H^\dagger S_2 S_2}^{\lambda S_2}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                                                                 | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                  | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                  | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                  | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                     | —                                                                  | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+2\lambda_{H^\dagger H^\dagger S_2 S_2}^{\lambda S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                     | —                                                                  | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                     | —                                                                  | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+2\lambda_{H^\dagger H^\dagger S_2 S_2}^{\lambda S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                     | —                                                                  | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                  | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                  | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                  | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                                                  | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                                                  | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                                                  | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                                                  | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                                                  | —                                                       |

$$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                                                     |
|--------|-----------------------------------------------------------------------------------|
| T1     | $\frac{3g_2^4}{4}$                                                                |
| T2     | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$ |
| T3     | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                 | T2                                                                                | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                  | —                                                                                 | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                  | —                                                                                 | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                  | —                                                                                 | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                  | —                                                                                 | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                  | —                                                                                 | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                  | —                                                                                 | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                  | —                                                                                 | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                  | —                                                                                 | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                  | —                                                                                 | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                  | —                                                                                 | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                  | —                                                                                 | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                  | —                                                                                 | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                  | —                                                                                 | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                  | —                                                                                 | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                  | —                                                                                 | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3g_2^4}{4}$ | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3g_2^4}{4}$ | —                                                                                 | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3g_2^4}{4}$ | —                                                                                 | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3g_2^4}{4}$ | —                                                                                 | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3g_2^4}{4}$ | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3g_2^4}{4}$ | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3g_2^4}{4}$ | —                                                                                 | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3g_2^4}{4}$ | —                                                                                 | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3g_2^4}{4}$ | —                                                                                 | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                 | T2                                                                                | T3                                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|--------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $\frac{3g_2^4}{4}$ | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$ |

# Terms 2 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+ \frac{3g_Y^4}{2}$          |
| T2     | $+ \frac{\lambda_{12}^2}{4}$  |
| T3     | $+ \frac{\lambda_{H1}^2}{2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                    | T2                           | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------|------------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                            | —                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                            | —                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                            | —                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                            | —                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                            | —                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                     | —                            | —                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                     | —                            | —                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                     | —                            | —                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                     | —                            | —                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                     | —                            | —                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                     | —                            | —                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                     | —                            | —                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                     | —                            | —                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                     | —                            | —                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                     | —                            | —                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3g_Y^4}{2}$  | $+ \frac{\lambda_{12}^2}{4}$ | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+ \frac{3g_Y^4}{32}$ | $+ \frac{\lambda_{12}^2}{4}$ | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                     | $+ \frac{\lambda_{12}^2}{4}$ | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3g_Y^4}{32}$ | $+ \frac{\lambda_{12}^2}{4}$ | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3g_Y^4}{2}$  | $+ \frac{\lambda_{12}^2}{4}$ | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3g_Y^4}{2}$  | —                            | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+ \frac{3g_Y^4}{32}$ | —                            | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                            | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3g_Y^4}{32}$ | —                            | $+ \frac{\lambda_{H1}^2}{2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3g_Y^4}{2}$  | —                            | $+ \frac{\lambda_{H1}^2}{2}$ |

# Terms 3 of 5

| Column | Representative term structure              |  |
|--------|--------------------------------------------|--|
| T1     | $+3\lambda_{H1}^{\text{Adj}2}$             |  |
| T2     | $+\frac{7\lambda_{S1}^2}{4}$               |  |
| T3     | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                             | T2                           | T3                                         |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|--------------------------------|------------------------------|--------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                              | —                            | —                                          |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                              | —                            | —                                          |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                              | —                            | —                                          |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                              | —                            | —                                          |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                              | —                            | —                                          |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                              | —                            | —                                          |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                              | —                            | —                                          |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                              | —                            | —                                          |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                              | —                            | —                                          |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                              | —                            | —                                          |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                              | —                            | —                                          |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                              | —                            | —                                          |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                              | —                            | —                                          |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                              | —                            | —                                          |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                              | —                            | —                                          |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+13\lambda_{S1}\lambda_{S1}^{\text{Adj}}$ |

# Terms 4 of 5

| Column | Representative term structure                                                                     |
|--------|---------------------------------------------------------------------------------------------------|
| T1     | $+22\lambda_{S_1}^{\text{Adj}^2}$                                                                 |
| T2     | $+g_Y^2 \left( \frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$ |
| T3     | $+\frac{3\lambda_{12}^2}{4}$                                                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                                                                                | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                                                                 | —                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                                                                 | —                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                                                                 | —                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                                                                 | —                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                                                                 | —                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                                                                                 | —                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                                                                                 | —                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                                                                                 | —                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                                                                                 | —                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                                                                                 | —                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                                                                                 | —                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                                                                                 | —                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                                                                                 | —                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                                                                                 | —                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                                                                                 | —                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+22\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                                 | —                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+22\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2 \left( \frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$ | —                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+22\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                                 | —                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+22\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2 \left( \frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$ | —                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+22\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                                 | —                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+22\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                                 | $+\frac{3\lambda_{12}^2}{4}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+22\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2 \left( \frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$ | $+\frac{3\lambda_{12}^2}{4}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+22\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                                 | $+\frac{3\lambda_{12}^2}{4}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+22\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2 \left( \frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$ | $+\frac{3\lambda_{12}^2}{4}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+22\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                                 | $+\frac{3\lambda_{12}^2}{4}$ |

# Terms 5 of 5

| Column | Representative term structure              |  |
|--------|--------------------------------------------|--|
| T1     | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |  |
| T2     | $+12\lambda_{12}^{\text{Adj}^2}$           |  |
| T3     | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                               | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|----------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                          | —                                | —                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                          | —                                | —                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                          | —                                | —                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                          | —                                | —                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                          | —                                | —                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | —                                | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | —                                | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | —                                | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | —                                | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | —                                | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |

$$\beta_{\lambda_{S_1 S_1^\dagger S_2^\dagger}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1 S_1^\dagger S_2^\dagger}}{d\mu}=\beta_{\lambda_{S_1 S_1^\dagger S_2^\dagger}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $4\lambda_{S_1^\dagger S_2^\dagger S_2^\dagger}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}\right)^*)$ |
| T2     | $-9g_2^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$                                                                                                               |
| T3     | $-3g_Y^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$                                                                                                               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                           | T2                                             | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                            | —                                              | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                            | —                                              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                            | —                                              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                            | —                                              | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $4\lambda_{S_1^\dagger S_2^\dagger S_2^\dagger}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}\right)^*)$ | $-9g_2^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$ | $-3g_Y^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                            | —                                              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                            | —                                              | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $4\lambda_{S_1^\dagger S_2^\dagger S_2^\dagger}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}\right)^*+5\left(\lambda_{S_1 S_1^\dagger S_2^\dagger}\right)^*)$ | $-9g_2^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$ | $-3g_Y^2\lambda_{S_1 S_1^\dagger S_2^\dagger}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                            | —                                              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                            | —                                              | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                            | —                                              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                            | —                                              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                            | —                                              | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                            | —                                              | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                            | —                                              | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                            | —                                              | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                            | —                                              | —                                              |



## Terms 2 of 3

| Column | Representative term structure                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$                                               |
| T2     | $+\lambda_{12}^{\text{Adj}} \left( \frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$ |
| T3     | $+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                   | T2                                                                                                                                                 | T3                                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$ | $+\lambda_{12}^{\text{Adj}} \left( \frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$ | $+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$ | $+\lambda_{12}^{\text{Adj}} \left( \frac{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6} \right)$ | $+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                    | —                                                                                                                                                  | —                                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                    | —                                                                                                                                                  | —                                                                           |

# Terms 3 of 3

| Column | Representative term structure                                                                     |  |
|--------|---------------------------------------------------------------------------------------------------|--|
| T1     | $-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$ |  |
| T2     | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$                            |  |
| T3     | $+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$                                       |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                | T2                                                                     | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                                      | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                                      | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$ | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$ | $+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                 | —                                                                      | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$ | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$ | $+12\lambda_{S_1}\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                 | —                                                                      | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                                      | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                                      | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                                      | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                                      | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                           |

$$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

# Terms 1 of 4

| Column | Representative term structure                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------|
| T1     | $4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ |
| T2     | $-9g_2^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$                                                           |
| T3     | $-3g_Y^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                           | T2                                                 | T3                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                            | —                                                  | —                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                            | —                                                  | —                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                            | —                                                  | —                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                            | —                                                  | —                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ | $-9g_2^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ | $-3g_Y^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                            | —                                                  | —                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                            | —                                                  | —                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ | $-9g_2^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ | $-3g_Y^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                            | —                                                  | —                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                            | —                                                  | —                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                            | —                                                  | —                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                            | —                                                  | —                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                            | —                                                  | —                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                            | —                                                  | —                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                            | —                                                  | —                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                            | —                                                  | —                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                            | —                                                  | —                                                  |

# Terms 2 of 4

| Column | Representative term structure                                                   |
|--------|---------------------------------------------------------------------------------|
| T1     | $+8\lambda_{12}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$                        |
| T2     | $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}}{3}$ |
| T3     | $+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}^2}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                       | T2                                                                              | T3                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                                               | —                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                                               | —                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                                               | —                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                        | —                                                                               | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{12}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$ | $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}}{3}$ | $+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}^2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                        | —                                                                               | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                        | —                                                                               | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{12}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$ | $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}}{3}$ | $+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}^2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                        | —                                                                               | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                                               | —                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                                               | —                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                                               | —                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                                               | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                                               | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                                               | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                                               | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                                               | —                                                                    |

Terms 3 of 4

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$ |
| T2     | $+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$ |
| T3     | $+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                      | T2                                                      | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                       | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                       | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                       | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                       | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                       | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                       | —                                                       | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$ | $+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$ | $+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                       | —                                                       | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                       | —                                                       | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                       | —                                                       | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                       | —                                                       | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{S_1} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$ | $+2\lambda_{S_1^\dagger 2 S_2^\dagger 2} \lambda_{S_2}$ | $+5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                       | —                                                       | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                       | —                                                       | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                       | —                                                       | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                       | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                       | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                       | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                       | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                       | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                       | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                       | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                       | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                       | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                       | —                                             |

Terms 4 of 4

| Column | Representative term structure                                                       |
|--------|-------------------------------------------------------------------------------------|
| T1     | $+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$ |
| T2     | $+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                  | T2                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                   | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$ | $+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                   | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                   | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                   | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                   | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_1 S_1^\dagger 2 S_2^\dagger}$ | $+5\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                   | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                   | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                   | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   | —                                             |

$$\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$                                                                                |
| T2     | $+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)$ |
| T3     | $-9g_2^2\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$                                                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                 | T2                                                                                                                                                                                                                | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$ | $+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)-9g_2^2\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$ |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$ | $+4\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(5\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*)-9g_2^2\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$ |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                  | —                                                                                                                                                                                                                 | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                  | —                                                                                                                                                                                                                 | —  |

## Terms 2 of 3

| Column | Representative term structure                                                                                                                   |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-3g_Y{}^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$                                                                                              |
| T2     | $+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$                                            |
| T3     | $+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                                                                                   | T3                                                                                                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-3g_Y{}^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$ | $+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$ | $+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-3g_Y{}^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$ | $+\lambda_{12}(6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$ | $+\lambda_{12}^{\text{Adj}}\left(\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                  | —                                                                                                    | —                                                                                                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                  | —                                                                                                    | —                                                                                                                                               |



# Terms 3 of 3

| Column | Representative term structure                                                                     |
|--------|---------------------------------------------------------------------------------------------------|
| T1     | $+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$                       |
| T2     | $-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$ |
| T3     | $+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_2}$                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                          | T2                                                                                                | T3                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                           | —                                                                                                 | —                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                           | —                                                                                                 | —                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$ | $+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                           | —                                                                                                 | —                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$ | $+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \lambda_{S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                           | —                                                                                                 | —                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                           | —                                                                                                 | —                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                           | —                                                                                                 | —                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                           | —                                                                                                 | —                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                           | —                                                                                                 | —                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                          |

$$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 6

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{3g_2^4}{4}$                                     |
| T2     | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ |
| T3     | $+\frac{\lambda_{12}^2}{4}$                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                 | T2                                                     | T3                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------|--------------------------------------------------------|-----------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                  | —                                                      | —                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                  | —                                                      | —                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                  | —                                                      | —                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                  | —                                                      | —                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                  | —                                                      | —                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                  | —                                                      | —                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                  | —                                                      | —                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                  | —                                                      | —                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                  | —                                                      | —                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                  | —                                                      | —                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                  | —                                                      | —                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                  | —                                                      | —                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                  | —                                                      | —                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                  | —                                                      | —                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                  | —                                                      | —                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                 | T2                                                       | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|--------------------|----------------------------------------------------------|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2} - 24\lambda_{S_2}^{\text{Adj}})$ | —  |

# Terms 2 of 6

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $+\frac{\lambda_{H2}^2}{2}$     |
| T2     | $+3\lambda_{H2}^{\text{Adj}^2}$ |
| T3     | $+\frac{7\lambda_{S2}^2}{4}$    |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                          | T2                              | T3                           |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-----------------------------|---------------------------------|------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                           | —                               | —                            |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                           | —                               | —                            |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                           | —                               | —                            |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                           | —                               | —                            |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                           | —                               | —                            |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                           | —                               | —                            |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                           | —                               | —                            |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                           | —                               | —                            |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                           | —                               | —                            |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                           | —                               | —                            |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                           | —                               | —                            |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                           | —                               | —                            |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                           | —                               | —                            |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                           | —                               | —                            |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                           | —                               | —                            |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |

# Terms 3 of 6

| Column | Representative term structure                                                              |  |
|--------|--------------------------------------------------------------------------------------------|--|
| T1     | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$                                               |  |
| T2     | $+22\lambda_{S_2}^{\text{Adj}^2}$                                                          |  |
| T3     | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}}{8}-3\lambda_{S_2}^{\text{Adj}}\right)$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                | T3                                                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}}{8}-3\lambda_{S_2}^{\text{Adj}}\right)$    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{81g_2^2}{4}-\frac{27\lambda_{S_2}}{8}-27\lambda_{S_2}^{\text{Adj}}\right)$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                 | —                                                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                 | —                                                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                 | —                                                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                 | —                                                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                 | —                                                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                 | —                                                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                 | —                                                                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                 | —                                                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                 | —                                                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                 | —                                                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                 | —                                                                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                 | —                                                                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                 | —                                                                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                 | —                                                                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                 | —                                                                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}}{8}-3\lambda_{S_2}^{\text{Adj}}\right)$    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{81g_2^2}{4}-\frac{27\lambda_{S_2}}{8}-27\lambda_{S_2}^{\text{Adj}}\right)$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+13\lambda_{S_2}\lambda_{S_2}^{\text{Adj}}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             |

# Terms 4 of 6

| Column | Representative term structure                                                            |  |  |  |  |  |  |
|--------|------------------------------------------------------------------------------------------|--|--|--|--|--|--|
| T1     | $+ \frac{3g_Y^4}{32}$                                                                    |  |  |  |  |  |  |
| T2     | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$ |  |  |  |  |  |  |
| T3     | $+g_Y^2 \left( 36g_2^2 - 6\lambda_{S_2} - 48\lambda_{S_2}^{\text{Adj}} \right)$          |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                     | T2                                                                                       | T3                                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                      | —                                                                                        | —                                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{32}$   | —                                                                                        | —                                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3g_Y^4}{2}$    | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$ | —                                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{243g_Y^4}{32}$ | —                                                                                        | —                                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                      | —                                                                                        | $+g_Y^2 \left( 36g_2^2 - 6\lambda_{S_2} - 48\lambda_{S_2}^{\text{Adj}} \right)$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                      | —                                                                                        | —                                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                      | —                                                                                        | —                                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                      | —                                                                                        | —                                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                      | —                                                                                        | —                                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                      | —                                                                                        | —                                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                      | —                                                                                        | —                                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                      | —                                                                                        | —                                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                      | —                                                                                        | —                                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                      | —                                                                                        | —                                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                      | —                                                                                        | —                                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                      | —                                                                                        | —                                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                      | —                                                                                        | —                                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                      | —                                                                                        | —                                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                      | —                                                                                        | —                                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                      | —                                                                                        | —                                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                      | —                                                                                        | —                                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{32}$   | —                                                                                        | —                                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3g_Y^4}{2}$    | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_2}}{2} - 12\lambda_{S_2}^{\text{Adj}} \right)$ | —                                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{243g_Y^4}{32}$ | —                                                                                        | —                                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                      | —                                                                                        | $+g_Y^2 \left( 36g_2^2 - 6\lambda_{S_2} - 48\lambda_{S_2}^{\text{Adj}} \right)$ |

# Terms 5 of 6

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $+24g_Y^4$                                 |
| T2     | $+\frac{3\lambda_{12}^2}{4}$               |
| T3     | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1         | T2                           | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------|------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | —                            | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | —                            | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | —                            | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | —                            | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+24g_Y^4$ | —                            | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —          | —                            | —                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —          | —                            | —                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —          | —                            | —                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —          | —                            | —                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —          | —                            | —                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —          | —                            | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —          | —                            | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —          | —                            | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —          | —                            | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —          | —                            | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —          | —                            | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —          | —                            | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —          | —                            | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —          | —                            | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —          | —                            | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+24g_Y^4$ | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |

# Terms 6 of 6

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $+12\lambda_{12}^{\text{Adj}2}$           |
| T2     | $-\frac{\lambda_{12}^{\text{Cross}2}}{2}$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                              | T2                                        |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------------|-------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                         |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                               | —                                         |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                               | —                                         |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                               | —                                         |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                               | —                                         |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                               | —                                         |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                               | —                                         |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                               | —                                         |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                               | —                                         |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                               | —                                         |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                               | —                                         |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                               | —                                         |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                               | —                                         |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                               | —                                         |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                               | —                                         |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                         |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                               | —                                         |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                               | —                                         |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                               | —                                         |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                               | —                                         |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_{12}^{\text{Adj}2}$ | $-\frac{\lambda_{12}^{\text{Cross}2}}{2}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_{12}^{\text{Adj}2}$ | $-\frac{\lambda_{12}^{\text{Cross}2}}{2}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_{12}^{\text{Adj}2}$ | $-\frac{\lambda_{12}^{\text{Cross}2}}{2}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_{12}^{\text{Adj}2}$ | $-\frac{\lambda_{12}^{\text{Cross}2}}{2}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_{12}^{\text{Adj}2}$ | $-\frac{\lambda_{12}^{\text{Cross}2}}{2}$ |



# Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis  $C_{ijab}$ . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{lambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                | T2 | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------|----|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$        | —  | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$        | —  | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$        | —  | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$        | —  | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$        | —  | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —  | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|--------------------------------------------------------|
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{1134}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$$

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{ilambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                          | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|---------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|--------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}ilambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{1144}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{lambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                        | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|-------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                     | T2 | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------|----|-------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{2\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{1155}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1155}}{d\mu}=\beta_{C_{1155}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_1}^2}{MF}$           |
| T2     | $\frac{\lambda S1Inv1\overline{y_1}^2}{MF}$  |
| T3     | $\frac{ye\overline{y_1}^2\overline{ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                            | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|-----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_1}^2}{4MF}$ | $-\frac{\lambda S1Inv1\overline{y_1}^2}{2MF}$ | $\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                             | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                             | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                             | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |

# Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda daS1Inv1\overline{y_1}^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                      | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                       | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                       | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$ | $-\frac{\sqrt{3}\lambda daS1Inv1\overline{y_1}^2}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                       | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |



$$\beta_{C_{1156}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1156}}{d\mu}=\beta_{C_{1156}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$      |
| T2     | $\frac{ig_2^2\overline{y_1}^2}{MF}$             |
| T3     | $\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                    | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$ | $\frac{3ig_2^2\overline{y_1}^2}{4MF}$ | $-\frac{iy_e\overline{y_1}^2\overline{y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                     | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                     | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                     | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |

Terms 2 of 2

| Column | Representative term structure                         |  |
|--------|-------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$           |  |
| T2     | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$    |  |
| T3     | $\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                  | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                   | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                   | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                   | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                   | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$ | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$ | $-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                   | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                   | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                   | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                   | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                   | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |

$$\beta_{C_{1157}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1157}}{d\mu}=\beta_{C_{1157}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                              | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                               | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                               | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                               | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                               | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                               | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                               | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                               | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                               | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                               | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                               | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                               | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                               | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                               | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                               | —                                                           |

Terms 2 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                               |

$$\beta_{C_{1158}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1158}}{d\mu}=\beta_{C_{1158}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{12 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{12 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{12 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{12 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y e}}{12 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        |

$$\beta_{C_{11511}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$$

### Terms 1 of 6

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |

## Terms 2 of 6

| Column | Representative term structure                                        |  |  |  |  |  |  |  |  |  |
|--------|----------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$ |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{MF}$            |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{g_2^2 \overline{y_1 y_2}}{MF}$                                |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                     | T2                                                          | T3                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                      | —                                                           | —                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                      | —                                                           | —                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                      | —                                                           | —                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                      | —                                                           | —                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                      | —                                                           | —                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                      | —                                                           | —                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                      | —                                                           | —                                        |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|----|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | —  | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | —  | —  |

# Terms 3 of 6

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $\frac{g_1^2 \overline{y_1 y_2}}{MF}$                             |  |
| T2     | $\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{MF}$ |  |
| T3     | $\frac{y e \overline{y_1 y_2} y \overline{e}}{MF}$                |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                        | T2                                                                  | T3                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                                   | —                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                                   | —                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                         | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3g_1^2 \overline{y_1 y_2}}{2MF}$  | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{27g_1^2 \overline{y_1 y_2}}{8MF}$ | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                         | —                                                                   | —                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                         | —                                                                   | —                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                                   | —                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                                   | —                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                                   | —                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                                   | —                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |

# Terms 4 of 6

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{MF}$ |
| T2     | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{MF}$       |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                            | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                             | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                             | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                   | —                                             | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                             | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{8MF}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{8MF}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{2MF}$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1 y_2}}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                             | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                             | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                             | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                             | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |

# Terms 5 of 6

| Column | Representative term structure                                   |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$           |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$              |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                       | T2                                                             | T3                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                              | —                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                              | —                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                              | —                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                        | —                                                              | —                                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                        | —                                                              | —                                                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                        | —                                                              | —                                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                              | —                                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                              | —                                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                              | —                                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                              | —                                                                 |

Terms 6 of 6

| Column | Representative term structure                                |
|--------|--------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{MF}$                      |
| T2     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$        | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$        | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}}g_1^2y_1y_2}{2MF}$        | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2y_1y_2}{8MF}$ | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                              |

$$\beta_{C_{11512}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Terms 1 of 6

| Column | Representative term structure                                   |
|--------|-----------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$              |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                         | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                          | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                          | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                            |

## Terms 2 of 6

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye y_1 y_2 y \bar{e}}{MF}$          |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{MF}$ |
| T3     | $\frac{ig_2^2 y_1 y_2}{MF}$                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                                           | T3                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|--------------------------------------------------------------|--------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y \bar{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$  | —                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y \bar{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$  | —                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y \bar{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$  | —                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y \bar{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$  | —                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y \bar{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$  | —                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                                            | $-\frac{3ig_2^2 y_1 y_2}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                                            | $-\frac{3ig_2^2 y_1 y_2}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                                            | $-\frac{3ig_2^2 y_1 y_2}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                                            | $-\frac{3ig_2^2 y_1 y_2}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                                            | $-\frac{3ig_2^2 y_1 y_2}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                                            | —                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                                            | —                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                                            | —                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                                            | —                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                                            | —                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$ | —                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$ | —                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$ | —                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$ | —                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 y_1 y_2}{3MF}$ | —                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                                            | —                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                                            | —                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|----|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | —  | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | —  | —  |



# Terms 3 of 6

| Column | Representative term structure                        |
|--------|------------------------------------------------------|
| T1     | $\frac{ig_1^2 \overline{y_1 y_2}}{MF}$               |
| T2     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{MF}$ |
| T3     | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                                                     | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|--------------------------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                                      | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                                      | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                                      | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                                      | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                                      | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3ig_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                          | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3ig_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3ig_1^2 \overline{y_1 y_2}}{2MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{27ig_1^2 \overline{y_1 y_2}}{8MF}$ | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | —                                                      | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | —                                                      | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | —                                                      | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | —                                                      | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | —                                                      | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                                      | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                                      | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                                      | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                                      | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                                      | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                                      | —                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                                      | —                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                                      | —                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                                      | —                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                                      | —                                               |

# Terms 4 of 6

| Column | Representative term structure                 |  |
|--------|-----------------------------------------------|--|
| T1     | $\frac{ilambda12Inv2\overline{y_1 y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{MF}$ |  |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y_1 y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                             | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                              | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                              | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                              | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                              | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                              | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                              | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                              | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                              | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                              | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                              | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1 y_2}}{8MF}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{4MF}$ | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1 y_2}}{8MF}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1 y_2}}{2MF}$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{ilambda12Inv2\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y_1 y_2}}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                              | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                              | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                              | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                              | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                              | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                              | —                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                              | —                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                              | —                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                              | —                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                              | —                                               |

# Terms 5 of 6

| Column | Representative term structure                                               |  |
|--------|-----------------------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$                         |  |
| T2     | $\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$                    |  |
| T3     | $\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                      | T2                                                                    | T3                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                     | —                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                     | —                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                     | —                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                     | —                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                     | —                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                       | —                                                                     | —                                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                       | —                                                                     | —                                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                       | —                                                                     | —                                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                       | —                                                                     | —                                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                       | —                                                                     | —                                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                     | —                                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                     | —                                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                     | —                                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                     | —                                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                     | —                                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | $-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                       | $-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                       | $-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                       | $-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                       | $-\frac{17 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                     | —                                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                     | —                                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                     | —                                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                     | —                                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                     | —                                                                              |

# Terms 6 of 6

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$             |
| T2     | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$         | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$         | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{2MF}$         | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |

$$\beta_{C_{11513}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11513}}{d\mu}=\beta_{C_{11513}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                           | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                            | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                            | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                            | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                            | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |  |
| T2     | $\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$                 |  |
| T3     | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{M F}$                 |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                                   | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |

$\beta_{C_{11514}}^{(1)}$

$16\pi^2\mu\frac{dC_{11514}}{d\mu}=\beta_{C_{11514}}^{(1)}$

Terms 1 of 2

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$          |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                    | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|---------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                     | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                     | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                     | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                     | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                     | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                     | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                     | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                     | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                     | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                     | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                     | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                     | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                     | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                     | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                     | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                     | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                     | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                     | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                     | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                     | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |

## Terms 2 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$        |
| T2     | $\frac{\sqrt{3}ilambda12Invl\overline{y_1}y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{MF}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                   | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                    | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                    | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                    | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                    | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                    | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                    | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                    | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                    | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                    | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                    | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                    | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                    | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                    | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                    | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                    | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                    | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                    | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                    | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                    | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                    | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12Invl\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | $\frac{\sqrt{3}ilambda12Invl\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12Invl\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{2MF}$  | $\frac{\sqrt{3}ilambda12Invl\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$ | $\frac{\sqrt{3}ilambda12Invl\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |



$$\beta_{C_{1166}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1166}}{d\mu}=\beta_{C_{1166}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$    |
| T2     | $\frac{g_2^2\overline{y_1}^2}{MF}$           |
| T3     | $\frac{ye\overline{y_1}^2\overline{ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                                   | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|--------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                    | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                    | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$ | $\frac{3g_2^2\overline{y_1}^2}{4MF}$ | $-\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                          | —                                    | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | —                                    | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | —                                    | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                    | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                    | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                    | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                    | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |

# Terms 2 of 2

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$             |
| T2     | $\frac{\sqrt{3}\lambda bdaS1Inv1\overline{y_1}^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                                      | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | —                                                       | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                                       | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$ | $\frac{\sqrt{3}\lambda bdaS1Inv1\overline{y_1}^2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                                       | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |

$$\beta_{C_{1167}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1167}}{d\mu}=\beta_{C_{1167}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                                |
|--------|--------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                            | T3                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                             | $\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{2MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                             | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                             | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                             | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                             | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                             | —                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                             | —                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                             | —                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                             | —                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                              |

Terms 2 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \bar{y}_1 y_2 y \bar{e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} i y e \bar{y}_1 y_2 y \bar{e}}{12 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                               |

$$\beta_{C_{1168}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1168}}{d\mu}=\beta_{C_{1168}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{3^{\frac{3}{4}}\lambda\overline{12Inv1y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                           | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{12Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{12Inv1y_1y_2}}{6MF}$ | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{12Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{12Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{12Inv1y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                            | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                            | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                            | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                            | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                            | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                            | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                            | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                            | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                            | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                            | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                            | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                            | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                            | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                            | —                                                           |

Terms 2 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                               |

$$\beta_{C_{11611}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

# Terms 1 of 6

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$          |
| T2     | $\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                            | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                             | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                             | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                             | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                             | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                             | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                             | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                             | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                             | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |

## Terms 2 of 6

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$                  |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$ |
| T3     | $\frac{ig_2^2 \overline{y_1 y_2}}{MF}$                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                     | T2                                                                      | T3                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$ | —                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$ | —                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$ | —                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$ | —                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$ | —                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                      | —                                                                       | $\frac{3ig_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                      | —                                                                       | $\frac{3ig_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                      | —                                                                       | $\frac{3ig_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                      | —                                                                       | $\frac{3ig_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                      | —                                                                       | $\frac{3ig_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                      | —                                                                       | —                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                      | —                                                                       | —                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                      | —                                                                       | —                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                      | —                                                                       | —                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                      | —                                                                       | —                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$  | —                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                      | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$  | —                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                      | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$  | —                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                      | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$  | —                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                      | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{3MF}$  | —                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | —                                                                       | —                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                      | —                                                                       | —                                        |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|----|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | —  | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | —  | —  |

# Terms 3 of 6

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $\frac{ig_1^2 \overline{y_1 y_2}}{MF}$                            |  |
| T2     | $\frac{iy e \overline{y_1 y_2 y \bar{e}}}{MF}$                    |  |
| T3     | $\frac{\sqrt{3} i \lambda d a 12 I n v 2 \overline{y_1 y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                        | T2                                               | T3                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3ig_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ | $\frac{\sqrt{3} i \lambda d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                         | $-\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ | $\frac{\sqrt{3} i \lambda d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3ig_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ | $\frac{\sqrt{3} i \lambda d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3ig_1^2 \overline{y_1 y_2}}{2MF}$  | $-\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ | $\frac{\sqrt{3} i \lambda d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{27ig_1^2 \overline{y_1 y_2}}{8MF}$ | $-\frac{iy e \overline{y_1 y_2 y \bar{e}}}{4MF}$ | $\frac{\sqrt{3} i \lambda d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                         | —                                                | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                         | —                                                | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                         | —                                                | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                         | —                                                | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                         | —                                                | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                | —                                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                | —                                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                | —                                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                | —                                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                | —                                                                  |

# Terms 4 of 6

| Column | Representative term structure     |  |
|--------|-----------------------------------|--|
| T1     | $\frac{ilambda12Inv2y_1y_2}{MF}$  |  |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$ |  |
| T3     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                  | T3                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|-------------------------------------|--------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                   | —                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                   | —                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                   | —                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                   | —                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                   | —                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                   | —                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                   | —                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                   | —                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                   | —                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                   | —                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{ilambda12Inv2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{ilambda12Inv2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ | —                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{ilambda12Inv2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{ilambda12Inv2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{ilambda12Inv2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ | $-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                   | —                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                   | —                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                   | —                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                   | —                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                   | —                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                   | —                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                   | —                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                   | —                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                   | —                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                   | —                                    |

# Terms 5 of 6

| Column | Representative term structure                                             |  |
|--------|---------------------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$                       |  |
| T2     | $\frac{3^{\frac{3}{4}} i \lambda d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |  |
| T3     | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$                  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                     | T2                                                                          | T3                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | —                                                                           | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                      | —                                                                           | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                      | —                                                                           | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                      | —                                                                           | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                      | —                                                                           | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                      | —                                                                           | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                      | —                                                                           | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                      | —                                                                           | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                      | —                                                                           | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                      | —                                                                           | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                           | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                           | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                           | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                           | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                           | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | $\frac{3^{\frac{3}{4}} i \lambda d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                      | $\frac{3^{\frac{3}{4}} i \lambda d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                      | $\frac{3^{\frac{3}{4}} i \lambda d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                      | $\frac{3^{\frac{3}{4}} i \lambda d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{2 M F}$         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                      | $\frac{3^{\frac{3}{4}} i \lambda d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $\frac{9 \cdot 3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | —                                                                           | —                                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                      | —                                                                           | —                                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                      | —                                                                           | —                                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                      | —                                                                           | —                                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                      | —                                                                           | —                                                                  |

Terms 6 of 6

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$             |
| T2     | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                | T2                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------|-----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                 | —                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                 | —                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                 | —                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                 | —                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                 | —                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                 | —                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                 | —                                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                 | —                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                 | —                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                 | —                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                 | —                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                 | —                                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                 | —                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                 | —                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                 | —                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{17 \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                 | —                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                 | —                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                 | —                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                 | —                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                 | —                                                                     |

$$\beta_{C_{11612}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$$

# Terms 1 of 6

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |

## Terms 2 of 6

| Column | Representative term structure                                        |  |  |  |  |  |  |  |  |  |
|--------|----------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$ |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{MF}$            |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{g_2^2 \overline{y_1 y_2}}{MF}$                                |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                     | T2                                                          | T3                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ | —                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                      | —                                                           | $-\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                      | —                                                           | —                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                      | —                                                           | —                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                      | —                                                           | —                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                      | —                                                           | —                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                      | —                                                           | —                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | —                                                           | —                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                      | —                                                           | —                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                      | —                                                           | —                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|----|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | —  | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | —  | —  |



# Terms 3 of 6

| Column | Representative term structure                                      |  |
|--------|--------------------------------------------------------------------|--|
| T1     | $\frac{g_1^2 \overline{y_1 y_2}}{M F}$                             |  |
| T2     | $\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$ |  |
| T3     | $\frac{y e \overline{y_1 y_2} y \overline{e}}{M F}$                |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                        | T2                                                                  | T3                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                                   | —                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                                   | —                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                         | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3g_1^2 \overline{y_1 y_2}}{2MF}$  | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{27g_1^2 \overline{y_1 y_2}}{8MF}$ | $-\frac{\sqrt{3} \lambda b d a 12 I n v 2 \overline{y_1 y_2}}{3MF}$ | $\frac{y e \overline{y_1 y_2} y \overline{e}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                         | —                                                                   | —                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                         | —                                                                   | —                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                                   | —                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                                   | —                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                                                   | —                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                                                   | —                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                                                   | —                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                                                   | —                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                                                   | —                                                   |

# Terms 4 of 6

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{MF}$ |
| T2     | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{MF}$       |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                            | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                             | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                             | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                   | —                                             | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                             | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{8MF}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{8MF}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1 y_2}}{2MF}$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{\text{lambda12Inv2}\overline{y_1 y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1 y_2}}{4MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1 y_2}}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                             | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                             | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                             | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                             | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                             | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                             | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                             | —                                              |

# Terms 5 of 6

| Column | Representative term structure                                   |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$           |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$              |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                       | T2                                                             | T3                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                              | —                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                              | —                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                              | —                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                        | —                                                              | —                                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                        | —                                                              | —                                                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                        | —                                                              | —                                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$ | —                                                              | —                                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                        | $-\frac{17 \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1}y_2}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                              | —                                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                              | —                                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                              | —                                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                              | —                                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                              | —                                                                 |

Terms 6 of 6

| Column | Representative term structure                                |
|--------|--------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$           |
| T2     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                          | T2                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                           | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                           | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                           | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                           | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                           | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                           | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                           | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                           | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                           | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                           | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                           | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                           | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                           | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                           | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                           | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | —                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                           | —                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                           | —                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                           | —                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                           | —                                                              |

$$\beta_{C_{11613}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11613}}{d\mu}=\beta_{C_{11613}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                                    | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|-------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                                     | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                                     | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             | —                                                     | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                                     | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{\sqrt{3}i g_2^2 \overline{y_1 y_2}}{M F}$                    |
| T3     | $\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{M F}$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                      | T2                                                   | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$\beta_{C_{11614}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11614}}{d\mu}=\beta_{C_{11614}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                           | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                            | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                            | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                            | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                            | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |  |
| T2     | $\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$                 |  |
| T3     | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{M F}$                 |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                                   | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |



$$\beta_{C_{11711}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11711}}{d\mu}=\beta_{C_{11711}}^{(1)}$$

# Terms 1 of 3

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                             | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                             | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                             | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                             | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                             | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |

## Terms 2 of 3

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2 y_1 y_2}{MF}$  |
| T2     | $\frac{\sqrt{3}ye y_1 y_2 y_e}{MF}$ |
| T3     | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                     | T3                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|----------------------------------------|--------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                      | —                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                      | —                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                      | —                                      | —                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                      | —                                      | —                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                      | —                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                      | —                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                      | —                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                      | —                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |

$$\beta_{C_{11712}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11712}}{d\mu}=\beta_{C_{11712}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2y_1y_2}{MF}$                    |
| T2     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$                    |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                 | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                  | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                  | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                  | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                  | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                  | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                  | —                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                  | —                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                  | —                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                  | —                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                  | —                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  | —                                  | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  | —                                  | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  | —                                  | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  | —                                  | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  | —                                  | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                  | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                  | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                  | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                  | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                  | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 2 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$         |
| T2     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  | T2                                           | T3                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|----------------------------------------------|---------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                            | —                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                            | —                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                            | —                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                            | —                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                            | —                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                   | —                                            | —                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                   | —                                            | —                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                   | —                                            | —                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                   | —                                            | —                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                   | —                                            | —                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                   | —                                            | —                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                   | —                                            | —                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                   | —                                            | —                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                   | —                                            | —                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                   | —                                            | —                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                            | —                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                            | —                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                            | —                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                            | —                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                            | —                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |

Terms 3 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |

$$\beta_{C_{11715}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11715}}{d\mu}=\beta_{C_{11715}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$                    |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                                    | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|-------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                     | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                     | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                     | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                     | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                     | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                                     | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                                     | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                                     | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                                     | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                                     | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                                     | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                                     | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                                     | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                                     | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                                     | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                     | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                     | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                     | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                     | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                     | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y}_1y_2}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$    |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$            |
| T3     | $\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                   | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|--------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                    | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                    | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                    | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                    | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                    | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                    | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                    | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                    | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |



$$\beta_{C_{11716}}^{(1)}$$

$$_{16\pi^2}\mu\frac{dC_{11716}}{d\mu}=\beta_{C_{11716}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                           | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                            | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                            | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                            | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                            | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                        |  |
|--------|------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}\lambda 12Inv1\overline{y_1y_2}}{MF}$ |  |
| T2     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$           |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                                            | T3                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|-----------------------------------------------|----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                             | —                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                             | —                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                             | —                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                             | —                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                             | —                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                     | —                                             | —                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                     | —                                             | —                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                     | —                                             | —                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                     | —                                             | —                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                     | —                                             | —                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                     | —                                             | —                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                     | —                                             | —                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                     | —                                             | —                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                     | —                                             | —                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                     | —                                             | —                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                                             | —                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                                             | —                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                                             | —                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                                             | —                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                                             | —                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$ |

$$\beta_{C_{11811}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11811}}{d\mu}=\beta_{C_{11811}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2y_1y_2}{MF}$  |
| T2     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$  |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                | T3                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|-----------------------------------|----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                 | —                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                 | —                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                 | —                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                 | —                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                 | —                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                 | —                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                 | —                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                 | —                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                 | —                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                 | —                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                 | —                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                 | —                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                 | —                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                 | —                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                 | —                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                 | —                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                 | —                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                 | —                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                 | —                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                 | —                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |

## Terms 2 of 3

| Column | Representative term structure                  |
|--------|------------------------------------------------|
| T1     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$      |
| T2     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$              |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                   | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|--------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                    | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                    | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                    | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                    | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                    | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                    | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                    | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                    | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                    | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                    | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                    | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             | —                                    | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                    | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                    | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                    | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                    | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                    | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                    | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                    | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                    | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | —                                    | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{11812}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11812}}{d\mu}=\beta_{C_{11812}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                             | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                             | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                             | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                             | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                             | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |

## Terms 2 of 3

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2 y_1 y_2}{MF}$  |
| T2     | $\frac{\sqrt{3}ye y_1 y_2 y_e}{MF}$ |
| T3     | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                     | T3                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|----------------------------------------|--------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                      | —                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                      | —                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                      | —                                      | —                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                      | —                                      | —                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                      | —                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                      | —                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                      | —                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                      | —                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                      | —                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                      | —                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                      | —                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{49\sqrt{3}g_2^2 y_1 y_2}{24MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 y_e}{12MF}$ | $\frac{\sqrt{6}ig_2^2 y_1 y_2}{3MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |



$$\beta_{C_{11815}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11815}}{d\mu}=\beta_{C_{11815}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                             | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                             | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                             | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                             | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                             | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |

## Terms 2 of 2

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$            |
| T2     | $\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$           |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                             | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                              | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                              | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                              | —                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                              | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |

$$\beta_{C_{11816}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11816}}{d\mu}=\beta_{C_{11816}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                    | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                     | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                     | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                     | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                     | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                     | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                     | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                     | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                     | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                     | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                     | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                     | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                                     | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                     | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                     | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                     | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                     | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                     | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                     | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                     | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                     | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{\sqrt{3}i g_2^2 \overline{y_1 y_2}}{M F}$                    |
| T3     | $\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{M F}$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                      | T2                                                   | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$\beta_{C_{11911}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{11911}}{d\mu}=\beta_{C_{11911}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |

Terms 2 of 2

| Column | Representative term structure                                         |  |  |  |  |  |  |
|--------|-----------------------------------------------------------------------|--|--|--|--|--|--|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{M F}$ |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        |

$$\beta_{C_{11912}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11912}}{d\mu}=\beta_{C_{11912}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                         | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                          | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                          | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                          | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                          | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                           |

Terms 2 of 2

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2ye}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2ye}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2ye}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2ye}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     |



$$\beta_{C_{11913}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11913}}{d\mu}=\beta_{C_{11913}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure             |  |
|--------|-------------------------------------------|--|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$          |  |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |  |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                     | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|----------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                      | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                      | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                      | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                      | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  | —                                      | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  | —                                      | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                      | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                      | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |

## Terms 2 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                              | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                               | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                               | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                               | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                               | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | —                                               | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{11914}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{11914}}{d\mu}=\beta_{C_{11914}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                      |  |
|--------|----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$ |  |
| T2     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$        |  |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$        |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                            | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{MF}$ |
| T2     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$               |
| T3     | $\frac{\sqrt{3}ig_2^2\bar{y}_1\bar{y}_2}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                               | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |

$$\beta_{C_{111011}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{111011}}{d\mu}=\beta_{C_{111011}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$          |
| T2     | $\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                            | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                             | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                             | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                             | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                             | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                             | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                             | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                             | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                             | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |

| Column | Representative term structure                     |
|--------|---------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   |



$$\beta_{C_{111012}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{111012}}{d\mu}=\beta_{C_{111012}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |

Terms 2 of 2

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 \overline{y} e}{12 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        |

$$\beta_{C_{111013}}^{(1)}$$

$$16\pi^2\,_{\mu}\frac{dC_{111013}}{d\mu}=\beta_{C_{111013}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                | —                                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                | —                                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                | —                                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                | —                                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | —                                                            | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                | —                                                            | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                | —                                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                | —                                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ |

## Terms 2 of 3

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$           |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                             | T2                                                      | T3                                            |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                             |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                             |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                       | —                                             |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                       | —                                             |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                                       | —                                             |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                                       | —                                             |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                             |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                             |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                              | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |

$$\beta_{C_{111014}}^{(1)}$$

$$16\pi^2\,_{\mu}\frac{dC_{111014}}{d\mu}=\beta_{C_{111014}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$          |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                     | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|----------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                      | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                      | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                      | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                      | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  | —                                      | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  | —                                      | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                      | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                      | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                      | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                      | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                      | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |

## Terms 2 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                              | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                               | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                               | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                               | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                               | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | —                                               | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2\bar{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |



$$\beta_{C_{111111}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{111111}}{d\mu}=\beta_{C_{111111}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                 |
|--------|-----------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_2}^2}{MF}$            |
| T2     | $\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$     |
| T3     | $\frac{ye\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                          | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|---------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_2}^2}{4MF}$ | $-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$ | $\frac{ye\overline{y_2}^2\overline{y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                           | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                           | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                           | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |

# Terms 2 of 2

| Column | Representative term structure                                  |
|--------|----------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y}_2^2}{MF}$                     |
| T2     | $\frac{\sqrt{3}\lambda\text{ambda}S2Inv1\overline{y}_2^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}e}{MF}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                               | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                                | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                                | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                                | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                                | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                                | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                                | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                                | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                                | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                                | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                                | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                                | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y}_2^2}{4MF}$ | $-\frac{\sqrt{3}\lambda\text{ambda}S2Inv1\overline{y}_2^2}{6MF}$ | $\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}e}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                                | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                                | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                                | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                                | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                                | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                                | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                                | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                                | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                                | —                                                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                                | —                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                                | —                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                                | —                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                                | —                                                      |

$$\beta_{C_{111112}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{111112}}{d\mu}=\beta_{C_{111112}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_2}^2}{MF}$             |
| T2     | $\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$      |
| T3     | $\frac{iy_e\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                           | T3                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|----------------------------------------------|--------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                            | —                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                            | —                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3ig_2^2\overline{y_2}^2}{4MF}$ | $-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$ | $\frac{iy_e\overline{y_2}^2\overline{y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                      | —                                            | —                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                            | —                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                            | —                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                            | —                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                            | —                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                            | —                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                            | —                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |

Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$           |
| T2     | $\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$    |
| T3     | $\frac{\sqrt{3}iye\overline{y_2}^2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                                   | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                    | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                    | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                                    | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                                    | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$ | $-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_2}^2\overline{y}e}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                                    | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                    | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                    | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                    | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                    | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |

$$\beta_{C_{111212}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{111212}}{d\mu}=\beta_{C_{111212}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{lambda S2Inv1\overline{y_2}^2}{MF}$   |
| T2     | $\frac{g_2^2\overline{y_2}^2}{MF}$           |
| T3     | $\frac{ye\overline{y_2}^2\overline{ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                   | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|--------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                    | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                    | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                    | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                    | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                    | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                    | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{lambda S2Inv1\overline{y_2}^2}{2MF}$ | $\frac{3g_2^2\overline{y_2}^2}{4MF}$ | $-\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                    | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                    | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                    | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                    | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                    | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                    | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                    | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                    | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                    | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                    | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                    | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                    | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                    | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                    | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                    | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                    | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                    | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                    | —                                              |

Terms 2 of 2

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y}_2^2}{MF}$             |
| T2     | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y}_2^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}e}{MF}$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                                      | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | —                                                       | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                                       | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}g_2^2\overline{y}_2^2}{4MF}$ | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y}_2^2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}e}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                                       | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                       |

$$\beta_{C_{1213}}^{(1)}$$

$$16\pi^2 \mu \frac{dC_{1213}}{d\mu} = \beta_{C_{1213}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{lambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                | T2                                       | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------|------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                 | $\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                 | $\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                 | $\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                 | $\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                 | $\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                        | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —  | $\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —  | $\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —  | $\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —  | $\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —  | $\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |



$$\beta_{C_{1214}}^{(1)}$$

$$16\pi^2 \mu \frac{dC_{1214}}{d\mu} = \beta_{C_{1214}}^{(1)}$$

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{ilambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                        | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|-------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ilambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —  | $\frac{\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —  | $\frac{\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —  | $\frac{\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —  | $\frac{\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —  | $\frac{\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{1223}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$$

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{ilambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                        | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|-------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | $\frac{ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2 | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|----|-----------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3i\lambda}T_3\bar{y}_1y_2}{3MF}$ | —  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —  | $\frac{\sqrt{6i\lambda}T_3\bar{y}_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —  | $\frac{\sqrt{6i\lambda}T_3\bar{y}_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —  | $\frac{\sqrt{6i\lambda}T_3\bar{y}_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —  | $\frac{\sqrt{6i\lambda}T_3\bar{y}_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —  | $\frac{\sqrt{6i\lambda}T_3\bar{y}_1y_2}{3MF}$ |

$$\beta_{C_{1224}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$$

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ |
| T2     | $\frac{lambdaT_3\overline{y_1y_2}}{3MF}$         |
| T3     | $\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                        | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|-------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                | $-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | $-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                | $-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                | $-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                | $-\frac{lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2 | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|----|-------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —  | $-\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —  | $-\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —  | $-\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —  | $-\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —  | $-\frac{\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{1257}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{1257}}{d\mu}=\beta_{C_{1257}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_1}^2}{MF}$           |
| T2     | $\frac{\lambda S1Inv1\overline{y_1}^2}{MF}$  |
| T3     | $\frac{ye\overline{y_1}^2\overline{ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                            | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|-----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_1}^2}{8MF}$ | $-\frac{\lambda S1Inv1\overline{y_1}^2}{4MF}$ | $\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                             | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                             | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                             | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             | —                                             |

Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda daS1Inv1\overline{y_1}^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                       | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                        | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                        | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                        | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                        | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                        | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                        | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                        | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                        | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                        | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                        | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                        | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$ | $-\frac{\sqrt{3}\lambda daS1Inv1\overline{y_1}^2}{12MF}$ | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                        | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                        | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                        | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                        | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                        | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                        | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                        | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                        | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                        | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                        | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                        | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                        | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                        | —                                                       |



$$\beta_{C_{1258}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1258}}{d\mu}=\beta_{C_{1258}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$      |
| T2     | $\frac{ig_2^2\overline{y_1}^2}{MF}$             |
| T3     | $\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                    | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$ | $\frac{3ig_2^2\overline{y_1}^2}{8MF}$ | $-\frac{iy_e\overline{y_1}^2\overline{y_e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                     | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                     | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                     | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |

Terms 2 of 2

| Column | Representative term structure                         |  |
|--------|-------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$           |  |
| T2     | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$    |  |
| T3     | $\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                   | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                    | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                    | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$ | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$ | $-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                    | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |

$$\beta_{C_{1259}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{1259}}{d\mu}=\beta_{C_{1259}}^{(1)}$$

### Terms 1 of 5

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                         | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |

## Terms 2 of 5

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{M F}$                |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$   |
| T3     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                                      | T3                                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |

# Terms 3 of 5

| Column | Representative term structure            |
|--------|------------------------------------------|
| T1     | $\frac{g_2^2 \overline{y_1 y_2}}{M F}$   |
| T2     | $\frac{g_1^2 \overline{y_1 y_2}}{M F}$   |
| T3     | $\frac{y e \overline{y_1 y_2 y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                      | T2                                        | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------|-------------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{16MF}$  | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | —                                         | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{16MF}$  | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{4MF}$   | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{27g_1^2 \overline{y_1 y_2}}{16MF}$ | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                       | —                                         | —                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                       | —                                         | —                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |

# Terms 4 of 5

| Column | Representative term structure                             |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{MF}$ |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{MF}$         |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$              |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                         | T2                                                  | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                   | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                   | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                   | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                   | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                   | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                   | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                   | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                   | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                   | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                   | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                   | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                   | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                   | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                   | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                   | —                                              |

Terms 5 of 5

| Column |  | Representative term structure                          |
|--------|--|--------------------------------------------------------|
| T1     |  | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$            |
| T2     |  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{24MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{24MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{4MF}$   | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{24MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$ | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{24MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                        |

$$\beta_{C_{12510}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12510}}{d\mu}=\beta_{C_{12510}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{MF}$       |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                                  | T3                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{6MF}$ | —                                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                                   | —                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                                   | —                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                                   | —                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                                   | —                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                                   | —                                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                                   | —                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                                   | —                                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                                   | —                                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                                   | —                                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                                   | —                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                                   | —                                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                                   | —                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                                   | —                                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                                   | —                                                             |



## Terms 2 of 5

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$                |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |
| T3     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                   | T2                                                             | T3                                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                    | —                                                              | —                                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                    | —                                                              | —                                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                    | —                                                              | —                                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                    | —                                                              | —                                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                    | —                                                              | —                                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                    | —                                                              | —                                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                              | —                                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                              | —                                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                              | —                                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                              | —                                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                              | —                                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                              | —                                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                              | —                                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                              | —                                                                         |

# Terms 3 of 5

| Column | Representative term structure            |  |
|--------|------------------------------------------|--|
| T1     | $\frac{ig_2^2 \overline{y_1 y_2}}{MF}$   |  |
| T2     | $\frac{ig_1^2 \overline{y_1 y_2}}{MF}$   |  |
| T3     | $\frac{iy e \overline{y_1 y_2 y e}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                       | T2                                         | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------|--------------------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                          | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                          | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                          | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                          | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                          | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3ig_1^2 \overline{y_1 y_2}}{16MF}$  | $-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$ | —                                          | $-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3ig_1^2 \overline{y_1 y_2}}{16MF}$  | $-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3ig_1^2 \overline{y_1 y_2}}{4MF}$   | $-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3ig_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{27ig_1^2 \overline{y_1 y_2}}{16MF}$ | $-\frac{iy e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                        | —                                          | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                        | —                                          | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                        | —                                          | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                        | —                                          | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                        | —                                          | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                          | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                          | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                          | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                          | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                          | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                        | —                                          | —                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                        | —                                          | —                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                        | —                                          | —                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                        | —                                          | —                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                        | —                                          | —                                          |

# Terms 4 of 5

| Column | Representative term structure                       |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$ |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$         |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$        |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |

Terms 5 of 5

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$           |
| T2     | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                     | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{16MF}$  | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{24MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ig_1^2y_1y_2}{16MF}$  | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{24MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}ig_1^2y_1y_2}{4MF}$   | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{24MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}ig_1^2y_1y_2}{16MF}$ | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{24MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                             |

$$\beta_{C_{12511}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12511}}{d\mu}=\beta_{C_{12511}}^{(1)}$$

Column Representative term structure

$$\begin{array}{c} \text{T1} \\ \frac{\sqrt{6}g_2^2y_1y_2}{MF} \end{array}$$

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12512}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12512}}{d\mu}=\beta_{C_{12512}}^{(1)}$$

Column Representative term structure

|    |                                   |
|----|-----------------------------------|
| T1 | $\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$ |
|----|-----------------------------------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12515}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12515}}{d\mu}=\beta_{C_{12515}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                                    | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                                     | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                                     | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             | —                                                     | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                                     | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |

Terms 2 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$         |
| T2     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  | T2                                           | T3                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|----------------------------------------------|--------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                            | —                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                            | —                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                   | —                                            | —                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                   | —                                            | —                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                   | —                                            | —                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                   | —                                            | —                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                            | —                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                            | —                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |



Terms 3 of 3

| Column | Representative term structure                              |  |
|--------|------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 y \bar{e}}{M F}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 y \bar{e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 y \bar{e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 y \bar{e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 y \bar{e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 y \bar{e}}{12 M F}$ |

$$\beta_{C_{12516}}^{(1)}$$

$$_{16\pi^2}\mu\frac{dC_{12516}}{d\mu}=\beta_{C_{12516}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{MF}$ |
| T3     | $\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                            | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{15\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{15\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{15\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{15\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{15\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{6MF}$ |

## Terms 2 of 3

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                      | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|---------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                       | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                       | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                                       | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                                       | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |

$$\beta_{C_{1267}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1267}}{d\mu}=\beta_{C_{1267}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$      |
| T2     | $\frac{ig_2^2\overline{y_1}^2}{MF}$             |
| T3     | $\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                    | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ilambdaS1Inv1\overline{y_1}^2}{4MF}$ | $\frac{3ig_2^2\overline{y_1}^2}{8MF}$ | $-\frac{iy_e\overline{y_1}^2\overline{y_e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                     | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                     | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                     | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |

Terms
2 of 2

| Column | Representative term structure                         |  |
|--------|-------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$           |  |
| T2     | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$    |  |
| T3     | $\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                   | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                    | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                    | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{8MF}$ | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{12MF}$ | $-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                    | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |

$$\beta_{C_{1268}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1268}}{d\mu}=\beta_{C_{1268}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$    |
| T2     | $\frac{g_2^2\overline{y_1}^2}{MF}$           |
| T3     | $\frac{ye\overline{y_1}^2\overline{ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                                   | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|--------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                    | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                    | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{lambdaS1Inv1\overline{y_1}^2}{4MF}$ | $\frac{3g_2^2\overline{y_1}^2}{8MF}$ | $-\frac{ye\overline{y_1}^2\overline{ye}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                          | —                                    | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | —                                    | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | —                                    | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                    | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                    | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                    | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                    | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                    | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                    | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                    | —                                              |

Terms 2 of 2

| Column | Representative term structure                          |  |  |  |  |  |  |  |  |  |
|--------|--------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$             |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\sqrt{3}\lambda bdaS1Inv1\overline{y_1}^2}{MF}$ |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$  |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                                       | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                        | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                        | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | —                                                        | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                                        | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{8MF}$ | $\frac{\sqrt{3}\lambda bdaS1Inv1\overline{y_1}^2}{12MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                                        | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                        | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                        | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                        | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                        | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |



$$\beta_{C_{1269}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{1269}}{d\mu}=\beta_{C_{1269}}^{(1)}$$

### Terms 1 of 5

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$       |
| T3     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                           | T3                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{25\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                            | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                            | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                            | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                            | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                                  |

# Terms 2 of 5

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$                |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{MF}$    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                                      | T3                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$         | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$         | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{2MF}$         | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2 y \overline{e}}}{12MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                                       | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                                       | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                                       | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                                       | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                                       | —                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                                       | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                       | —                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                       | —                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                       | —                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                       | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                       | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                       | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                       | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                       | —                                                                    |

Terms 3 of 5

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_1y_2}}{MF}$                |
| T2     | $\frac{ig_1^2\overline{y_1y_2}}{MF}$                |
| T3     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                      | T2                                        | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------|-------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | —                                         | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{4MF}$   | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{27ig_1^2\overline{y_1y_2}}{16MF}$ | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                       | —                                         | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                       | —                                         | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |

# Terms 4 of 5

| Column | Representative term structure     |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{iyey_1y_2ye}{MF}$          |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{ilambda12Inv2y_1y_2}{MF}$  |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                        | T2                                | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------|-----------------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |

Terms 5 of 5

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$                      |
| T2     | $\frac{\sqrt{3}iye\overline{y}_1y_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                   | T2                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                    | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                    | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                    | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_1^2y_1y_2}{16MF}$  | $-\frac{\sqrt{3}iye\overline{y}_1y_2\overline{y}e}{24MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                    | $-\frac{\sqrt{3}iye\overline{y}_1y_2\overline{y}e}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_1^2y_1y_2}{16MF}$  | $-\frac{\sqrt{3}iye\overline{y}_1y_2\overline{y}e}{24MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\sqrt{3}ig_1^2y_1y_2}{4MF}$   | $-\frac{\sqrt{3}iye\overline{y}_1y_2\overline{y}e}{24MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{9\sqrt{3}ig_1^2y_1y_2}{16MF}$ | $-\frac{\sqrt{3}iye\overline{y}_1y_2\overline{y}e}{24MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                                         |

$$\beta_{C_{12610}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12610}}{d\mu}=\beta_{C_{12610}}^{(1)}$$

### Terms 1 of 5

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                         | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{25\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |

## Terms 2 of 5

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{M F}$                |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$   |
| T3     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                                      | T3                                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                                       | —                                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                                       | —                                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                                       | —                                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                                       | —                                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                                       | —                                                                          |

# Terms 3 of 5

| Column | Representative term structure            |
|--------|------------------------------------------|
| T1     | $\frac{g_2^2 \overline{y_1 y_2}}{M F}$   |
| T2     | $\frac{g_1^2 \overline{y_1 y_2}}{M F}$   |
| T3     | $\frac{y e \overline{y_1 y_2 y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                      | T2                                        | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------|-------------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{16MF}$  | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | —                                         | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{16MF}$  | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{4MF}$   | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3g_2^2 \overline{y_1 y_2}}{8MF}$ | $\frac{27g_1^2 \overline{y_1 y_2}}{16MF}$ | $-\frac{y e \overline{y_1 y_2 y e}}{8MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                       | —                                         | —                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                       | —                                         | —                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                         |



# Terms 4 of 5

| Column | Representative term structure                             |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{MF}$ |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{MF}$         |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$              |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                         | T2                                                  | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                   | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                   | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                   | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                   | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                   | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | —                                                   | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                          | $-\frac{\lambda\text{Inv}2\bar{y}_1\bar{y}_2}{6MF}$ | $-\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                   | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                   | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                   | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                   | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                   | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                   | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                   | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                   | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                   | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                   | —                                              |

Terms 5 of 5

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$            |
| T2     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{4MF}$   | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$ | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                        |

$$\beta_{C_{12611}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12611}}{d\mu}=\beta_{C_{12611}}^{(1)}$$

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12612}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12612}}{d\mu}=\beta_{C_{12612}}^{(1)}$$

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |

$\beta_{C_{12615}}^{(1)}$

$16\pi^2\mu\frac{dC_{12615}}{d\mu}=\beta_{C_{12615}}^{(1)}$

Terms 1 of 3

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                            | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{MF}$ |
| T2     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$               |
| T3     | $\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                              | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                               | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                               | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                               | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                               | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                               | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                               | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                               | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                               | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                               | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                               | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                               | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                               | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                               | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                               | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                               | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                               | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |

$$\beta_{C_{12616}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12616}}{d\mu}=\beta_{C_{12616}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                                    | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                                     | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                                     | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             | —                                                     | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                                     | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |



Terms 2 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$         |
| T2     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  | T2                                           | T3                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|----------------------------------------------|--------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                            | —                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                            | —                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                   | —                                            | —                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                   | —                                            | —                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                   | —                                            | —                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                   | —                                            | —                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                            | —                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                            | —                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                            | —                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                            | —                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                            | —                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{15\sqrt{3}ig_2^2y_1y_2}{8MF}$ |

Terms 3 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |

$$\beta_{C_{12711}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$$

### Terms 1 of 5

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_1y_2}}{MF}$                      |
| T2     | $\frac{g_1^2\overline{y_1y_2}}{MF}$                      |
| T3     | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                       | T3                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|------------------------------------------|------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3g_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | —                                        | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3g_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3g_1^2\overline{y_1y_2}}{4MF}$   | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{27g_1^2\overline{y_1y_2}}{16MF}$ | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                        | —                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                        | —                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |

# Terms 2 of 5

| Column | Representative term structure                |  |  |  |  |  |  |  |  |  |
|--------|----------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{MF}$    |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{MF}$  |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                            | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                            | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                            | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                            | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                            | —                                             |

# Terms 3 of 5

| Column | Representative term structure                                 |  |  |  |  |  |  |  |  |  |
|--------|---------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$                   |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$                    |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                            | T3                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                             | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                             | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                             | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                             | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                             | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                             | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                             | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                             | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                             | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                             | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$ | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | $-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$ | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$ | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{4MF}$   | $-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$ | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{16MF}$ | $-\frac{\sqrt{3}ye\overline{y_1y_2ye}}{24MF}$ | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                             | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                             | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                             | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                             | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                             | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                             | —                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                             | —                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                             | —                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                             | —                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                             | —                                                              |

# Terms 4 of 5

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{MF}$   |
| T2     | $\frac{3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$   |
| T3     | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                               | T3                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                | —                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                | —                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                | —                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                                | —                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                                | —                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                                | —                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                                | —                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                                | —                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                                | —                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{2MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8MF}$ | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                | —                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                | —                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                | —                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                | —                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                | —                                                          |

Terms 5 of 5

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$   |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                         | T2                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                            |

$$\beta_{C_{12712}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_1y_2}}{MF}$                |
| T2     | $\frac{ig_1^2\overline{y_1y_2}}{MF}$                |
| T3     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                      | T2                                        | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------|-------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | —                                         | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{4MF}$   | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{27ig_1^2\overline{y_1y_2}}{16MF}$ | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                       | —                                         | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                       | —                                         | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                         | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                         | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                         | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                         | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                         | —                                                     |



## Terms 2 of 5

| Column | Representative term structure     |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{iyey_1y_2ye}{MF}$          |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{ilambda12Inv2y_1y_2}{MF}$  |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                        | T2                                | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------|-----------------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{iyey_1y_2ye}{8MF}$ | —                                 | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                         | $\frac{ilambda12Inv2y_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |

# Terms 3 of 5

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$                      |  |
| T2     | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{MF}$                       |  |
| T3     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                             | T3                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                              | —                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                              | —                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                              | —                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                              | —                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                              | —                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                              | —                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                              | —                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                              | —                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                              | —                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                              | —                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$  | $-\frac{\sqrt{3}iye\overline{y_1}y_2ye}{24MF}$ | —                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | $-\frac{\sqrt{3}iye\overline{y_1}y_2ye}{24MF}$ | —                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$  | $-\frac{\sqrt{3}iye\overline{y_1}y_2ye}{24MF}$ | —                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{4MF}$   | $-\frac{\sqrt{3}iye\overline{y_1}y_2ye}{24MF}$ | —                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{9\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$ | $-\frac{\sqrt{3}iye\overline{y_1}y_2ye}{24MF}$ | —                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}g_2^2\overline{y_1}y_2}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                              | —                                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                              | —                                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                              | —                                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                              | —                                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                              | —                                                                   |

# Terms 4 of 5

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}} ig_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                            | T3                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                             | —                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                             | —                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                             | —                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                             | —                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                             | —                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                             | —                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1y_2}}{8MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                             | $\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1y_2}}{8MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1y_2}}{2MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{25 \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                             | —                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                             | —                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                               |

# Terms 5 of 5

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                         |

$$\beta_{C_{12713}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12713}}{d\mu}=\beta_{C_{12713}}^{(1)}$$

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12714}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12714}}{d\mu}=\beta_{C_{12714}}^{(1)}$$

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12811}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure         |
|--------|---------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_1y_2}}{MF}$  |
| T2     | $\frac{ig_1^2\overline{y_1y_2}}{MF}$  |
| T3     | $\frac{iy e\overline{y_1y_2y e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                       | T3                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|------------------------------------------|-----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $\frac{3ig_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{iy e\overline{y_1y_2y e}}{8MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | —                                        | $-\frac{iy e\overline{y_1y_2y e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $\frac{3ig_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{iy e\overline{y_1y_2y e}}{8MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $\frac{3ig_1^2\overline{y_1y_2}}{4MF}$   | $-\frac{iy e\overline{y_1y_2y e}}{8MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3ig_2^2\overline{y_1y_2}}{8MF}$ | $\frac{27ig_1^2\overline{y_1y_2}}{16MF}$ | $-\frac{iy e\overline{y_1y_2y e}}{8MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                        | —                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                        | —                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                        | —                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                        | —                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                        | —                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                       |

## Terms 2 of 5

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$ |  |
| T2     | $\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$        |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{6MF}$ | —                                             | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |



# Terms 3 of 5

| Column | Representative term structure                          |  |
|--------|--------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$           |  |
| T2     | $\frac{\sqrt{3}iye\overline{y_1}y_2\overline{ye}}{MF}$ |  |
| T3     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{MF}$    |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                                       | T3                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                        | —                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                        | —                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                        | —                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                        | —                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                        | —                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                | —                                                        | —                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                | —                                                        | —                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                | —                                                        | —                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                | —                                                        | —                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                | —                                                        | —                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$  | $\frac{\sqrt{3}iye\overline{y_1}y_2\overline{ye}}{24MF}$ | —                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | $\frac{\sqrt{3}iye\overline{y_1}y_2\overline{ye}}{24MF}$ | —                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$  | $\frac{\sqrt{3}iye\overline{y_1}y_2\overline{ye}}{24MF}$ | —                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{4MF}$   | $\frac{\sqrt{3}iye\overline{y_1}y_2\overline{ye}}{24MF}$ | —                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}ig_1^2\overline{y_1}y_2}{16MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2\overline{ye}}{24MF}$ | —                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                        | $-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{24MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                        | $-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{24MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                        | $-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{24MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                        | $-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{24MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                        | $-\frac{25 \cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{24MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                        | —                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                        | —                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                        | —                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                        | —                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                        | —                                                               |

# Terms 4 of 5

| Column | Representative term structure                                               |
|--------|-----------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$                    |
| T3     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                             | T2                                                                  | T3                                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                              | —                                                                   | —                                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                              | —                                                                   | —                                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                              | —                                                                   | —                                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                              | —                                                                   | —                                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                              | —                                                                   | —                                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                              | —                                                                   | —                                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                              | —                                                                   | —                                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                              | —                                                                   | —                                                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                              | —                                                                   | —                                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | —                                                                   | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{2 M F}$         | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{9 \cdot 3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                              | —                                                                   | —                                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                              | —                                                                   | —                                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                              | —                                                                   | —                                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                              | —                                                                   | —                                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                              | —                                                                   | —                                                                       |

# Terms 5 of 5

| Column | Representative term structure                                          |
|--------|------------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       | T2                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                        | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                        | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                           |

$$\beta_{C_{12812}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_1y_2}}{MF}$                      |
| T2     | $\frac{g_1^2\overline{y_1y_2}}{MF}$                      |
| T3     | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                       | T3                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|------------------------------------------|------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3g_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | —                                        | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3g_1^2\overline{y_1y_2}}{16MF}$  | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3g_1^2\overline{y_1y_2}}{4MF}$   | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{27g_1^2\overline{y_1y_2}}{16MF}$ | $-\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{6MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                        | —                                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                        | —                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                        | —                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                        | —                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                        | —                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                        | —                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                        | —                                                          |

# Terms 2 of 5

| Column | Representative term structure                |  |  |  |  |  |  |  |  |  |
|--------|----------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{MF}$    |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{MF}$  |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{ye\bar{y}_1\bar{y}_2\bar{y}e}{8MF}$ | —                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          | $\frac{lambda12Inv2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                            | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                            | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                            | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                            | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                            | —                                             |

# Terms 3 of 5

| Column | Representative term structure                                 |  |  |  |  |  |  |  |  |  |
|--------|---------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$                   |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$        |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1}y_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                        | T3                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                         | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                         | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                         | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                         | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                         | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                         | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                         | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                         | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                         | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                         | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$  | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$  | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{4MF}$   | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{16MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{24MF}$ | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                         | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1}y_2}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                         | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1}y_2}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                         | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1}y_2}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                         | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1}y_2}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                         | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1}y_2}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                         | —                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                         | —                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                         | —                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                         | —                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                         | —                                                              |

# Terms 4 of 5

| Column | Representative term structure               |  |
|--------|---------------------------------------------|--|
| T1     | $\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{MF}$  |  |
| T2     | $\frac{3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$  |  |
| T3     | $\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                                    | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|-------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                                     | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                                     | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                                     | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                                     | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                                     | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                                     | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                   | —                                                     | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                                     | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                                     | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                                     | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                   | —                                                     | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                   | —                                                     | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                   | —                                                     | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                   | —                                                     | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                   | —                                                     | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} g_1^2 y_1 y_2}{2MF}$         | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 y_1 y_2}{8MF}$ | $\frac{25 \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{24MF}$ | $-\frac{3^{\frac{3}{4}} ye y_1 y_2 y e}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                                     | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                                     | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                                     | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                                     | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                                     | —                                              |

Terms 5 of 5

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{MF}$   |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                         | T2                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 y_1 y_2}{6MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 y_1 y_2}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                          | —                                                            |



$$\beta_{C_{12813}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12813}}{d\mu}=\beta_{C_{12813}}^{(1)}$$

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12814}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12814}}{d\mu}=\beta_{C_{12814}}^{(1)}$$

Column Representative term structure

|    |                                  |
|----|----------------------------------|
| T1 | $\frac{\sqrt{6}g_2^2y_1y_2}{MF}$ |
|----|----------------------------------|

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ |

$$\beta_{C_{12911}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$$

### Terms 1 of 5

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\textit{lambdaS2Inv1}\overline{y_2}^2}{MF}$ |
| T2     | $\frac{g_2^2\overline{y_2}^2}{MF}$                 |
| T3     | $\frac{ye\overline{y_2}^2\overline{ye}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                   | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|--------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\textit{lambdaS2Inv1}\overline{y_2}^2}{4MF}$ | $\frac{3g_2^2\overline{y_2}^2}{8MF}$ | $-\frac{ye\overline{y_2}^2\overline{ye}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                    | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                   | —                                    | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                   | —                                    | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |

Terms 2 of 5

| Column | Representative term structure                          |  |
|--------|--------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$             |  |
| T2     | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y_2}^2}{MF}$ |  |
| T3     | $\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                                       | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                        | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                        | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | —                                                        | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                                        | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$ | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y_2}^2}{12MF}$ | $-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                                        | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                        | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                        | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                        | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                        | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                        | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                        | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                        | —                                                        |

# Terms 3 of 5

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2 y_1 y_2}{MF}$         |
| T2     | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{MF}$        |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                   | T2                                      | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------|-----------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                       | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                       | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                    | —                                       | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                    | —                                       | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                    | —                                       | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                    | —                                       | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                       | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                       | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |

# Terms 4 of 5

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                             | T2                                              | T3                                                   |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                               | —                                                    |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                               | —                                                    |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                               | —                                                    |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                               | —                                                    |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | —                                               | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{12912}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$$

Terms 1 of 5

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$      |
| T2     | $\frac{ig_2^2\overline{y_2}^2}{MF}$             |
| T3     | $\frac{iy_e\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                    | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$ | $\frac{3ig_2^2\overline{y_2}^2}{8MF}$ | $-\frac{iy_e\overline{y_2}^2\overline{y_e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                     | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                     | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                     | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |



Terms 2 of 5

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$            |
| T2     | $\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$     |
| T3     | $\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                   | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                    | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                    | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$ | $\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$ | $-\frac{\sqrt{3}iye\overline{y_2}^2\overline{y_e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                    | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                         |

# Terms 3 of 5

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $\frac{\sqrt{3}\lambda da12Inv1\bar{y}_1\bar{y}_2}{MF}$ |
| T2     | $\frac{\sqrt{6}g_2^2\bar{y}_1\bar{y}_2}{MF}$            |
| T3     | $\frac{\sqrt{3}g_1^2\bar{y}_1\bar{y}_2}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                        | T2                                            | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                             | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                             | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                             | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                             | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                             | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                         | —                                             | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                         | —                                             | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                         | —                                             | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                         | —                                             | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                         | —                                             | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                         | —                                             | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                         | —                                             | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                         | —                                             | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                         | —                                             | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                         | —                                             | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                             | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                             | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                             | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                             | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                             | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}\lambda da12Inv1\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{6}g_2^2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_1^2\bar{y}_1\bar{y}_2}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}\lambda da12Inv1\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{6}g_2^2\bar{y}_1\bar{y}_2}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}\lambda da12Inv1\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{6}g_2^2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_1^2\bar{y}_1\bar{y}_2}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}\lambda da12Inv1\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{6}g_2^2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{3}g_1^2\bar{y}_1\bar{y}_2}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}\lambda da12Inv1\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{\sqrt{6}g_2^2\bar{y}_1\bar{y}_2}{6MF}$ | $\frac{9\sqrt{3}g_1^2\bar{y}_1\bar{y}_2}{8MF}$ |

# Terms 4 of 5

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{MF}$ |
| T2     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$               |
| T3     | $\frac{\sqrt{6}ig_2^2\bar{y}_1\bar{y}_2}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                              | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                               | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                               | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                               | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                               | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                               | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                               | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                               | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                               | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                               | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                               | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                               | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                               | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                               | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                               | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                               | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                               | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                               | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{6}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{6}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{6}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{6}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{15\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{8MF}$ | $-\frac{\sqrt{6}ig_2^2\bar{y}_1\bar{y}_2}{6MF}$ |

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |

$$\beta_{C_{121011}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$$

Terms 1 of 5

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$      |
| T2     | $\frac{ig_2^2\overline{y_2}^2}{MF}$             |
| T3     | $\frac{iy_e\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                    | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ilambdaS2Inv1\overline{y_2}^2}{4MF}$ | $\frac{3ig_2^2\overline{y_2}^2}{8MF}$ | $-\frac{iy_e\overline{y_2}^2\overline{y_e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                     | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                     | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                     | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |

Terms 2 of 5

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$           |
| T2     | $\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$    |
| T3     | $\frac{\sqrt{3}iye\overline{y_2}^2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                   | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                    | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                    | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{8MF}$ | $\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{12MF}$ | $-\frac{\sqrt{3}iye\overline{y_2}^2\overline{y}e}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                    | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                    | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                    | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                    | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                    | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                    | —                                                        |

# Terms 3 of 5

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                            | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$ |

Terms 4 of 5

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                      | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                       | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                       | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                                       | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                                       | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ |



| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$ |

$$\beta_{C_{121012}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{121012}}{d\mu}=\beta_{C_{121012}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                 |
|--------|-----------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_2}^2}{MF}$            |
| T2     | $\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$     |
| T3     | $\frac{ye\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                          | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|---------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_2}^2}{8MF}$ | $-\frac{lambdaS2Inv1\overline{y_2}^2}{4MF}$ | $\frac{ye\overline{y_2}^2\overline{y_e}}{8MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                           | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                           | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                           | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |

Terms 2 of 5

| Column | Representative term structure                          |  |
|--------|--------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$             |  |
| T2     | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y_2}^2}{MF}$ |  |
| T3     | $\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                        | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                         | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                         | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                         | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                         | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                         | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                         | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                         | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                         | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y_2}^2}{8MF}$ | $-\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y_2}^2}{12MF}$ | $\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{24MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                         | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                         | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                         | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                         | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                         | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                         | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                         | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                         | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                         | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                         | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                         | —                                                       |

# Terms 3 of 5

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2 y_1 y_2}{MF}$         |
| T2     | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{MF}$        |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                   | T2                                      | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------|-----------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                       | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                       | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                    | —                                       | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                    | —                                       | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                    | —                                       | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                    | —                                       | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                       | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                       | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                       | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                       | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                       | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2 y_1 y_2}{6MF}$ | $-\frac{15\sqrt{3}ig_2^2 y_1 y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1 y_2}{22MF}$ |

# Terms 4 of 5

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                             | T2                                              | T3                                                   |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                               | —                                                    |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                               | —                                                    |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                               | —                                                    |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                               | —                                                    |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | —                                               | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{2211}}^{(1)}$$

$$16\pi^2 \mu \frac{dC_{2211}}{d\mu} = \beta_{C_{2211}}^{(1)}$$

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{MF}$ |
| T2     | $\frac{\text{lambda} T_3 \overline{y_1 y_2}}{MF}$          |
| T3     | $\frac{\sqrt{6} \text{lambda} T_3 \overline{y_1 y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | $-\frac{2 \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | $-\frac{2 \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | $-\frac{2 \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | $-\frac{2 \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | $-\frac{2 \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3} \text{lambda} T_3 \overline{y_1 y_2}}{3MF}$ | —                                                     | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|--------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |



$$\beta_{C_{2212}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$$

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{ilambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}ilambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                          | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|---------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | $-\frac{2ilambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{2\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$ | —                                           | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|--------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{2\sqrt{3}ilambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —  | $-\frac{2\sqrt{6}ilambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{2222}}^{(1)}$$

$$16\pi^2 \mu \frac{dC_{2222}}{d\mu} = \beta_{C_{2222}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{lambdaT_3\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                        | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|-------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | $\frac{2lambdaT_3\overline{y_1y_2}}{3MF}$ | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ | —                                         | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                     | T2 | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------|----|-------------------------------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{2\sqrt{3}\lambda T_3 \overline{y_1 y_2}}{3MF}$ | —  | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                      | —  | $\frac{2\sqrt{6}\lambda T_3 \overline{y_1 y_2}}{3MF}$ |

$$\beta_{C_{2257}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2257}}{d\mu}=\beta_{C_{2257}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                         | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |

Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{MF}$       |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |

$$\beta_{C_{2258}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2258}}{d\mu}=\beta_{C_{2258}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                                     |
|--------|-------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,i\lambda bda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$             |
| T3     | $\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 | T2                                                            | T3                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\,i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                             | $\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\,i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\,i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{2MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\,i\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  | —                                                             | —                                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  | —                                                             | —                                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  | —                                                             | —                                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  | —                                                             | —                                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  | —                                                             | —                                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  | —                                                             | —                                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                             | —                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                             | —                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                             | —                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                             | —                                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                             | —                                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                             | —                                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                             | —                                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                             | —                                                               |

Terms 2 of 2

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                         |



$$\beta_{C_{22511}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22511}}{d\mu}=\beta_{C_{22511}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |

## Terms 2 of 3

| Column | Representative term structure                                        |
|--------|----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$           |
| T2     | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{M F}$ |
| T3     | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                                     | T3                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                                      | —                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                                      | —                                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                                      | —                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                                      | —                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                                      | —                                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                                      | —                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | —                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{2 M F}$         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                                      | —                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                                      | —                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                                      | —                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                                      | —                                                                |

Terms 3 of 3

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2y_1y_2}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                | T2                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                 | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                 | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                 | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                 | —                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                 | —                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                 | —                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                 | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                 | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                 | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                 | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                 | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                 | —                                                     |

$$\beta_{C_{22512}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22512}}{d\mu}=\beta_{C_{22512}}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                                   |
|--------|-----------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$              |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                         | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                          | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                          | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                          | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                          | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                          | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                          | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                          | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                          | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                          | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                          | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                          | —                                                           |

## Terms 2 of 3

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1}y_2}{MF}$   |
| T3     | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1}y_2}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                           | T3                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | —                                                            | —                                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | —                                                            | —                                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | —                                                            | —                                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | —                                                            | —                                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | —                                                            | —                                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                            | —                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                            | —                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                            | —                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                            | —                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                            | —                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                            | —                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                            | —                                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                            | —                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                            | —                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                            | —                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1}y_2}{6MF}$ | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1}y_2}{8MF}$         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1}y_2}{6MF}$ | —                                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1}y_2}{6MF}$ | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1}y_2}{8MF}$         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1}y_2}{6MF}$ | $\frac{3^{\frac{3}{4}} ig_1^2\overline{y_1}y_2}{2MF}$         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                | $\frac{3^{\frac{3}{4}} ilambda12Inv1\overline{y_1}y_2}{6MF}$ | $\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2\overline{y_1}y_2}{8MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                            | —                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                            | —                                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                            | —                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                            | —                                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                            | —                                                             |

# Terms 3 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$   |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                              |

$$\beta_{C_{22513}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22513}}{d\mu}=\beta_{C_{22513}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                                           | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                            | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                            | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                | —                                                            | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                | —                                                            | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | —                                                            | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                | —                                                            | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                            | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                            | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$ |

## Terms 2 of 3

| Column | Representative term structure                                                |
|--------|------------------------------------------------------------------------------|
| T1     | $\frac{\sqrt{3\lambda d a_1 2 \operatorname{Inv} 1 \overline{y_1 y_2}}}{MF}$ |
| T2     | $\frac{\sqrt{6 i g_2^2 \overline{y_1 y_2}}}{MF}$                             |
| T3     | $\frac{\sqrt{3 i g_2^2 \overline{y_1 y_2}}}{MF}$                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                            | T2                                                | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                             | —                                                 | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                             | —                                                 | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                             | —                                                 | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                             | —                                                 | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                             | —                                                 | —                                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                             | —                                                 | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                             | —                                                 | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                             | —                                                 | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                             | —                                                 | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                             | —                                                 | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                             | —                                                 | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                             | —                                                 | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                             | —                                                 | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3\lambda d a_1 2 \operatorname{Inv} 1 \overline{y_1 y_2}}}{6MF}$ | $\frac{\sqrt{6 i g_2^2 \overline{y_1 y_2}}}{3MF}$ | $\frac{\sqrt{3 i g_2^2 \overline{y_1 y_2}}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3\lambda d a_1 2 \operatorname{Inv} 1 \overline{y_1 y_2}}}{6MF}$ | $\frac{\sqrt{6 i g_2^2 \overline{y_1 y_2}}}{3MF}$ | $\frac{\sqrt{3 i g_2^2 \overline{y_1 y_2}}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3\lambda d a_1 2 \operatorname{Inv} 1 \overline{y_1 y_2}}}{6MF}$ | $\frac{\sqrt{6 i g_2^2 \overline{y_1 y_2}}}{3MF}$ | $\frac{\sqrt{3 i g_2^2 \overline{y_1 y_2}}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3\lambda d a_1 2 \operatorname{Inv} 1 \overline{y_1 y_2}}}{6MF}$ | $\frac{\sqrt{6 i g_2^2 \overline{y_1 y_2}}}{3MF}$ | $\frac{\sqrt{3 i g_2^2 \overline{y_1 y_2}}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3\lambda d a_1 2 \operatorname{Inv} 1 \overline{y_1 y_2}}}{6MF}$ | $\frac{\sqrt{6 i g_2^2 \overline{y_1 y_2}}}{3MF}$ | $\frac{\sqrt{3 i g_2^2 \overline{y_1 y_2}}}{6MF}$ |



| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |

$$\beta_{C_{22514}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22514}}{d\mu}=\beta_{C_{22514}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2y_1y_2}{MF}$  |
| T2     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$  |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                 | T3                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|------------------------------------|----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                  | —                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                  | —                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                  | —                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                  | —                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                  | —                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                  | —                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                  | —                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                  | —                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                  | —                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                  | —                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  | —                                  | —                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  | —                                  | —                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  | —                                  | —                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  | —                                  | —                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  | —                                  | —                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                  | —                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                  | —                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                  | —                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                  | —                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                  | —                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ |

## Terms 2 of 3

| Column | Representative term structure                  |
|--------|------------------------------------------------|
| T1     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$      |
| T2     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$              |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                         | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|--------------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                          | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                          | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                          | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                          | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                          | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                          | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                   | —                                          | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                          | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                          | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                          | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                   | —                                          | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                   | —                                          | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                   | —                                          | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                   | —                                          | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                   | —                                          | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                          | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                          | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                          | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                          | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                          | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3\sqrt{11}ilambda12Inv2\bar{y}_1y_2}{22MF}$ | $-\frac{\sqrt{3}ig_1^2\bar{y}_1y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3\sqrt{11}ilambda12Inv2\bar{y}_1y_2}{22MF}$ | —                                          | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3\sqrt{11}ilambda12Inv2\bar{y}_1y_2}{22MF}$ | $-\frac{\sqrt{3}ig_1^2\bar{y}_1y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3\sqrt{11}ilambda12Inv2\bar{y}_1y_2}{22MF}$ | $-\frac{\sqrt{3}ig_1^2\bar{y}_1y_2}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3\sqrt{11}ilambda12Inv2\bar{y}_1y_2}{22MF}$ | $-\frac{9\sqrt{3}ig_1^2\bar{y}_1y_2}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\bar{y}_1y_2}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{2267}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2267}}{d\mu}=\beta_{C_{2267}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                            | —                                                            |

Terms 2 of 2

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                         | T2                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------|---------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                          | —                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                          | —                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                          | —                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                          | —                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                          | —                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                          | —                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                          | —                                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                          | —                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                          | —                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                          | —                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                          | —                                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                          | —                                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                          | —                                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                          | —                                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                          | —                                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                          | —                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                          | —                                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                          | —                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                          | —                                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                          | —                                                             |

$$\beta_{C_{2268}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2268}}{d\mu}=\beta_{C_{2268}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                         | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |

Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{MF}$       |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{y}e}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1}y_2}{3MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |



$$\beta_{C_{22611}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22611}}{d\mu}=\beta_{C_{22611}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$          |
| T2     | $\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                                            | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | —                                                             | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | —                                                             | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | —                                                             | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | —                                                             | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                             | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                             | —                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | —                                                             | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | —                                                             | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                            | —                                                             | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                            | —                                                             | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                            | —                                                             | —                                                           |

## Terms 2 of 3

| Column | Representative term structure                                               |  |
|--------|-----------------------------------------------------------------------------|--|
| T1     | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{M F}$                    |  |
| T2     | $\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$                    |  |
| T3     | $\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                          | T2                                                                   | T3                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                    | —                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                    | —                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                    | —                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                    | —                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                    | —                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                           | —                                                                    | —                                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                           | —                                                                    | —                                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                           | —                                                                    | —                                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                           | —                                                                    | —                                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                           | —                                                                    | —                                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                           | —                                                                    | —                                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                           | —                                                                    | —                                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                           | —                                                                    | —                                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                           | —                                                                    | —                                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                           | —                                                                    | —                                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | $-\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                           | $-\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                           | $-\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                           | $-\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                           | $-\frac{11 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{3^{\frac{3}{4}} i \lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | —                                                                    | —                                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                           | —                                                                    | —                                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                           | —                                                                    | —                                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                           | —                                                                    | —                                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                           | —                                                                    | —                                                                              |

# Terms 3 of 3

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$              |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} \overline{y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|--------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} \overline{y e}}{12 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} \overline{y e}}{12 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} \overline{y e}}{12 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{2 M F}$         | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} \overline{y e}}{12 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9 \cdot 3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} \overline{y e}}{12 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                                        |

$$\beta_{C_{22612}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22612}}{d\mu}=\beta_{C_{22612}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                           | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{2MF}$        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                            | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                            | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                            | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                            | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                            | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                            | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                            | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                            | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                            | —                                                            |

## Terms 2 of 3

| Column | Representative term structure                                        |
|--------|----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$           |
| T2     | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{M F}$ |
| T3     | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{M F}$               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                                     | T3                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | —                                                                      | —                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                                      | —                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                                      | —                                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                                      | —                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                                      | —                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                                      | —                                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                                      | —                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | —                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{2 M F}$         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | $\frac{3^{\frac{3}{4}} \lambda \text{Inv1} \overline{y_1 y_2}}{6 M F}$ | $\frac{9 \cdot 3^{\frac{3}{4}} g_1^2 \overline{y_1 y_2}}{8 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                                      | —                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                                      | —                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                                      | —                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                                      | —                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                                      | —                                                                |

Terms 3 of 3

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2y_1y_2}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                | T2                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                 | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                 | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                 | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                 | —                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                 | —                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                 | —                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                 | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                 | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                 | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{11 \cdot 3^{\frac{3}{4}}g_2^2y_1y_2}{8MF}$ | $-\frac{3^{\frac{3}{4}}ye\bar{y}_1y_2\bar{y}e}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                 | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                 | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                 | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                 | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                 | —                                                     |

$$\beta_{C_{22613}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22613}}{d\mu}=\beta_{C_{22613}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$         |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                           | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|----------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                            | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                            | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                            | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                            | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                            | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                            | —                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                            | —                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                            | —                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                            | —                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                            | —                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                            | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                            | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                            | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                            | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                            | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                            | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                            | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                            | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                            | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                            | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 2 of 3

| Column | Representative term structure                        |
|--------|------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                      | T3                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|---------------------------------------------------------|--------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                       | —                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                       | —                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                       | —                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                       | —                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                                       | —                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                                       | —                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                                       | —                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                                       | —                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$ | $\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | $\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$ | $\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$ | $\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$ | $\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$ | $\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$ |



Terms 3 of 3

| Column | Representative term structure                              |  |
|--------|------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{M F}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3} i y e \bar{y}_1 \bar{y}_2 \bar{y} e}{12 M F}$ |

$$\beta_{C_{22614}}^{(1)}$$

$$_{16\pi^2}\mu\frac{dC_{22614}}{d\mu}=\beta_{C_{22614}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                                           | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                            | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                            | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                | —                                                            | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                | —                                                            | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | —                                                            | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                | —                                                            | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                            | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                            | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                            | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                            | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                            | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $\frac{\sqrt{3\lambda da12Inv1y_1y_2}}{MF}$ |
| T2     | $\frac{\sqrt{6ig_2^2y_1y_2}}{MF}$           |
| T3     | $\frac{\sqrt{3ig_2^2y_1y_2}}{MF}$           |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                           | T2                                 | T3                                 |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|----------------------------------------------|------------------------------------|------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                  | —                                  |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                  | —                                  |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                  | —                                  |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                  | —                                  |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                  | —                                  |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                  | —                                  |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                  | —                                  |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                  | —                                  |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                  | —                                  |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                  | —                                  |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                  | —                                  |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                  | —                                  |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                  | —                                  |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                  | —                                  |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                  | —                                  |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                  | —                                  |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                  | —                                  |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                  | —                                  |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                  | —                                  |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                  | —                                  |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3\lambda da12Inv1y_1y_2}}{6MF}$ | $\frac{\sqrt{6ig_2^2y_1y_2}}{3MF}$ | $\frac{\sqrt{3ig_2^2y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3\lambda da12Inv1y_1y_2}}{6MF}$ | $\frac{\sqrt{6ig_2^2y_1y_2}}{3MF}$ | $\frac{\sqrt{3ig_2^2y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3\lambda da12Inv1y_1y_2}}{6MF}$ | $\frac{\sqrt{6ig_2^2y_1y_2}}{3MF}$ | $\frac{\sqrt{3ig_2^2y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3\lambda da12Inv1y_1y_2}}{6MF}$ | $\frac{\sqrt{6ig_2^2y_1y_2}}{3MF}$ | $\frac{\sqrt{3ig_2^2y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3\lambda da12Inv1y_1y_2}}{6MF}$ | $\frac{\sqrt{6ig_2^2y_1y_2}}{3MF}$ | $\frac{\sqrt{3ig_2^2y_1y_2}}{6MF}$ |

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |

$$\beta_{C_{2277}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{2277}}{d\mu}=\beta_{C_{2277}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                 |
|--------|-----------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_1}^2}{MF}$            |
| T2     | $\frac{lambdaS1Inv1\overline{y_1}^2}{MF}$     |
| T3     | $\frac{ye\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                          | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|---------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_1}^2}{4MF}$ | $-\frac{lambdaS1Inv1\overline{y_1}^2}{2MF}$ | $\frac{ye\overline{y_1}^2\overline{y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                           | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                           | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                           | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                              |

# Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda daS1Inv1\overline{y_1}^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                      | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                       | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                       | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$ | $-\frac{\sqrt{3}\lambda daS1Inv1\overline{y_1}^2}{6MF}$ | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                       | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                       |

$$\beta_{C_{2278}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2278}}{d\mu}=\beta_{C_{2278}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ilambdaS1Inv1\overline{y_1}^2}{MF}$      |
| T2     | $\frac{ig_2^2\overline{y_1}^2}{MF}$             |
| T3     | $\frac{iy_e\overline{y_1}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                    | T3                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------|---------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{ilambdaS1Inv1\overline{y_1}^2}{2MF}$ | $\frac{3ig_2^2\overline{y_1}^2}{4MF}$ | $-\frac{iy_e\overline{y_1}^2\overline{y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                     | —                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | —                                     | —                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                     | —                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                     | —                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                     | —                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                     | —                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                     | —                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                     | —                                                 |

Terms 2 of 2

| Column | Representative term structure                         |  |
|--------|-------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{MF}$           |  |
| T2     | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{MF}$    |  |
| T3     | $\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                  | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                   | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                   | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                   | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                   | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1}^2}{4MF}$ | $\frac{\sqrt{3}ilambdaS1Inv1\overline{y_1}^2}{6MF}$ | $-\frac{\sqrt{3}iye\overline{y_1}^2\overline{ye}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                   | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                   | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                   | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                   | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                   | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                   | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                   | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                   | —                                                        |



$$\beta_{C_{2279}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2279}}{d\mu}=\beta_{C_{2279}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $\frac{g_2^2 \overline{y_1 y_2}}{MF}$       |
| T2     | $\frac{g_1^2 \overline{y_1 y_2}}{MF}$       |
| T3     | $\frac{ye\overline{y_1 y_2 y \bar{e}}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                      | T2                                       | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------|------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                        | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                        | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                        | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                        | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                        | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{ye\overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ | —                                        | $-\frac{ye\overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{8MF}$  | $-\frac{ye\overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ | $\frac{3g_1^2 \overline{y_1 y_2}}{2MF}$  | $-\frac{ye\overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3g_2^2 \overline{y_1 y_2}}{4MF}$ | $\frac{27g_1^2 \overline{y_1 y_2}}{8MF}$ | $-\frac{ye\overline{y_1 y_2 y \bar{e}}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                       | —                                        | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                       | —                                        | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                       | —                                        | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                       | —                                        | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                       | —                                        | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                        | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                        | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                        | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                        | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                        | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                        | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                        | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                        | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                        | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                        | —                                             |

# Terms 2 of 3

| Column | Representative term structure                            |  |  |  |  |  |  |  |  |  |
|--------|----------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$ |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\lambda\text{Inv}2\overline{y_1y_2}}{MF}$         |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                        | T2                                                 | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------|----------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                                  | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                                  | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                                  | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                                  | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                                  | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                                  | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                                  | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                                  | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                                  | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                                  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                                  | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                                  | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                                  | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                                  | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                                  | —                                             |

Terms 3 of 3

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$            |
| T2     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{2MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$ | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                        |

$$\beta_{C_{22710}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22710}}{d\mu}=\beta_{C_{22710}}^{(1)}$$

# Terms 1 of 3

| Column | Representative term structure         |
|--------|---------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_1y_2}}{MF}$  |
| T2     | $\frac{ig_1^2\overline{y_1y_2}}{MF}$  |
| T3     | $\frac{iy_e\overline{y_1y_2y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                      | T3                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|-----------------------------------------|-----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                       | —                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                       | —                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                       | —                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                       | —                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                       | —                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $\frac{3ig_1^2\overline{y_1y_2}}{8MF}$  | $-\frac{iy_e\overline{y_1y_2y_e}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | —                                       | $-\frac{iy_e\overline{y_1y_2y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $\frac{3ig_1^2\overline{y_1y_2}}{8MF}$  | $-\frac{iy_e\overline{y_1y_2y_e}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $\frac{3ig_1^2\overline{y_1y_2}}{2MF}$  | $-\frac{iy_e\overline{y_1y_2y_e}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $\frac{27ig_1^2\overline{y_1y_2}}{8MF}$ | $-\frac{iy_e\overline{y_1y_2y_e}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                       | —                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                       | —                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                       | —                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                       | —                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                       | —                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                       | —                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                       | —                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                       | —                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                       | —                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                       | —                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                       | —                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                       | —                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                       | —                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                       | —                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                       | —                                       |

## Terms 2 of 3

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$ |  |
| T2     | $\frac{ilambda12Inv2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$        |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ | —                                             | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ | —                                             | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ | —                                             | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ | —                                             | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ | —                                             | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | $-\frac{ilambda12Inv2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{4MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                              |

Terms 3 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$           |
| T2     | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                   | T2                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                    | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                    | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                    | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                    | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                    | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                    | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ | $\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                             |

$$\beta_{C_{22711}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22711}}{d\mu}=\beta_{C_{22711}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                             | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                             | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                             | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                             | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                             | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |

## Terms 2 of 2

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$            |
| T2     | $\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$           |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                             | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                              | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                              | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                              | —                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                              | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |



$$\beta_{C_{22712}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22712}}{d\mu}=\beta_{C_{22712}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{3}i\lambda bda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                        | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-----------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                         | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                         | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                         | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                         | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                         | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                         | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                         | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                         | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                                         | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                         | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                         | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                         | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                         | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                         | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                         | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                         | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                         | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}\lambda bda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{\sqrt{3}i g_2^2 \overline{y_1 y_2}}{M F}$                    |
| T3     | $\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{M F}$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                      | T2                                                   | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$\beta_{C_{22715}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22715}}{d\mu}=\beta_{C_{22715}}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$          |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                     | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|----------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                      | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                      | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                      | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                      | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                      | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                      | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                      | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                      | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |

## Terms 2 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                              | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                               | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                               | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                               | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                               | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | —                                               | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{22716}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22716}}{d\mu}=\beta_{C_{22716}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                      |  |
|--------|----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$ |  |
| T2     | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$        |  |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$        |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                            | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                            | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                            | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                            | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{MF}$ |
| T2     | $\frac{\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{MF}$               |
| T3     | $\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                               | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}\lambda a_{12}Inv2\bar{y}_1\bar{y}_2}{22MF}$ | $\frac{49\sqrt{3}g_2^2\bar{y}_1\bar{y}_2}{24MF}$ | $-\frac{\sqrt{3}ye\bar{y}_1\bar{y}_2\bar{y}e}{12MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |



$$\beta_{C_{2288}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2288}}{d\mu}=\beta_{C_{2288}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\textit{lambdaS1Inv1}\overline{y_1}^2}{MF}$ |
| T2     | $\frac{g_2^2\overline{y_1}^2}{MF}$                 |
| T3     | $\frac{ye\overline{y_1}^2\overline{ye}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                   | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|--------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\textit{lambdaS1Inv1}\overline{y_1}^2}{2MF}$ | $\frac{3g_2^2\overline{y_1}^2}{4MF}$ | $-\frac{ye\overline{y_1}^2\overline{ye}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                    | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                   | —                                    | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                   | —                                    | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |

# Terms 2 of 2

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{MF}$             |
| T2     | $\frac{\sqrt{3}\lambda bdaS1Inv1\overline{y_1}^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{MF}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                                      | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | —                                                       | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                                       | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_1}^2}{4MF}$ | $\frac{\sqrt{3}\lambda bdaS1Inv1\overline{y_1}^2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}^2\overline{y_e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                                       | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |

$$\beta_{C_{2289}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{2289}}{d\mu}=\beta_{C_{2289}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_1y_2}}{MF}$                |
| T2     | $\frac{ig_1^2\overline{y_1y_2}}{MF}$                |
| T3     | $\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                      | T2                                       | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------|------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                        | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                        | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                        | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                        | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                        | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{8MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | —                                        | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{8MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $-\frac{3ig_1^2\overline{y_1y_2}}{2MF}$  | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3ig_2^2\overline{y_1y_2}}{4MF}$ | $-\frac{27ig_1^2\overline{y_1y_2}}{8MF}$ | $-\frac{\sqrt{3}ilambda12Inv2\overline{y_1y_2}}{3MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                       | —                                        | —                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                       | —                                        | —                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                       | —                                        | —                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                       | —                                        | —                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                       | —                                        | —                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                        | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                        | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                        | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                        | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                        | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                       | —                                        | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                       | —                                        | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                       | —                                        | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                       | —                                        | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                       | —                                        | —                                                     |

## Terms 2 of 3

| Column | Representative term structure     |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{iyey_1y_2ye}{MF}$          |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{ilambda12Inv2y_1y_2}{MF}$  |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$ |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                        | T2                                | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------|-----------------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{iyey_1y_2ye}{4MF}$ | —                                 | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{iyey_1y_2ye}{4MF}$ | —                                 | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{iyey_1y_2ye}{4MF}$ | —                                 | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{iyey_1y_2ye}{4MF}$ | —                                 | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{iyey_1y_2ye}{4MF}$ | —                                 | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                         | $\frac{ilambda12Inv2y_1y_2}{3MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                         | $\frac{ilambda12Inv2y_1y_2}{3MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                         | $\frac{ilambda12Inv2y_1y_2}{3MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                         | $\frac{ilambda12Inv2y_1y_2}{3MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                         | $\frac{ilambda12Inv2y_1y_2}{3MF}$ | $\frac{\sqrt{3}ig_2^2y_1y_2}{4MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                 | —                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | —                                 | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                 | —                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                 | —                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                 | —                                  |

Terms 3 of 3

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2y_1y_2}{MF}$                      |
| T2     | $\frac{\sqrt{3}iye\overline{y_1y_2}y\overline{e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                   | T2                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                    | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                    | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                    | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $-\frac{\sqrt{3}iye\overline{y_1y_2}y\overline{e}}{12MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                    | $-\frac{\sqrt{3}iye\overline{y_1y_2}y\overline{e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ig_1^2y_1y_2}{8MF}$  | $-\frac{\sqrt{3}iye\overline{y_1y_2}y\overline{e}}{12MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}ig_1^2y_1y_2}{2MF}$  | $-\frac{\sqrt{3}iye\overline{y_1y_2}y\overline{e}}{12MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}ig_1^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\overline{y_1y_2}y\overline{e}}{12MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                    | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                    | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                    | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                    | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                    | —                                                         |

$$\beta_{C_{22810}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22810}}{d\mu}=\beta_{C_{22810}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure            |
|--------|------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_1y_2}}{MF}$      |
| T2     | $\frac{g_1^2\overline{y_1y_2}}{MF}$      |
| T3     | $\frac{ye\overline{y_1y_2y\bar{e}}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                     | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|----------------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                      | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                      | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                      | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                      | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                      | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{3g_2^2\overline{y_1y_2}}{4MF}$ | $\frac{3g_1^2\overline{y_1y_2}}{8MF}$  | $-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{3g_2^2\overline{y_1y_2}}{4MF}$ | —                                      | $-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{3g_2^2\overline{y_1y_2}}{4MF}$ | $\frac{3g_1^2\overline{y_1y_2}}{8MF}$  | $-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{3g_2^2\overline{y_1y_2}}{4MF}$ | $\frac{3g_1^2\overline{y_1y_2}}{2MF}$  | $-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{3g_2^2\overline{y_1y_2}}{4MF}$ | $\frac{27g_1^2\overline{y_1y_2}}{8MF}$ | $-\frac{ye\overline{y_1y_2y\bar{e}}}{4MF}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                      | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                      | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                      | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                      | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                      | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                      | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                      | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                      | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                      | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                      | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                      | —                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                      | —                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                      | —                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                      | —                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                      | —                                          |

# Terms 2 of 3

| Column | Representative term structure                            |  |
|--------|----------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{MF}$ |  |
| T2     | $\frac{\lambda\text{Inv}2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                        | T2                                                 | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------|----------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                                  | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                                  | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                                  | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                                  | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                                  | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | —                                                  | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                         | $-\frac{\lambda\text{Inv}2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{4MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                                  | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                                  | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                                  | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                                  | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                                  | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                         | —                                                  | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                         | —                                                  | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                         | —                                                  | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                         | —                                                  | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                         | —                                                  | —                                             |

Terms 3 of 3

| Column |  | Representative term structure                          |
|--------|--|--------------------------------------------------------|
| T1     |  | $\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{MF}$            |
| T2     |  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1}y_2}{2MF}$  | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1}y_2}{8MF}$ | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                        |



$$\beta_{C_{22811}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22811}}{d\mu}=\beta_{C_{22811}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$          |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                    | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|---------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                     | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                     | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                     | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                     | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                     | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                     | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                     | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                     | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                     | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                     | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  | —                                     | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  | —                                     | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  | —                                     | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  | —                                     | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  | —                                     | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                     | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                     | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                     | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                     | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                     | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |

## Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$          |
| T2     | $\frac{\sqrt{3}ilambda12In\nu1\overline{y_1}y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{MF}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                     | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                      | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                      | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                      | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                      | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                      | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                      | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                      | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                      | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                      | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                      | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                      | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                      | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                      | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                      | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                      | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                      | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                      | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                      | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                      | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                      | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12In\nu1\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | $\frac{\sqrt{3}ilambda12In\nu1\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$  | $\frac{\sqrt{3}ilambda12In\nu1\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{2MF}$  | $\frac{\sqrt{3}ilambda12In\nu1\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$ | $\frac{\sqrt{3}ilambda12In\nu1\overline{y_1}y_2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_1}y_2ye}{12MF}$ |

$$\beta_{C_{22812}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22812}}{d\mu}=\beta_{C_{22812}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                             | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                             | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                             | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                             | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                             | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |

## Terms 2 of 2

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$            |
| T2     | $\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$           |
| T3     | $\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                             | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                              | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                              | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                              | —                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                              | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                              | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                              | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                              | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                              | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                              | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y_e}}{12MF}$ |

$$\beta_{C_{22815}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22815}}{d\mu}=\beta_{C_{22815}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$                |
| T2     | $\frac{\sqrt{11}\text{lambda}12Inv2\overline{y_1}y_2}{MF}$ |
| T3     | $\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{MF}$                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                                                             | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                              | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                              | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                              | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                              | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                              | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                | —                                                              | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                | —                                                              | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                | —                                                              | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                | —                                                              | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                | —                                                              | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                | —                                                              | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                | —                                                              | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                | —                                                              | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                | —                                                              | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                | —                                                              | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                | —                                                              | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                | —                                                              | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                | —                                                              | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                | —                                                              | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                | —                                                              | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\text{lambda}12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\text{lambda}12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\text{lambda}12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\text{lambda}12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$ | $-\frac{3\sqrt{11}\text{lambda}12Inv2\overline{y_1}y_2}{22MF}$ | $-\frac{\sqrt{6}g_2^2\overline{y_1}y_2}{3MF}$ |

## Terms 2 of 3

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                      | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|---------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                                       | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                       | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                                       | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                                       | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                       | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                       | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                       | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                       | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                       | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$ |

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}ye\overline{y}_1\overline{y}_2\overline{y}e}{12MF}$ |

$\beta_{C_{22816}}^{(1)}$

$16\pi^2\mu\frac{dC_{22816}}{d\mu}=\beta_{C_{22816}}^{(1)}$

Terms 1 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$          |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$         |
| T3     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                     | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|----------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                      | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                      | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                      | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                      | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                      | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                      | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                      | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                      | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                      | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                      | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                      | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$ | $-\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ |



## Terms 2 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                              | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              | —                                               | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                               | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              | —                                               | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                               | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                               | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                               | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                               | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                               | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                               | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | —                                               | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}iye\bar{y}_1\bar{y}_2ye}{12MF}$ |

$$\beta_{C_{2299}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{2299}}{d\mu}=\beta_{C_{2299}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{g_2^2\overline{y_2}^2}{MF}$           |
| T2     | $\frac{lambdaS2Inv1\overline{y_2}^2}{MF}$    |
| T3     | $\frac{ye\overline{y_2}^2\overline{ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                          | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|---------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                           | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_2^2\overline{y_2}^2}{4MF}$ | $-\frac{lambdaS2Inv1\overline{y_2}^2}{2MF}$ | $\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                           | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                           | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                           | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                           | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                           | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                           | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                           | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                           | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                           | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                           | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                           | —                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                           | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                           | —                                             |

# Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y}_2^2}{MF}$            |
| T2     | $\frac{\sqrt{3}\lambda daS2Inv1\overline{y}_2^2}{MF}$ |
| T3     | $\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}e}{MF}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                                      | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|---------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                                       | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                                       | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y}_2^2}{4MF}$ | $-\frac{\sqrt{3}\lambda daS2Inv1\overline{y}_2^2}{6MF}$ | $\frac{\sqrt{3}ye\overline{y}_2^2\overline{y}e}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                                       | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                                       | —                                                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                                       | —                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                                       | —                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                                       | —                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                                       | —                                                      |

$$\beta_{C_{22910}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22910}}{d\mu}=\beta_{C_{22910}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{ig_2^2\overline{y_2}^2}{MF}$             |
| T2     | $\frac{ilambdaS2Inv1\overline{y_2}^2}{MF}$      |
| T3     | $\frac{iy_e\overline{y_2}^2\overline{y_e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                     | T2                                           | T3                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------|----------------------------------------------|--------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                            | —                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                            | —                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3ig_2^2\overline{y_2}^2}{4MF}$ | $-\frac{ilambdaS2Inv1\overline{y_2}^2}{2MF}$ | $\frac{iy_e\overline{y_2}^2\overline{y_e}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                      | —                                            | —                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                      | —                                            | —                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                      | —                                            | —                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                            | —                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                            | —                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                            | —                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                            | —                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                            | —                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                            | —                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                            | —                                                |

Terms 2 of 2

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{MF}$           |
| T2     | $\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{MF}$    |
| T3     | $\frac{\sqrt{3}iye\overline{y_2}^2\overline{y}e}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                                   | T3                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|------------------------------------------------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                    | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                    | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                                    | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                                    | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\sqrt{3}ig_2^2\overline{y_2}^2}{4MF}$ | $-\frac{\sqrt{3}ilambdaS2Inv1\overline{y_2}^2}{6MF}$ | $\frac{\sqrt{3}iye\overline{y_2}^2\overline{y}e}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                                    | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                    | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                    | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                    | —                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                    | —                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                    | —                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                    | —                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                    | —                                                       |

$$\beta_{C_{22911}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22911}}{d\mu}=\beta_{C_{22911}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                                 | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |

## Terms 2 of 2

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{MF}$             |
| T2     | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$         | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$         | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{2MF}$         | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9 \cdot 3^{\frac{3}{4}} ig_1^2 \overline{y_1 y_2}}{8MF}$ | $\frac{3^{\frac{3}{4}} iye \overline{y_1 y_2} \overline{y e}}{12MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                    |



$$\beta_{C_{22912}}^{(1)}$$

$$16\pi^2\mu\frac{dC_{22912}}{d\mu}=\beta_{C_{22912}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                         | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                          | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                          | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                          | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                          | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | —                                                          | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}}\lambda^{12}Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                          | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                          | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                          | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                          | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                          | —                                                            |

## Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{MF}$       |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                               | T2                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------|-----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                | —                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                | —                                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                | —                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                | —                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                | —                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                | —                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                | —                                                                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                | —                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                | —                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                | —                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2\overline{ye}}{12MF}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                | —                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                | —                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                | —                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                | —                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                | —                                                                     |

$$\beta_{C_{22913}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22913}}{d\mu}=\beta_{C_{22913}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$         |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                                    | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|-------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                                     | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                                     | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             | —                                                     | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                                     | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                                     | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                                     | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                                     | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                                     | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                                     | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $\frac{\sqrt{11}ilambda12Inv2y_1y_2}{MF}$    |
| T2     | $\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$            |
| T3     | $\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                                   | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|--------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                    | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                    | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                                    | —                                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                                    | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                                    | —                                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                                    | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                                    | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                                    | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                                    | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                                    | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                                    | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}ilambda12Inv2y_1y_2}{22MF}$ | $\frac{11\sqrt{3}ig_2^2y_1y_2}{8MF}$ | $-\frac{\sqrt{3}iye\bar{y}_1y_2y\bar{e}}{12MF}$ |

$$\beta_{C_{22914}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{22914}}{d\mu}=\beta_{C_{22914}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                           | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                               | —                                                            | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                               | —                                                            | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                            | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                            | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                            | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                            | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                            | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                            | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                            | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3\sqrt{11}\lambda bda12Inv2\overline{y_1y_2}}{22MF}$ | $-\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                     |  |
|--------|-------------------------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{M F}$ |  |
| T2     | $\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{M F}$                 |  |
| T3     | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{M F}$                 |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                                   | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                    | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                    | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                    | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                    | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{\sqrt{3}\lambda b d a 12 I n v 1 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$\beta_{C_{221010}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{221010}}{d\mu}=\beta_{C_{221010}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\textit{lambdaS2Inv1}\overline{y_2}^2}{MF}$ |
| T2     | $\frac{g_2^2\overline{y_2}^2}{MF}$                 |
| T3     | $\frac{ye\overline{y_2}^2\overline{ye}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                  | T2                                   | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------|--------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{\textit{lambdaS2Inv1}\overline{y_2}^2}{2MF}$ | $\frac{3g_2^2\overline{y_2}^2}{4MF}$ | $-\frac{ye\overline{y_2}^2\overline{ye}}{4MF}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                   | —                                    | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                   | —                                    | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                   | —                                    | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                   | —                                    | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                   | —                                    | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                   | —                                    | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                   | —                                    | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                   | —                                    | —                                              |

Terms 2 of 2

| Column | Representative term structure                          |  |  |  |  |  |  |  |  |  |
|--------|--------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\frac{\sqrt{3}g_2^2\overline{y_2}^2}{MF}$             |  |  |  |  |  |  |  |  |  |
| T2     | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y_2}^2}{MF}$ |  |  |  |  |  |  |  |  |  |
| T3     | $\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{MF}$  |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                                                      | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | —                                                       | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | —                                                       | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{\sqrt{3}g_2^2\overline{y_2}^2}{4MF}$ | $\frac{\sqrt{3}\lambda bdaS2Inv1\overline{y_2}^2}{6MF}$ | $-\frac{\sqrt{3}ye\overline{y_2}^2\overline{y_e}}{12MF}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | —                                                       | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                                                       | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                                                       | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                                                       | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                                                       | —                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                                                       | —                                                        |



$$\beta_{C_{221011}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{221011}}{d\mu}=\beta_{C_{221011}}^{(1)}$$

### Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$          |
| T2     | $\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                            | T2                                                            | T3                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                             | —                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                             | —                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                             | —                                                             | —                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                             | —                                                             | —                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                             | —                                                             | —                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                             | —                                                             | —                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | —                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{2MF}$        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda 12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{9\cdot 3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                             | —                                                             | —                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                             | —                                                             | —                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                             | —                                                             | —                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                             | —                                                             | —                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                             | —                                                             | —                                                           |

Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{MF}$       |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                              | T2                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                               | —                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                               | —                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                               | —                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                               | —                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                               | —                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                               | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                               | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                               | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                               | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                               | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                               | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                               | —                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                               | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                               | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                               | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{3^{\frac{3}{4}} ye\overline{y_1}y_2y\overline{e}}{12MF}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1}y_2}{3MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                               | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                               | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                               | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                               | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                               | —                                                                    |

$$\beta_{C_{221012}}^{(1)}$$

$$_{16\pi^2}\,_{\mu}\frac{dC_{221012}}{d\mu}=\beta_{C_{221012}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                                 | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                                  | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                                  | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                                  | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                                  | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                                  | —                                                            |

## Terms 2 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$   |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y e}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|---------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   | —                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | —                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | —                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | —                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | —                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   | —                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | —                                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | —                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | —                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | —                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y e}{12 M F}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y e}{12 M F}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$         | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y e}{12 M F}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{2 M F}$         | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y e}{12 M F}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9 \cdot 3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y e}{12 M F}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                                             |

$$\beta_{C_{221013}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dC_{221013}}{d\mu}=\beta_{C_{221013}}^{(1)}$$

## Terms 1 of 2

| Column | Representative term structure                       |  |
|--------|-----------------------------------------------------|--|
| T1     | $\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$  |  |
| T2     | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$         |  |
| T3     | $\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                            | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                    | —                                             | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                    | —                                             | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                             | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                             | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                             | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                             | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                             | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                             | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                             | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | —                                             | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{2MF}$  | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{9\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$ |

## Terms 2 of 2

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2 y_1 y_2}{MF}$  |
| T2     | $\frac{\sqrt{3}ye y_1 y_2 ye}{MF}$  |
| T3     | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                    | T2                                    | T3                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------|---------------------------------------|--------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                     | —                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                     | —                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                     | —                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                     | —                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                     | —                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                     | —                                     | —                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                     | —                                     | —                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                     | —                                     | —                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                     | —                                     | —                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                     | —                                     | —                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                     | —                                     | —                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                     | —                                     | —                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                     | —                                     | —                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                     | —                                     | —                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                     | —                                     | —                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                     | —                                     | —                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                     | —                                     | —                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                     | —                                     | —                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                     | —                                     | —                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                     | —                                     | —                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{11\sqrt{3}g_2^2 y_1 y_2}{8MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 ye}{12MF}$ | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{6MF}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{11\sqrt{3}g_2^2 y_1 y_2}{8MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 ye}{12MF}$ | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{6MF}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{11\sqrt{3}g_2^2 y_1 y_2}{8MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 ye}{12MF}$ | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{6MF}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{11\sqrt{3}g_2^2 y_1 y_2}{8MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 ye}{12MF}$ | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{6MF}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{11\sqrt{3}g_2^2 y_1 y_2}{8MF}$ | $-\frac{\sqrt{3}ye y_1 y_2 ye}{12MF}$ | $\frac{\sqrt{3}ig_2^2 y_1 y_2}{6MF}$ |

$$\beta_{C_{221014}}^{(1)}$$

$$16\pi^2\,\mu\frac{dC_{221014}}{d\mu}=\beta_{C_{221014}}^{(1)}$$

# Terms 1 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\frac{\sqrt{3}g_2^2y_1y_2}{MF}$                    |
| T2     | $\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{MF}$ |
| T3     | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                                    | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|-------------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                     | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                     | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                     | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                     | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                                     | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                  | —                                                     | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                  | —                                                     | —                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                  | —                                                     | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                  | —                                                     | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                  | —                                                     | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                  | —                                                     | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                  | —                                                     | —                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                  | —                                                     | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                  | —                                                     | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                  | —                                                     | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                     | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                     | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                     | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                     | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                                     | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{8MF}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{8MF}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{\sqrt{3}ig_1^2\overline{y}_1y_2}{2MF}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$ | $-\frac{\sqrt{3}ilambda12Inv1\overline{y}_1y_2}{6MF}$ | $\frac{9\sqrt{3}ig_1^2\overline{y}_1y_2}{8MF}$ |

## Terms 2 of 2

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $\frac{\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{\sqrt{3}i g_2^2 \overline{y_1 y_2}}{M F}$                    |
| T3     | $\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{M F}$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                      | T2                                                   | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                    | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                    | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                    | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{3\sqrt{11}i\lambda b d a 12 I n v 2 \overline{y_1 y_2}}{22 M F}$ | $\frac{11\sqrt{3}i g_2^2 \overline{y_1 y_2}}{8 M F}$ | $-\frac{\sqrt{3}i y e \overline{y_1 y_2 y e}}{12 M F}$ |